-->

Infertility

Number: 0327

Table Of Contents

Policy
Applicable CPT / HCPCS / ICD-10 Codes
Background
References

Note: This policy has updates that may not be in effect for certain plans until plan renewal; please check plan documents.

Policy

Scope of Policy

This Clinical Policy Bulletin addresses services for the management of infertility for commercial medical plans. For Medicare criteria, see Medicare Part B Criteria.

The American Society for Reproductive Medicine (ASRM, 2023) has defined infertility as a "disease, condition, or status characterized by the inability to achieve a successful pregnancy based on a patient's medical, sexual, and reproductive history, age, physical findings, diagnostic testing, or any combination of those factors; or the need for medical intervention, including, but not limited to, the use of donor gametes or donor embryos in order to achieve a successful pregnancy either as an individual or with a partner; or in patients having regular, unprotected intercourse and without any known etiology for either partner suggestive of impaired reproductive ability, evaluation should be initiated at 12 months when the female partner is under 35 years of age and at 6 months when the female partner is 35 years of age or older. Nothing in this definition shall be used to deny or delay treatment to any individual, regardless of relationship status or sexual orientation." Please see below for medical necessity criteria for specific infertility related procedures.

Note: Requires Precertification:

Precertification of Cetrotide (cetrorelix acetate), ganirelix acetate, Follistim AQ (follitropin beta), Gonal-F (follitropin alfa), Menopur (menotropins), Novarel (chorionic gonadotropin), Pregnyl (chorionic gonadotropin), Ovidrel (choriogonadotropin alfa), and chorionic gonadotropin is required of all Aetna participating providers and members in applicable plan designs. For precertification, call (866) 782-2779 or fax (860) 754-2515. For Statement of Medical Necessity (SMN) precertification forms, see Specialty Pharmacy Precertification.

Note: Medical/Pharmacy Benefit Alignment of Coverage for Infertility Drugs and Procedures:

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

Medical necessity review of infertility drugs by Aetna Specialty Pharmacy Guideline Management may be bypassed for infertility ART drugs that are for use with infertility medical procedures if the infertility procedure has been approved for coverage under the member's Aetna medical benefit plan. During precertification, a medical authorization number and confirmation of the approval of the infertility procedures will be required to bypass medical necessity review by Specialty Pharmacy Guideline Management. **Note:** Some plans may require medical necessity review of all infertility drugs by Aetna Specialty Guideline Management. Members of these plans must undergo Specialty Pharmacy Guideline Management medical necessity review of all infertility drugs regardless of whether the drugs are for use with approved infertility medical procedures.

Notes:

1. For purposes of this entire policy, coverage is subject to the terms and conditions of the member's benefit plan. Coverage may vary due to state mandates and plan customization; please check plan documents.
2. For plans with an Advanced Reproductive Technology (ART) benefit (please check benefit plan descriptions), less invasive therapeutic approaches should be attempted prior to undertaking more invasive procedures, as directed by an appropriate, licensed medical specialist. The medical necessity criteria in the Advanced Reproductive Technology section below are met for persons who are unable to conceive after an appropriate trial of egg-sperm contact. For women under 35 years of age, an appropriate trial of egg-sperm contact requires 12 months of regular intravaginal inseminations or 4 cycles of timed intrauterine or intracervical inseminations, documented in the medical record. For women 35 years of age and older, an appropriate trial of egg-sperm contact requires 6 months of regular intravaginal inseminations or 3 cycles of timed intrauterine or intracervical insemination, documented in the medical record. This requirement applies to all individuals regardless of sexual orientation or the availability of a reproductive partner. For IVF procedures under the ART benefit, these requirements for a trial of egg-sperm contact are waived for persons for whom intravaginal, intrauterine, or intracervical inseminations would not be expected to be effective and IVF procedures are the only effective treatment (see In Vitro Fertilization (IVF) Procedures subsection below.)
3. Most plans exclude coverage of infertility services for persons who have had a previous sterilization procedure, including tubal sterilization and vasectomy, with or without surgical reversal, and for persons who have undergone a hysterectomy. Please check benefit plan descriptions for details. In addition, infertility services for persons who have undergone voluntary sterilization procedures are not covered because such services are the result of an elective procedure intended to prevent conception.
4. Some plans exclude coverage of Advanced Reproductive Technology (ART) using a woman's own eggs for women with poor ovarian reserve. Ovarian reserve is determined by measurement of menstrual cycle day 3 serum follicle-stimulating hormone (FSH) drawn after the normal onset of menstruation, or after progesterone induced menstruation for women who do not reliably menstruate. For women 39 years of age and older, ovarian responsiveness is determined by measurement of day 3 FSH obtained within the prior 6 months. For women who are less than 40 years of age, the day 3 FSH must be less than 19 mIU/mL in their most recent laboratory test to use their own eggs. For women 40 years of age and older, their unmedicated day 3 FSH must be less than 19 mIU/mL in all prior tests to use their own eggs. Please check benefit plan descriptions.
5. ART services for women 40 years of age and older with natural menopause is not covered because it is not considered medically necessary treatment of disease; natural menopause is not considered a disease. For women 40 years of age and older, their unmedicated day 3 FSH must be less than 19 mIU/mL in all prior tests to document that they are not menopausal and eligible for coverage of ART. Women who are less than 40 years of age who have a day 3 FSH greater than 19 mIU/L are considered to have the disease of premature ovarian failure (also known as premature ovarian insufficiency, primary ovarian insufficiency, or hypergonadotropic hypogonadism). For women with premature ovarian failure, ART (in vitro fertilization) services are considered medically necessary until they reach 45 years of age. Women 40 years of age and older with premature ovarian failure may submit a new unmedicated D3 FSH level to utilize her own oocytes (even if she has had an elevated D3 FSH level above 40 years of age in the past). For women 40 years of and older with premature ovarian failure, the day 3 FSH must be less than 19 mIU/mL in their most recent laboratory test to use their own eggs.

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

Please check benefit plan descriptions.

I. Medical Necessity

A. Females: Basic Infertility Services

The following services are considered medically necessary:

1. History and Physical Examination

Basal body temperature;

2. Laboratory Studies

a. Anti-adrenal antibodies for apparently spontaneous primary ovarian insufficiency (premature ovarian failure);
b. Anti-sperm antibodies (e.g., immunobead or mixed antiglobulin method);
c. Chlamydia trachomatis screening (see CPB 0433 - Chlamydia Trachomatis - Screening and Diagnosis);
d. Fasting and 2 hours post 75 gram glucose challenge levels;
e. Lipid panel (total cholesterol, HDL cholesterol, triglycerides);
f. Post-coital testing (PCT) (Simms-Huhner test) of cervical mucus;
g. Rubella serology;
h. Testing for viral status (HIV, hepatitis B, hepatitis C);
i. Serum hormone levels:

i. Androgens (testosterone, androstenedione, dehydroepiandrosterone sulfate (DHEA-S) if there is evidence of hyperandrogenism (e.g., hirsuitism, acne, signs of virilization) or ovulatory dysfunction;
ii. Anti-mullerian hormone (AMH), for the following indications:

a. assessing menopausal status, including premature ovarian failure;
b. assessing ovarian status, including ovarian reserve and ovarian responsiveness, as part of an evaluation for infertility and assisted reproduction protocols such as in vitro fertilization;

iii. Gonadotropins (serum follicle-stimulating hormone [FSH], luteinizing hormone [LH]) for women with irregular menstrual cycles (see Appendix for medical necessity limitations) or age-related ovulatory dysfunction. **Note**: Aetna considers urinary FSH testing to be experimental, investigational, or unproven. Serum, not urinary, FSH is the standard of care for determination of menopausal status (AACE, 1999; NAMS, 2000; SOGC, 2002);
iv. Human chorionic gonadotrophin (hCG) (see Appendix for medical necessity limitations);
v. Prolactin for women with an ovulatory disorder, galactorrhea, or a pituitary tumor;
vi. Progestins (progesterone, 17-hydroxyprogesterone) (see Appendix for medical necessity limitations);
vii. Estrogens (estradiol) (see Appendix for medical necessity limitations);
viii. Thyroid stimulating hormone (TSH) for women with symptoms of thyroid disease;
ix. Adrenocortitrophic hormone (ACTH) for ruling out Cushing's syndrome or Addison's disease in women who are amenorrheic;
x. Clomiphene citrate challenge test;

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

j. Karyotype testing for couples with recurrent pregnancy loss (2 or more consecutive spontaneous abortions) (see CPB 0348 - Recurrent Pregnancy Loss);

3. **Artificial Insemination(s) (intrauterine, intracervical, or intravaginal)**

Note: Some Aetna benefit plans may exclude coverage of artificial insemination (AI). For Aetna benefit plans that cover artificial insemination, coverage may be limited to a maximum number of cycles per lifetime. Please check benefit plan descriptions.

4. **Diagnostic Procedures**

The following diagnostic procedures are considered medically necessary:

a. CT or MR imaging of sella turcica is considered medically necessary if prolactin is elevated;
b. Endometrial biopsy;
c. Hysterosalpingography (hysterosalpingogram [HSG]) or hysterosalpingo-contrast-ultrasonography (HyCoSy) to screen for tubal occlusion;

   The following are considered experimental, investigational, or unproven to screen for tubal occlusion (not an all-inclusive list) because of a lack of reliable evidence of effectiveness:

   i. Sonohysterosalpingography or saline hysterosalpingography (e.g., Femvue);
   ii. Hysterosalpingo-foam sonography (HyFoSy) (e.g. ExEm Foam);
d. Hysteroscopy, salpingoscopy (falloscopy), hydrotubation where clinically indicated;
e. Laparoscopy and chromotubation (contrast dye) to assess tubal and other pelvic pathology, and to follow-up on hysterosalpingography abnormalities;
f. Sonohysterography to evaluate the uterus;
g. Ultrasound (e.g., ovarian, transvaginal, pelvic) (see Appendix for medical necessity limitations);
h. Monitoring of ovarian response to ovulatory stimulants:

   i. Estradiol (see Appendix for medical necessity limitations);
   ii. FSH (see Appendix for medical necessity limitations);
   iii. hCG quantitative (see Appendix for medical necessity limitations);
   iv. LH assay (see Appendix for medical necessity limitations);
   v. Progesterone (see Appendix for medical necessity limitations);
   vi. Serial ovarian ultrasounds are considered medically necessary for cycle monitoring (see Appendix for medical necessity limitations);

5. **Non-Surgical Treatments**

The following non-surgical treatments are considered medically necessary:

a. Aromatase inhibitors (e.g., anastrozole [Arimidex], exemestane [Aromasin], and letrozole [Femara]);
b. Corticosteroids (e.g., dexamethasone, prednisone);
c. Estrogens (e.g., estrone and conjugated estrogens [Premarin]);
d. Hepatitis B vaccination of partners of people with hepatitis B;
e. Lutropin alfa (Luveris) for use in combination with human FSH to stimulate follicular development in infertile hypo-gonadotropic hypo-gonadal women or in women with a profound LH deficiency defined as LH less than 1.2 International Units/L;

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

f. Metformin (Glucophage) for women with WHO Group II anovulatory disorders such as polycystic ovarian syndrome;
g. Progestins (oral, topical gel (8 % progesterone) (Crinone 8 %, Prochieve 8 %) or intramuscular progestins and progesterone vaginal suppositories (Endometrin), see CPB 0510 - Progestins);
h. Prolactin inhibitors (bromocriptine (Parlodel), cabergoline (Dostinex), peroglide (Permax)) for women with ovulatory disorders due to hyperprolactinemia;
i. Rubella vaccination of women susceptible to rubella;
j. Tamoxifen (Novaldex) or oral clomiphene citrate (Clomid, Serophene) for ovulation induction;

Note: The medications listed above may not be covered for members without pharmacy benefit plans; in addition, some pharmacy benefit plans may exclude or limit coverage of some or all of these medications. Please check benefit plan descriptions for details.

6. **Infertility Surgery**

The following are considered medically necessary:

a. Hysteroscopic adhesiolysis for women with amenorrhea who are found to have intrauterine adhesions;
b. Hysteroscopic or fluoroscopic tubal cannulation (salpingostomy, fimbrioplasty), selective salpingography plus tubal catheterization, or transcervical balloon tuboplasty for women with proximal tubal obstruction (see CPB 0347 - Transcervical Balloon Tuboplasty);
c. Laparoscopic cystectomy for women with ovarian endometriomas;
d. Laparoscopy for treatment of pelvic pathology;
e. Open or laparoscopic resection, vaporization, or fulguration of endometriosis implants plus adhesiolysis in women with endometriosis;
f. Ovarian wedge resection or ovarian drilling for women with WHO Group II ovulation disorders such as polycystic ovarian syndrome who have not responded to clomiphene citrate;
g. Removal of myomas, uterine septa, cysts, ovarian tumors, and polyps;
h. Surgical tubal reconstruction (unilateral or bilateral tubal microsurgery, laparoscopic tubal surgery, tuboplasty and tubal anastomosis) for women with mid or distal tubal occlusion and for women with proximal tubal disease where tubal cannulation has failed or where severe proximal tubal disease precludes the likelihood of successful cannulation;
i. Tubal ligation (salpingectomy) for women with hydrosalpinges who are contemplating in vitro fertilization, as this has been demonstrated to improve the chance of a live birth before in-vitro fertilization treatment;
j. Cervicectomy/trachelectomy is an acceptable alternative to hysterectomy for treatment of early stage (IA2 or small IB1) cervical adenocarcinoma in women who wish to preserve their fertility.

B. **Male Infertility**

The following services are considered medically necessary:

1. **History and Physical Examination**

2. **Laboratory Studies**

a. Anti-sperm antibodies (e.g., immunobead or mixed antiglobulin method);

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

b. Cultures:

   i. Prostatic secretion;
   ii. Semen;
   iii. Urine;

c. Serum hormone levels:

   i. 17-hydroxyprogesterone;
   ii. Adrenal cortical stimulating hormone (ACTH);
   iii. Androgens (testosterone, free testosterone) - if initial testosterone level is low, a repeat measurement of total and free testosterone as well as serum luteinizing hormone (LH) and prolactin levels is medically necessary;
   iv. Estrogens (e.g., estradiol, estrone);
   v. Gonadotropins (FSH, LH);
   vi. Growth hormone (GH);
   vii. Prolactin for men with reduced sperm counts, galactorrhea, or pituitary tumors;
   viii. Sex hormone binding globulin (SHGB) for men with signs and symptoms of hypogonadism and low normal testosterone levels. (SHGB is not indicated in the routine evaluation of male infertility);
   ix. Thyroid stimulating hormone (TSH) for men with symptoms of thyroid disease;

d. Semen analysis - Semen analysis (volume, pH, liquefaction time, sperm concentration, total sperm number, motility (forward progression), motile sperm per ejaculate, vitality, round cell differentiation (white cells versus germinal), morphology, viscosity, agglutination) is considered medically necessary for the evaluation of infertility in men. Because of the marked inherent variability of semen analyses, an abnormal result should be confirmed by at least one additional sample collected one or more weeks after the first sample;

   i. For men with abnormal semen analysis exposed to gonadotoxins, up to 4 semen analyses are considered medically necessary;
   ii. For men with a normal initial semen analysis, a repeat semen analysis is considered medically necessary if there is no pregnancy 4 months after the initial normal semen analysis;
   iii. If the result of the first semen analysis is abnormal and the man has not been exposed to gonadotoxins, up to 2 repeat confirmatory tests may be considered medically necessary;

e. Semen leukocyte analysis (e.g., Endtz test, immunohistochemical staining);
f. Seminal fructose; **Note**: Seminal alpha-glucosidase, zinc, citric acid, and acid phosphatase are considered experimental, investigational, or unproven.
g. Blood test for cytogenetic analysis (karyotype and FISH) in men with severe deficits of semen quality or azoospermia (for consideration of intracytoplasmic sperm injection [ICSI]);
h. Y chromosome microdeletion analysis in men with severe deficits of semen quality or azoospermia (for consideration of intracytoplasmic sperm injection [ICSI]); **Note**: Y chromosome microdeletion analysis is not routinely indicated before ICSI, and is subject to medical necessity review.
i. Post-coital test (PCT) (Simms-Huhner test) of cervical mucus;
j. Sperm penetration assay (zona-free hamster egg penetration test);
k. Karyotyping for persons with recurrent pregnancy loss (defined as 2 or more consecutive spontaneous abortions) (See CPB 0348 - Recurrent Pregnancy Loss) and for men with severe deficits in semen quality or nonobstructive azoospermia (for consideration of intracytoplasmic sperm injection [ICSI]);

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

l. Testing for viral status (HIV, hepatitis B, hepatitis C);
m. Genetic testing of CFTR mutations for a man and his female partner if the man has congenital absence of the vas deferens (CAVD);

3. **Diagnostic procedures**

a. CT or MR imaging of sella turcica if prolactin is elevated;
b. Scrotal exploration;
c. Scrotal (testicular) ultrasound (See CPB 0532 - Scrotal Ultrasonography);
d. Testicular biopsy;
e. Transrectal ultrasound (See CPB 0001 - Transrectal Ultrasound);
f. Vasography;
g. Venography;

4. **Treatments**

a. Endocrine Management

i. Androgens (testosterone) for persons with documented androgen deficiency;
ii. Anti-estrogens (tamoxifen (Nolvadex)) for men with elevated estrogen levels;
iii. Clomiphene (Clomid, Serophene);
iv. Corticosteroids (e.g., dexamethasone, prednisone);
v. Prolactin inhibitors (bromocriptine (Parlodel), cabergoline (Dostinex)) for persons with hyperprolactinemia;
vi. Thyroid hormone replacement for men with thyroid deficiency;

b. Injectable Endocrine Management

i. Gonadotropin releasing hormone analogs and antagonists

For gonadotropin-releasing hormone analogs and antagonists (GnRH; luteinizing hormone releasing hormone (LHRH)) see CPB 0501 - Gonadotropin-Releasing Hormone Analogs and Antagonists;

ii. Gonadotropins

For human chorionic gonadotropin (hCG) (e.g., Novarel, Ovidrel, Pregnyl, generic formulation), see CPB 1081 - Human Chorionic Gonadotropin (hCG).

For follitropins (e.g., follitropin alfa [Gonal-f]; follitropin beta [Follistim AQ]), see CPB 1082 - Follitropins and Menotropins.

Note: Many plans that otherwise cover infertility treatments exclude coverage for infertility injectable medications. Please check benefit plan descriptions.

c. Antibiotics for men with an identified infection; **Note:** Intra-prostatic antibiotic injection is considered experimental, investigational, or unproven;
d. Varicocelectomy (spermatic vein ligation) - see CPB 0413 - Varicocele: Selected Treatments;
e. Spermatocelectomy and hydrocelectomy;
f. Surgical repair of vas deferens: vasovasostomy; **Note**: Most plans exclude coverage for reversal of sterilization procedures. This would include vasectomy. Please check benefit plan descriptions for details.
g. Surgical correction of epididymal blockage for men with obstructive azoospermia:

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

i. Epididymectomy;
ii. Epididymovasostomy;
iii. Excision of epididymal tumors and cysts;
iv. Epididymostomy;

h. Transurethral resection of ejaculatory ducts (TURED) for obstruction of ejaculatory ducts;
i. Orchiopexy;
j. Alpha sympathomimetic agents for retrograde ejaculation (e.g., phenylephrine, imipramine);
k. Hepatitis B vaccination of partners of people with hepatitis B;
l. Impotence treatments - see CPB 0007 - Erectile Dysfunction.

Note: Under most Aetna benefit plans, self-administered prescription medications are covered under the pharmacy benefit. Please check benefit plan descriptions.

C. **Electroejaculation**

Aetna considers electroejaculation medically necessary DME to overcome total anejaculation secondary to neurologic impairment, which most commonly occurs among members with the following conditions:

1. Diabetic neuropathy;
2. Prior retroperitoneal surgery (most commonly retroperitoneal lymphadenectomy as a treatment of testicular cancer);
3. Spinal cord injury.

D. **Donor Insemination**

1. Donor insemination is considered medically necessary for the following indications:

a. Non-obstructive azoospermia;
b. Obstructive azoospermia;
c. Severe deficits in semen quality in couples who do not wish to undergo intracytoplasmic sperm injection (ICSI);
d. Severe rhesus isoimmunization;
e. Where there is a high risk of transmitting a genetic disorder in the male partner to the offspring;
f. Where there is a high risk of transmitting an infectious disease (such as HIV) to the partner or offspring.

E. **Advanced Reproductive Technology**

Note: Coverage is limited to plans with an ART benefit; please check benefit plan descriptions.

1. **Injectable Medications**

See CPB 0020 - Injectable Medications.

a. Gonadotropin releasing hormone analogs and antagonists

For gonadotropin-releasing hormone analogs and antagoinists (GnRH; luteinizing hormone releasing hormone [LHRH]) (e.g., generic leuprolide acetate injection, leuprolide acetate for depot suspension [Lupron Depot], goserelin [Zoladex], histrelin [Supprelin LA], triptorelin [Trelstar; Triptodur], ganirelix acetate/cetrorelix

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

acetate [Cetrotide]), see CPB 0501 - Gonadotropin-Releasing Hormone Analogs and Antagonists;

b. Gonadotropins:

For human chorionic gonadotropin (hCG) (e.g., Novarel, Ovidrel, Pregnyl, generic formulation), see CPB 1081 - Human Chorionic Gonadotropin (hCG).

For follitropins (e.g., follitropin alfa [Gonal-f]; follitropin beta [Follistim AQ]), see CPB 1082 - Follitropins and Menotropins.

For menotropins for injection (Menopur), see CPB 1082 - Follitropins and Menotropins.

Note: Many plans exclude coverage for infertility injectable medications; other plans may limit coverage of ovulation induction cycles with menotropins to a maximum number per lifetime. Please check plan documents for details.

Note: Under most Aetna benefit plans, self-administered prescription medications are covered under the pharmacy benefit. Please check benefit plan descriptions.

2. **Ovulation Induction**

Aetna considers oral or injectable ovulation induction (OI) medically necessary for women 37 years of age or younger who are unable to conceive or produce conception after an appropriate trial of egg-sperm contact. Oral ovulation induction may occur concurrently with the cycles of egg-sperm contact required to establish medical necessity for ART. For women under 35 years of age, an appropriate trial of egg-sperm contact requires 12 months of regular intravaginal inseminations or 4 cycles of timed intrauterine or intracervical inseminations documented in the medical record. For women 35 years of age and older, an appropriate trial of egg-sperm contact requires 6 months of regular intravaginal inseminations or 3 cycles of timed intrauterine or intracervical insemination documented in the medical record.

3. **In Vitro Fertilization (IVF) procedures**

Aetna considers the following in vitro fertilization (IVF) procedures medically necessary for persons who meet *any* of the following criteria:

a. Women who have failed to conceive after a trial of ovulation induction:

i. For women 37 years of age or younger, three cycles of oral or injectable ovulation induction (with or without intravaginal, intrauterine, or intracervical inseminations); *or*
ii. For women 38 years of age or older, no trial of ovulation induction is required; *or*

b. Persons for whom intravaginal, intrauterine, or intracervical inseminations would not be expected to be effective and IVF would be expected to be the only effective treatment, including:

i. Men with azoospermia or severe deficits in semen quality or quantity (see Appendix); or
ii. Women with tubal factor infertility:

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

a. Bilateral tubal disease (e.g., salpingitis isthmica nodosum, tubal obstruction, absence, or hydrosalpinges).
b. Endometriosis stage 3 or 4 (see Appendix).
c. Failure to conceive after pelvic surgery with restoration of normal pelvic anatomy (e.g., myomectomy of cavitary-obscuring myomata, resection of intrauterine adhesions or uterine septum, or surgical reconstruction of tubal disease):

i. After regular egg-sperm contact for 6 months if less than 40 years of age;
ii. After regular egg-sperm contact for 3 months if 40 years of age or older.

d. Unilateral hydrosalpinx with failure to conceive:

i. After regular egg-sperm contact for 12 months if less than 40 years of age;
ii. After regular egg-sperm contact for 6 months if 40 years of age or older.

iii. Inadvertent ovarian hyperstimulation (estradiol level was greater than 1,000 pg/ml plus greater than 3 follicles greater than 16 mm or 4 to 8 follicles greater than 14 mm or a larger number of smaller follicles) during preparation for a planned stimulated cycle in women less than 38 years of age.
iv. Women who have had a hysterectomy, or who have a medical contraindication to pregnancy such as severe cardiac disease, or have a medical condition that requires the mother to ingest a fetotoxic agent. **Note:** Some plans limit and/or exclude coverage for gestational surrogacy; please check benefit plan descriptions.

Note on coverage of advanced reproduction technology (ART) for preimplantation genetic testing for monogenic disorders (PGT-M) (formerly called preimplantation genetic diagnosis [PGD]): embryo biopsy, and testing of the biopsied material, is covered for persons with an ART benefit when medical necessity criteria for PGT-M are met as set forth in CPB 0358 - Invasive Prenatal Diagnosis of Genetic Diseases.

c. IVF with embryo transfer is considered medically necessary when criteria for ART are met. IVF with embryo transfer includes:

i. Embryo transfer (transcervical transfer back to the donor) (including cryopreserved embryo transfer);
ii. Frozen embryo transfer (FET); (**Note**: It may be considered medically necessary to freeze embryos not transferred during a stimulated IVF treatment cycle, and to transfer the embryos before the next stimulated treatment cycle because this will minimize ovulation induction and egg collection, both of which carry risks for the woman and use more resources. Before proceeding to a fresh ART cycle, previously frozen oocytes must be used (i.e. fertilized and transferred). Similarly, Before proceeding to the next fresh ART cycle, FET using cryopreserved embryos must be used if there are reasonable quality (grade B or its equivalent) cryopreserved embryo(s) available.
iii. Oocyte (egg) insemination in laboratory dish;
iv. Oocyte (egg) retrieval via laparoscope or transvaginal needle aspiration of follicles;
v. Sperm preparation and capacitation;

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

vi. Intracytoplasmic sperm injection (ICSI) is considered medically necessary for the following:

a. Severe deficits in semen quality or quantity, including azoospermia or oligospermia (obstructive or non-obstructive) (see Appendix on Semen Quality and Quantity);
ii. To fertilize frozen oocytes for in vitro fertilization;
iii. persons facing iatrogenic infertility due to chemotherapy, pelvic radiotherapy, other gonadotoxic therapies, or ovary or testicle removal for treatment of disease; or
iv. For couples where a previous IVF treatment cycle has resulted in failed or poor fertilization (IVF cycles that resulted in less than 50% fertilization).

Notes:

Intracytoplasmic sperm injection (ICSI) is considered not medically necessary in men whose abnormal sperm quality or quantity had been rectified by varicocelectomy. For use of ICSI in preimplantation genetic testing for monogenic disorders (PGT-M) (formerly called preimplantation genetic diagnosis [PGD]) and preimplantation genetic testing for aneuploidy (PGT-A) (formerly called preimplantation genetic screening [PGS]), see CPB 0358 - Invasive Prenatal Diagnosis of Genetic Diseases.

Physiological, hyaluronan-selected intracytoplasmic sperm injection (PICSI) is a variation of intracytoplasmic sperm injection (ICSI) that uses hyaluronic acid (HA) to select sperm. The PICSI method is considered experimental, investigational, or unproven because of a lack of reliable evidence of effectiveness compared to standard ICSI.

vii. Assisted hatching is considered medically necessary when the plan in the cycle is to transfer the embryos into the uterus and the member meets any of the following criteria:

a. Age of embryo recipient is 38 years or older; *or*
b. Multiple (2 or more) failed embryo transfer attempts; *or*
c. Thickened zona pellucida.

Note: Assisted hatching is a process to assist in the implantation of the embryo; unless the cycle involves that transfer of the embryo assisted hatching is considered not medically necessary.

Note on IVF cycles for embryo banking: IVF cycles for the sole purpose of embryo banking (where none of the embryos that are suitable for transfer are used in the current cycle in which they are created, but are frozen for use in a future cycle) is not considered treatment of disease and is not covered.

Note on oocytes used in ART cycles: IVF cycles using either fresh or previously frozen oocytes are considered medically necessary when the ART cycle is considered medically necessary.

4. **Other Advanced Reproductive Technology (ART) Procedures**

a. Gamete intra-fallopian transfer (GIFT) is considered medically necessary as an alternative to IVF for women with female factor infertility. GIFT includes:

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

i. Immediate loading of the eggs into a transfer catheter with sperm and insertion into the member's fallopian tube via the same laparoscope (the member must have at least 1 patent fallopian tube for this method to be an effective treatment for infertility)
ii. Oocyte (egg) retrieval via laparoscope.

GIFT is considered experimental, investigational, or unproven for person with male factor infertility or unexplained infertility problems because there is insufficient evidence to recommend GIFT over IVF for these indications.

b. Zygote intra-fallopian transfer (ZIFT), tubal embryo transfer (TET), pronuclear stage tubal embryo transfer (PROUST) is considered medically necessary as an alternative to IVF for women with female factor infertility.

ZIFT is considered experimental, investigational, or unproven for persons with male factor infertility or unexplained infertility problems because there is insufficient evidence to recommend ZIFT over IVF for these indications.

c. Specialized sperm retrieval techniques (including vasal sperm aspiration, microsurgical epididymal sperm aspiration (MESA), percutaneous epididymal sperm aspiration (PESA), electroejaculation, testicular sperm aspiration (TESA), microsurgical testicular sperm extraction (TESE), seminal vesicle sperm aspiration, and sperm recovery from bladder or urine for retrograde ejaculation) is considered medically necessary to overcome anejaculation or azoospermia that is not due to a previous elective sterilization. (see Appendix on Semen Quality and Quantity)

Note: Most plans exclude coverage of infertility services for persons who have undergone sterilization. This would include sperm retrieval for men who have undergone vasectomy. Please check benefit plan descriptions for details.

d. Oocyte donation is considered medically necessary for managing infertility problems associated with the following conditions, when the infertile member is the intended recipient of the resulting embryos:

i. Bilateral oophorectomy;
ii. Gonadal dysgenesis including Turner syndrome;
iii. High-risk of transmitting a genetic disorder from the female partner to the offspring;
iv. IVF treatment failure
v. Ovarian failure following chemotherapy or radiotherapy; *or*
vi. Premature ovarian failure (failure of ovulation in woman younger than 40 years of age) (considered medically necessary until the woman with POF is 45 years of age).

Note: Many Aetna plans that otherwise cover ART exclude coverage of fees associated with oocyte donation, including recruitment and selection of donors, ovarian stimulation of donors, collection of oocytes from donors, and screening and storage of donor oocytes. Please check benefit plan descriptions for details. Under plans with benefits for ART that have this exclusion, medically necessary ART services are covered only once an embryo is created from the donor egg.

e. Cryopreservation of mature gametes (oocytes or sperm) or embryos is considered medically necessary for use in persons facing iatrogenic infertility due to chemotherapy, pelvic radiotherapy, other gonadotoxic therapies, or ovary or testicle removal for treatment of disease.

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

Routine use of gamete cryopreservation in lieu of embryo cryopreservation, gamete cryopreservation to circumvent reproductive aging in healthy persons, cryopreservation of immature gametes, and laser-assisted necrotic blastomere removal from cryopreserved embryos are considered experimental, investigational, or unproven.

Note: Some Aetna plans have a specific contractual exclusion of coverage of any charges associated with embryo cryopreservation or storage of cryopreserved embryos. Please check benefit plan descriptions. In addition, cryopreservation of embryos and gametes (other than short-term cryopreservation of embryos that are necessary for contemporaneous use in infertile persons currently under active fertility treatment, or use of cryopreserved embryos or mature gametes in persons facing infertility due to chemotherapy or other gonadotoxic therapies or gonad removal) is not considered treatment of disease and is not covered.

f. Sperm cryopreservation to circumvent reproductive aging in healthy men is considered experimental, investigational, or unproven.

Note: Some Aetna plans have a specific contractual exclusion of coverage of any charges associated with sperm cryopreservation or storage. Please check benefit plan descriptions. In addition, cryopreservation of sperm (other than cryopreserved sperm in men facing infertility due to chemotherapy or other gonadotoxic therapies or gonad removal) is not considered treatment of disease and is not covered.

Note: A cycle of ART defined in the CPB may be any of the following: IVF (with fresh embryos), IVF/frozen embryo transfer, GIFT or ZIFT.

Note on elective single embryo transfer: In order to reduce the number of high-order multiple pregnancies, the American Society for Reproductive Medicine (ASRM, 2021) recommends elective single embryo transfer (eSET) for women under 35 years of age, or for transfer of a euploid embryo regardless of member age. For women who meet these criteria and undergo transfer of a single fresh embryo that does not result in a pregnancy (defined as absence of a documented fetal heartbeat), and who are enrolled in a benefit plan that limits the number of covered IVF cycles, Aetna will consider the transfer of one cryopreserved embryo immediately following the unsuccessful fresh embryo transfer as part of the same IVF cycle. Please check benefit plan description for details.

II. Experimental, Investigational, or Unproven

The following are considered experimental, investigational, or unproven for infertility (not an all-inclusive list):

- Acupuncture (see CPB 0135 - Acupuncture)
- Bariatric surgery (see CPB 0157 - Obesity Surgery);
- Dehydroepiandrosterone (DHEA)
- Direct intra-peritoneal insemination, fallopian tube sperm transfusion, intra-follicular insemination, and the use of sperm precursors (i.e., round or elongated spermatid nuclei, immature sperm)
- Drainage of ovarian cyst
- DuoStim IVF protocol
- Early Embryo Viability Assessment (Eeva) test
- EmbryoGlue
- Evaluation of CYP1A1 rs4646903 T > C genetic variations for risk of male infertility
- Evaluation of *FAS/FASL* genetic variations for risk of male infertility
- Evaluation of telomere length
- Fine needle aspiration ("mapping") of testes

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

- FSH manipulation of women with elevated FSH levels (an elevated FSH level is a marker of reduced ovarian reserve, as occurs with advancing age. Elevated FSH-related (i.e., age-related) infertility has not been proven to be affected by interventions to reduce FSH levels)
- Germ cell transplantation or cultured testicular stem cells
- Growth hormone: there is inadequate evidence that the use of adjuvant growth hormone treatment during ovulation induction improves pregnancy rates. See CPB 0170 - Growth Hormone (GH) and Growth Hormone Antagonists.
- GIFT for person with male factor infertility or unexplained infertility problems
- Hyperbaric oxygen therapy for the treatment of male infertility
- Intralipid infusion for the treatment of female infertility
- Intrauterine injection/infusion of platelet rich plasma for the treatment of female infertility
- Intravenous immunoglobulins for treatment of infertility (see CPB 0348 - Recurrent Pregnancy Loss and CPB 0206 - Parenteral Immunoglobulins)
- In-vitro maturation of oocytes
- Leokocyte immunization (immunizing the female partner with the male partner's leukocytes (see CPB 0348 - Recurrent Pregnancy Loss)
- Microdissection of the zona
- Parenteral administration of lipids
- Partial zonal dissection (PZD)
- Preimplantation genetic testing for aneuploidy (PGT-A) (formerly called preimplantation genetic screening (PGS)) for IVF optimization (see CPB 0358 - Invasive Prenatal Diagnosis of Genetic Diseases)
- Stem cell therapy
- Subzonal sperm insertion (SUZI)
- Uterine transplant
- Vaginal sildenafil
- Vasodilators for women undergoing fertility treatment
- ZIFT for persons with male factor infertility or unexplained infertility problems
- The following sperm function tests are considered experimental, investigational, or unproven:

    - Acrosome reaction test
    - Cap-Score Test
    - Comet assay
    - Computer-assisted sperm analysis (CASA)/computer-assisted sperm motion analysis
    - Hemizona assay
    - Hyaluronan binding assay
    - Hypoosmotic swelling test
    - In vitro testing of sperm penetration
    - Reactive oxygen species (ROS) test
    - Sperm chromatin assay
    - Spern DNA condensation test
    - Sperm DNA fragmentation assay
    - Sperm nucleus maturation
    - TUNEL assay

- For follitropins and menotropins, see CPB xxxx - Follitropins and Menotropins
- For human chorionic gonadotropin (hCG), see CPB xxxx - Human Chorionic Gonadotropin (hCG)
- The following laboratory studies are considered experimental, investigational, or unproven for infertility:

    - Analysis of Infectious Chronic Endometritis (ALICE) (Igenomix)
    - Anti-CarP (anti-carbamylated proteins) panel
    - Antinuclear antibodies
    - Antiovarian antibodies
    - Antiphospholipid antibodies

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

- Antiphosphatidic acid antibodies
- Antiphosphatidylethanolamine antibodies
- Antiphosphatidylglycerol antibodies
- Antiphosphatidylinositol antibodies
- Antiphosphatidylserine antibodies
- Antiprothrombin antibodies (see CPB 0662 - Antiprothrombin Antibody Testing)
- Antithrombin III (ATIII) activity
- Antithrombin III (ATIII) antigen
- Antithyroglobulin antibodies
- Embryotoxicity assay (see CPB 0348 - Recurrent Pregnancy Loss)
- Endometrial Microbiome Metagenomic Analysis (EMMA) (Igenomix)
- Endometrial receptivity testing (e.g., endometrial receptivity analysis (Igenomix), endometrial receptivity array (ERA), integrin testing, beta-3 integrin test)
- Evaluation of telomere length
- Factor V Leiden coagulation
- Factor V Leiden mutation (see CPB 0140 - Genetic Testing)
- HLA genotyping (A, B, C, DR, DQ)
- Homocysteine (see CPB 0763 - Homocysteine Testing)
- Methylenetetrahydrofolate reductase (MTHFR)
- Oxidative Stress Adduct Test (OSA)
- Plasminogen Activator Inhibitor-I activity
- Plasminogen Activator Inhibitor-I (PAI-1) antigen
- Protein C activity
- Protein C antigen
- Protein S activity
- Protein S antigen (free or total)
- Prothrombin (Factor II) mutation (see CPB 0140 - Genetic Testing)
- Uterine and endometrial receptivity testing (Endometrial function test (EFT) (cyclin E and p27) and E-tegrity)
- Measurement of natural killer (NK) cell activity
- Reproductive immunophenotyping
- Serum inhibin B measurement (value in assessing ovarian reserve is uncertain)
- Th1 (T Helper 1) and Th2 (T Helper 2) intracellular cytokine assay (Th1/Th2 ratio)
- uBiome SmartJane screen (see CPB 0650 - Polymerase Chain Reaction Testing: Selected Indications)
- Vaginal microbiota.

III. **Related Policies**

- CPB 0001 - Transrectal Ultrasound
- CPB 0007 - Erectile Dysfunction
- CPB 0020 - Injectable Medications
- CPB 0135 - Acupuncture
- CPB 0140 - Genetic Testing
- CPB 0157 - Obesity Surgery
- CPB 0170 - Growth Hormone (GH) and Growth Hormone Antagonists
- CPB 0189 - Genetic Counseling
- CPB 0206 - Parenteral Immunoglobulins
- CPB 0323 - Preconceptional Sex Selection Techniques
- CPB 0347 - Transcervical Balloon Tuboplasty
- CPB 0348 - Recurrent Pregnancy Loss
- CPB 0358 - Invasive Prenatal Diagnosis of Genetic Diseases
- CPB 0413 - Varicocele: Selected Treatments
- CPB 0433 - Herpes Simplex Virus - Screening and Diagnosis
- CPB 0501 - Gonadotropin-Releasing Hormone Analogs and Antagonists

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

- CPB 0510 - Progestins
- CPB 0532 - Scrotal Ultrasonography
- CPB 0650 - Polymerase Chain Reaction Testing: Selected Indications
- CPB 0662 - Antiprothrombin Antibody Testing
- CPB 0763 - Homocysteine Testing
- CPB 1081 - Human Chorionic Gonadotropin (hCG)
- CPB 1082 - Follitropins and Menotropins

Table:

Applicable CPT / HCPCS / ICD-10 Codes

Code	Code Description
CPT codes covered if selection criteria are met:	
0167U	Gonadotropin, chorionic (hCG), immunoassay with direct optical observation, blood
0353U	Infectious agent detection by nucleic acid (DNA), Chlamydia trachomatis and Neisseria gonorrhoeae, multiplex amplified probe technique, urine, vaginal, pharyngeal, or rectal, each pathogen reported as detected or not detected
49186	Excision or destruction, open, intra-abdominal (ie, peritoneal, mesenteric, retroperitoneal), primary or secondary tumor(s) or cyst(s), sum of the maximum length of tumor(s) or cyst(s); 5 cm or less
49187	5.1 to 10 cm
49188	10.1 to 20 cm
49189	20.1 to 30 cm
49190	greater than 30 cm
49203	Excision or destruction, open, intra-abdominal tumors, cysts or endometriomas, 1 or more peritoneal, mesenteric, or retroperitoneal primary or secondary tumors; largest tumor 5 cm diameter or less
49204	largest tumor 5.1 - 10.0 cm diameter
49205	largest tumor greater than 10.0 cm diameter
49320	Laparoscopy, abdomen, peritoneum, and omentum, diagnostic, with or without collection of specimen(s) by brushing or washing (separate procedure)
49321	Laparoscopy, surgical; with biopsy (single or multiple)
49322	with aspiration of cavity or cyst (eg, ovarian cyst) (single or multiple)
52402	Cystourethroscopy with transurethral resection or incision of ejaculatory ducts
54500	Biopsy of testis, needle (separate procedure)
54505	Biopsy of testis, incisional (separate procedure)
54640	Orchiopexy, inguinal approach, with or without hernia repair
54650	Orchiopexy, abdominal approach, for intra-abdominal testis (eg, Fowler-Stephens)
54692	Laparoscopy, surgical; orchiopexy for intra-abdominal testis
54800	Biopsy of epididymis, needle
54830	Excision of local lesion of epididymis
54840	Excision of spermatocele, with or without epididymectomy
54860	Epididymectomy; unilateral
54861	bilateral

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

Code	Code Description
54865	Exploration of epididymis, with or without biopsy
54900	Epididymovasostomy, anastamosis of epididymis to vas deferens; unilateral
54901	bilateral
55000	Puncture aspiration of hydrocele, tunica vaginalis, with or without injection of medication
55040	Excision of hydrocele; unilateral
55041	bilateral
55060	Repair of tunica vaginalis hydrocele (Bottle type)
55110	Scrotal exploration
55300	Vasotomy for vasograms, seminal vesiculograms, or epididymograms, unilateral or bilateral
55400	Vasovasostomy, vasovasorrhaphy
55500	Excision of hydrocele of spermatic cord, unilateral (separate procedure)
55530	Excision of varicocele or ligation of spermatic veins for varicocele; (separate procedure)
55535	abdominal approach
55540	with hernia repair
55870	Electroejaculation
57530	Trachelectomy (cervicectomy), amputation of cervix (separate procedure)
58100	Endometrial sampling (biopsy) with or without endocervical sampling (biopsy), without cervical dilation, any method (separate procedure)
58120	Dilation and curettage, diagnostic and/or therapeutic (nonobstetrical)
58140	Myomectomy, excision of fibroid tumor(s) of uterus, 1 to 4 intramural myoma(s) with total weight of 250 g or less and/or removal of surface myomas; abdominal approach
58145	vaginal approach
58146	Myomectomy, excision of fibroid tumor(s) of uterus, 5 or more intramural myomas and/or intramural myomas with total weight greater than 250 g, abdominal approach
58321	Artificial insemination; intra-cervical
58322	intra-uterine
58323	Sperm washing for artificial insemination
58340	Catheterization and introduction of saline or contrast material for saline infusion sonohysterography (SIS) or hysterosalpingography
58345	Transcervical introduction of fallopian tube catheter for diagnosis and/or re-establishing patency (any method), with or without hysterosalpingography
58350	Chromotubation of oviduct, including materials
58545	Laparoscopy, surgical, myomectomy, excision; 1 to 4 intramural myomas with total weight of 250 g or less and/or removal of surface myomas
58546	5 or more intramural myomas and/or intramural myomas with total weight greater than 250 g
58555	Hysteroscopy, diagnostic (separate procedure)
58558	Hysteroscopy, surgical; with sampling (biopsy) of endometrium and/or polypectomy, with or without D & C
58559	with lysis of intrauterine adhesions (any method)
58560	with division or resection of intrauterine septum (any method)
58561	with removal of leiomyomata
58562	with removal of impacted foreign body
58563	with endometrial ablation (eg, endometrial resection, electrosurgical ablation, thermoablation)

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

Code	Code Description
58600	Ligation or transection of fallopian tube(s), abdominal or vaginal approach, unilateral or bilateral
58660	Laparoscopy, surgical; with lysis of adhesions (salpingolysis, ovariolysis) (separate procedure)
58661	with removal of adnexal structures (partial or total oophorectomy and/or salpingectomy)
58662	with fulguration or excision of lesions of the ovary, pelvic viscera, or peritoneal surface by any method
58672	with fimbrioplasty
58673	with salpingostomy (salpingoneostomy)
58700	Salpingectomy, complete or partial, unilateral or bilateral (separate procedure)
58720	Salpingo-oophorectomy, complete or partial, unilateral or bilateral (separate procedure)
58740	Lysis of adhesions (salpingolysis, ovariolysis)
58750	Tubotubal anastomosis
58752	Tubouterine implantation
58760	Fimbrioplasty
58770	Salpingostomy (salpingoneostomy)
58800	Drainage of ovarian cyst(s), unilateral or bilateral (separate procedure); vaginal approach
58805	abdominal approach
58820	Drainage of ovarian abscess; vaginal approach, open
58822	abdominal approach
58825	Transposition, ovary(s)
58900	Biopsy of ovary, unilateral or bilateral (separate procedure)
58920	Wedge resection or bisection of ovary, unilateral or bilateral
58925	Ovarian cystectomy, unilateral or bilateral
58970	Follicle puncture for oocyte retrieval, any method
58974	Embryo transfer, intrauterine
58976	Gamete, zygote, or embryo intrafallopian transfer, any method
70480	Computed tomography, orbit, sella, or posterior fossa or outer, middle, or inner ear; without contrast material
70481	with contrast material(s)
70482	without contrast material, followed by contrast material(s) and further sections
70540	Magnetic resonance (eg, proton) imaging, orbit, face, and/or neck; without contrast material(s)
70542	with contrast material(s)
70543	without contrast material(s), followed by contrast material(s) and further sequences
74440	Vasography, vesiculography, or epididymography, radiological supervision and interpretation
74740	Hysterosalpingography, radiological supervision and interpretation
74742	Transcervical catheterization of fallopian tube, radiological supervision and interpretation
76830	Ultrasound, transvaginal
76831	Saline infusion sonohysterography(SIS), including color flow Doppler, when performed
76856	Ultrasound, pelvic (nonobstetric), real time with image documentation; complete
76857	limited or follow-up (e.g., for follicles)

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

Code	Code Description
76870	Ultrasound, scrotum and contents
76872	Ultrasound, transrectal
76948	Ultrasonic guidance for aspiration of ova, imaging supervision and interpretation
80400	ACTH stimulation panel; for adrenal insufficiency
80402	for 21 hydroxylase deficiency
80406	for 3 beta-hydroxydehydrogenase deficiency
80412	Corticotropic releasing hormone (CRH) stimulation panel
80414	Chorionic gonadotropin stimulation panel; testosterone response
80415	estradiol response
80418	Combined rapid anterior pituitary evaluation panel
80426	Gonadotropin releasing hormone stimulation panel
80428	Growth hormone stimulation panel (eg, arginine infusion, l-dopa administration)
80438	Thyrotropin releasing hormone (TRH) stimulation panel; one hour
80439	two hour
81224	Intron 8 poly-T analysis (eg,male infertility)
82024	Adrenocorticotropic hormone (ACTH)
82157	Androstenedione
82166	Anti-mullerian hormone (AMH)
82465	Cholesterol, serum or whole blood, total
82626	Dehydroepiandrosterone (DHEA)
82670	Estradiol
82671	Estrogens; fractionated
82672	total
82679	Estrone
82757	Fructose, semen
82951	Glucose: tolerance test (GTT), three specimens (includes glucose)
83001	Gonadotropin; follicle stimulating hormone (FSH)
83002	luteinizing hormone (LH)
83003	Growth hormone, human (HGH) (somatotropin)
83498	Hydroxyprogesterone, 17-d
83499	Hydroxyprogesterone, 20-
83519	Immunoassay for analyte other than infectious agent antibody or infectious agent antigen; quantitative, by radioimmunoassay (eg, RIA) [measurement of anti-adrenal antibodies]
83520	Immunoassay for analyte other than infectious agent antibody or infectious agent antigen; quantitative, not otherwise specified [covered for anti-mullerian hormone testing] [not covered for the Th1/Th2 ratio, or antiphosphatidic acid antibodies]
83718	Lipoprotein, direct measurement; high density cholesterol (HDL cholesterol)
84144	Progesterone
84146	Prolactin
84233	Receptor assay; estrogen
84234	progesterone
84270	Sex hormone binding globulin (SHBG)
84402	Testosterone; free
84403	total
84443	Thyroid stimulating hormone (TSH)
84478	Triglycerides

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

Code	Code Description
84702	Gonadotropin, chorionic (hCG); quantitative
84703	qualitative
86256	Fluorescent noninfectious agent antibody; titer, each antibody [measurement of anti-adrenal antibodies]
86277	Growth hormone, human (HGH), antibody
86631	Antibody; Chlamydia
86632	Chlamydia, IgM
86689	HTLV or HIV antibody, confirmatory test (eg, Western Blot)
86701	HIV-1
86702	HIV-2
86703	HIV-1 and HIV-2, single result
86704	Hepatitis B core antibody (HBcAb); total
86705	IgM antibody
86706	Hepatitis B surface antibody (HBsAb)
86762	Antibody; rubella
86803	Hepatitis C antibody;
86804	confirmatory test (eg, immunoblot)
87110	Culture, chlamydia, any source
87270	Chlamydia trachomatis
87340	Infectious agent antigen detection by immunoassay technique, (eg, enzyme immunoassay [EIA], enzyme-linked immunosorbent assay [ELISA], immunochemiluminometric assay [IMCA]) qualitative or semiquantitative, multiple-step method; hepatitis B surface antigen (HBsAg)
87341	hepatitis B surface antigen (HBsAg) neutralization
87491	Chlamydia trachomatis, amplified probe technique
87492	Chlamydia trachomatis, quantification
87810	Infectious agent detection by immunoassay with direct optical observation; Chlamydia trachomatis
88245	Chromosome analysis for breakage syndromes; baseline Sister Chromatid Exchange (SCE), 20-25 cells
88248	baseline breakage, score 50-100 cells, count 20 cells, 2 karyotypes (eg, for ataxia telangectasia, Fanconi anemia, fragile X)
88249	score 100 cells, clastogen stress (eg, diepoxybutane, mitomycin C, ionizing radiation, UV radiation)
88261	Chromosome analysis; count 5 cells, 1 karyotype, with banding
88262	count 15-20 cells, 2 karyotypes, with banding
88263	count 45 cells for mosaicism, 2 karyotypes, with banding
88264	analyze 20-25 cells
88271	Molecular cytogenetics: DNA probe, each (eg, FISH)
88272	chromosomal in situ hybridization, analyze 3-5 cells (eg, for derivatives and markers)
88273	chromosomal in situ hybridization, analyze 10-30 cells (eg, for microdeletions)
88274	interphase in situ hybridization, analyze 25-99 cells
88275	interphase in situ hybridization, analyze 100-300 cells
88280	Chromosome analysis; additional karyotypes, each study
88283	additional specialized banding technique (eg, NOR, C-banding)
88285	additional cells counted, each study
88289	additional high resolution study

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

Code	Code Description
88291	Cytogenics and molecular cytogenetics, interpretation and report
89250 - 89356	Reproductive Medicine Procedures
90739 - 90747	Hepatitis B vaccine
90748	Hepatitis B and Haemophilus influenzae type b vaccine (Hib-HepB), for intramuscular use
93975	Duplex scan of arterial inflow and venous outflow of abdominal, pelvic, scrotal contents and/or retroperitoneal organs; complete study
93976	limited study
96040	Medical genetics and genetic counseling services, each 30 minutes face-to-face with patient/family

CPT codes not covered for indications listed in the CPB:

***Evaluation of CYP1A1 rs4646903 T > C genetic variations for risk of male infertility, evaluation of FAS/FASL genetic variations for risk of male fertility, stem cell therapy for the treatment of female infertility, vaginal microbiota, physiological, hyaluronan-selected intracytoplasmic sperm injection (PICSI)*- - no specific code**

0087T	Sperm evaluation, Hyaluronan sperm binding test
0232T	Injection(s), platelet rich plasma, any site, including image guidance, harvesting and preparation when performed
0253U	Reproductive medicine (endometrial receptivity analysis), RNA gene expression profile, 238 genes by next-generation sequencing, endometrial tissue, predictive algorithm reported as endometrial window of implantation (eg, pre-receptive, receptive, post-receptive)
0255U	Andrology (infertility), sperm-capacitation assessment of ganglioside GM1 distribution patterns, fluorescence microscopy, fresh or frozen specimen, reported as percentage of capacitated sperm and probability of generating a pregnancy score
0357T	Cryopreservation; immature oocyte(s)
0607U	Reproductive medicine (endometrial microbiome assessment), real-time PCR analysis for 31 bacterial DNA targets from endometrial biopsy, reported with quantified levels of bacterial presence and targeted treatment recommendations
0608U	Reproductive medicine (endometrial microbiome assessment), real-time PCR analysis for 10 bacterial DNA targets from endometrial biopsy, reported with quantified levels of bacterial presence and targeted treatment recommendations
0664T	Donor hysterectomy (including cold preservation); open, from cadaver donor
0665T	Donor hysterectomy (including cold preservation); open, from living donor
0666T	Donor hysterectomy (including cold preservation); laparoscopic or robotic,from living donor
0667T	Donor hysterectomy (including cold preservation); laparoscopic or robotic,from living donor
10005	Fine needle aspiration biopsy, including ultrasound guidance first lesion
10006	Fine needle aspiration biopsy, including ultrasound guidance; each additional lesion (List separately in addition to code for primary procedure)
10021	Fine needle aspiration; without imaging guidance; first lesion
43631 - 43635 43644 - 43645 43770 - 43775 43842 - 43848 43886 - 43888	Bariatric surgery

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

Code	Code Description
81240	F2 (prothrombin, coagulation factor II) (eg, hereditary hypercoagulability) gene analysis, 20210G>A variant
81241	F5 (coagulation Factor V) (eg, hereditary hypercoagulability) gene analysis, Leiden variant
81291	MTHFR (5, 10-methylenetetrahydrofolate reductase) (eg, hereditary hypercoagulability) gene analysis, common variants (eg, 677T, 1298C)
81370	HLA Class I and II typing, low resolution (eg, antigen equivalents); HLA-A, -B, -C, -DRB1/3/4/5, and -DQB1
81400	Molecular pathology procedure, Level 1(eg, identification of single germline variant [eg, SNP] by techniques such as restriction enzyme digestion or melt curve analysis) [Plasminogen activator inhibitor-I (PAI-1) antigen]
81406	Molecular pathology procedure, Level 7 (eg, analysis of 11-25 exons by DNA sequence analysis, mutation scanning or duplication/deletion variants of 26-50 exons, cytogenomic array analysis for neoplasia) [determination of CAG-repeat polymorphisms in the polymerase γ (POLG) gene for evaluation of male infertility]
83001 - QW	Gonadotropin; follicle stimulating hormone (FSH) [urinary FSH - CLIA waived test]
83090	Homocysteine
85300	Clotting inhibitors or anticoagulants; antithrombin III, activity
85301	Clotting inhibitors or anticoagulants; antithrombin III, antigen assay
85302	Clotting inhibitors or anticoagulants; protein C, antigen
85303	Clotting inhibitors or anticoagulants; protein C, activity
85305	Clotting inhibitors or anticoagulants; protein S, total
85306	Clotting inhibitors or anticoagulants; protein S, free
86038	Antinuclear antibodies
86039	Antinuclear antibodies (ANA); titer
86146	Beta 2 Glycoprotein I antibody, each
86147	Cardiolipin (phospholipid) antibody, each Ig class
86148	Anti-phosphatidylserine (phospholipid) antibody
86255	Fluorescent noninfectious agent antibody; screen, each [antiovarain antibodies]
86357	Natural killer (NK) cells, total count[not covered for female infertility]
88184 - 88185	Flow cytometry, cell surface,
88187 - 88189	Flow cytometry, interpretation
97810 - 97814	Acupuncture
99183	Physician or other qualified health care professional attendance and supervision of hyperbaric oxygen therapy, per session

Other CPT codes related to the CPB:

38780	Retroperitoneal transabdominal lymphadenectomy, extensive, including pelvic, aortic, and renal nodes (separate procedure)
77261 - 77790	Radiation Oncology
89342	Storage (per year); embryo(s)
89343	sperm/semen
89344	reproductive tissue, testicular/ovarian
89346	oocyte(s)
90460 - 90461	Immunization administration through 18 years of age via any route of administration, with counseling by physician or other qualified health care professional

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

Code	Code Description
90471 - 90472	Immunization administration (includes percutaneous intradermal, subcutaneous, or intramuscular injections
96401 - 96548	Chemotherapy and other highly complex drug or highly complex biologic agent administration

There are no specific CPT codes for the Laboratory Test listed below:

Oxidative Stress Adduct Test (OSA), Plasminogen activator inhibitor-I, antiphosphatidylglycerol antibodies, antiphosphatidylinositol antibodies, antithyroglobulin antibodies, anti-CarP (anti-carbamylated proteins) panel, the DuoStim IVF protocol

HCPCS codes covered if selection criteria are met:

***Letrozole, Clomiphene* - no specific code**

A9574	Air polymer-type a intrauterine foam, 0.1 ml
G0010	Administration of hepatitis B vaccine
G0027	Semen analysis; presence and/or motility of sperm excluding huhner
G0123	Screening cytopathology, cervical or vaginal (any reporting system), collected in preservative fluid, automated thin layer preparation, screening by cytotechnologist under physician supervision
G0124	Screening cytopathology, cervical or vaginal (any reporting system), collected in preservative fluid, automated thin layer preparation, requiring interpretation by physician
G0141 - G0148	Screening cytopathology smears, cervical or vaginal
G0472	Hepatitis C antibody screening for individual at high risk and other covered indication(s)
J0900	Injection, testosterone enanthate and estradiol valerate, up to 1cc
J1000	Injection, depo-estradiol cypionate, up to 5 mg
J1060	Injection, testosterone cypionate and estradiol cypionate, up to 1 ml
J1071	Injection, testosterone cypionate, 1mg
J1072	Injection, testosterone cypionate (azmiro), 1 mg
J1094	Injection, dexamethasone acetate, 1 mg
J1100	Injection, dexamethasone sodium phosphate, 1 mg
J1380	Injection, estradiol valerate, up to 10 mg
J1410	Injection, estrogen conjugated, per 25 mg
J1620	Injection, gonadorelin HCI, per 100 mcg
J2371	Injection, phenylephrine hydrochloride, 20 micrograms
J2372	Injection, phenylephrine hydrochloride (biorphen), 20 micrograms
J2675	Injection, progesterone, per 50 mg
J3120	Injection, testosterone enanthate, up to 100 mg
J3121	Injection, testosterone enanthate, 1mg
J3130	Injection, testosterone enanthate, up to 200 mg
J3140	Injection, testosterone suspension, up to 50 mg
J3145	Injection, testosterone undecanoate, 1 mg
J3150	Injection, testosterone propionate, up to 100 mg
J3355	Injection, urofollitropin, 75 IU
J7512	Prednisone, immediate release or delayed release, oral, 1 mg
J8515	Cabergoline, oral, 0.25 mg

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

Code	Code Description
J8540	Dexamethasone, oral, 0.25 mg
J9202	Goserelin acetate implant, per 3.6 mg
J9218	Leuprolide acetate, per 1 mg
P3000	Screening Papanicolaou smear, cervical or vaginal, up to three smears, by technician under physician supervision
P3001	Screening Papanicolaou smear, cervical or vaginal, up to three smears, requiring interpretation by physician
Q0115	Post-coital direct, qualitative examinations of vaginal or cervical mucous
S0132	Injection, ganirelix acetate, 250 mcg [not covered for men]
S0187	Tamoxifen citrate, oral, 10 mg
S0265	Genetic counseling, under physician supervision, each 15 minutes
S2078	Laparoscopic supracervical hysterectomy (subtotal hysterectomy), with or without removal of tube(s), with or without removal of ovary(s)
S3655	Antisperm antibodies test (immunobead)
S4011	In vitro fertilization; including but not limited to identification and incubation of mature oocytes, fertilization with sperm, incubation of embryo(s), and subsequent visualization for determination of development
S4013	Complete cycle, gamete intrafallopian transfer (GIFT), case rate
S4014	Complete cycle, zygote intrafallopian transfer (ZIFT), case rate
S4015	Compete in vitro fertilization cycle, not otherwise specified, case rate
S4016	Frozen in vitro fertilization cycle, case rate
S4017	Incomplete cycle, treatment canceled prior to stimulation, case rate
S4018	Frozen embryo transfer procedure canceled before transfer, case rate
S4020	In vitro fertilization procedure canceled before aspiration, case rate
S4021	In vitro fertilization procedure canceled after aspiration, case rate
S4022	Assisted oocyte fertilization, case rate
S4023	Donor egg cycle, incomplete, case rate
S4025	Donor services for in vitro fertilization (sperm or embryo), case rate
S4026	Procurement of donor sperm from sperm bank
S4028	Microsurgical epididymal sperm aspiration (MESA)
S4030	Sperm procurement and cryopreservation services; initial visit
S4031	subsequent visit
S4035	Stimulated intrauterine insemination (IUI), case rate
S4037	Cryopreserved embryo transfer, case rate
S4993	Contraceptive pills for birth control
S9560	Home injectable therapy; hormonal therapy (e.g., leuprolide, goserelin), including administrative services, professional pharmacy services, care coordination, and all necessary supplies and equipment (drugs and nursing visits coded separately), per diem

HCPCS codes not covered for indications listed in the CPB:

Vasodilators for women undergoing fertility treatment - no specific code:

A4575	Topical hyperbaric oxygen chamber, disposable
B4185	Parenteral nutrition solution, per 10 grams lipids
G0277	Hyperbaric oxygen under pressure, full body chamber, per 30 minute

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

Code	Code Description
J1561	Injection, immune globulin, (Gamunex-C/Gammaked), nonlyophilized (e.g. liquid), 500 mg
J1566	Injection, immune globulin, intravenous, lyophilized (e.g., powder), not otherwised specified, 500 mg
J1568	Injection, immune globulin, (Octagam), intravenous, nonlyophilized (e.g., liquid), 500 mg
J1569	Injection, immune globulin, (Gammagard liquid), nonlyophilized (e.g. liquid), 500 mg
J2940	Injection, somatrem, 1 mg
J2941	Injection, somatropin, 1 mg
Q0515	Injection, sermorelin acetate, 1 mcg
S0090	Sildenafil citrate, 25 mg [phosphodiesterase 5 inhibitor] [vaginal Sildenafil]
S4024	Air polymer-type a intrauterine foam, per study dose
S8930	Electrical stimulation of auricular acupuncture points; each 15 minutes of personal one-on-one contact with the patient
S9558	Home injectable therapy; growth hormone, including administrative services, professional pharmacy services, care coordination, and all necessary supplies and equipment (drugs and nursing visits coded separately), per diem

Other HCPCS codes related to the CPB:

G0310	Immunization counseling by a physician or other qualified health care professional when the vaccine(s) is not administered on the same date of service, 5 to 15 mins time (this code is used for Medicaid billing purposes)
G0311	Immunization counseling by a physician or other qualified health care professional when the vaccine(s) is not administered on the same date of service, 16-30 mins time (this code is used for Medicaid billing purposes)
G0312	Immunization counseling by a physician or other qualified health care professional when the vaccine(s) is not administered on the same date of service for ages under 21, 5 to 15 mins time (this code is used for Medicaid billing purposes)
G0313	Immunization counseling by a physician or other qualified health care professional when the vaccine(s) is not administered on the same date of service for ages under 21, 16-30 mins time (this code is used for Medicaid billing purposes)
J0725	Injection, chorionic gonadotropin, per 1,000 USP units
S4027	Storage of previously frozen embryos
S4040	Monitoring and storage of cryopreserved embryos, per 30 days
S4042	Management of ovulation induction (interpretation of diagnostic tests and studies, non-face-to-face medical management of the patient), per cycle

ICD-10 codes covered if selection criteria are met (not all-inclusive):

B20	Human immunodeficiency virus [HIV] disease [HIV positive male undergoing sperm washing]
C00.0 - C69.92, C81.00 - C96.9	Malignant neoplasms, lip, oral cavity, and pharynx, digestive organs and peritoneum, respiratory and intrathoracic organs, bone, connective tissue, skin, and breast, genitourinary organs, other and unspecified sites, lymphatic and hematopoietic tissue
C4A.4- C4A.9	Merkel cell carcinoma, scalp, neck, trunk, upper limb including shoulder, lower limb including hip, overlapping sites and unspecified

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

Code	Code Description
C7A.00 - C7A.098	Malignant carcinoid tumors, small intestine, appendix, large intestine, rectum, and other sites
C7A.1	Malignant poorly differentiated neuroendocrine tumors
C7B.1	Secondary Merkel cell carcinoma
D25.0 - D25.9	Leiomyoma of uterus
D27.0 - D27.9	Benign neoplasm of ovary
D29.30 - D29.32	Benign neoplasm of epididymis
D35.2 - D35.3	Benign neoplasm of pituitary gland and craniopharyngeal duct
D39.0 - D39.2	Neoplasm of uncertain behavior of female genital organs
D40.8 - D40.9	Neoplasm of uncertain behavior of other and unspecified male genital organs
D44.3 - D44.4	Neoplasm of uncertain behavior of pituitary gland and craniopharyngeal duct
E01.8	Other iodine-deficiency related thyroid disorders and allied conditions
E02	Subclinical iodine-deficiency hypothyroidism
E03.0 - E03.8	Other hypothyroidism
E13.40 - E13.43	Other specified diabetes mellitus
E22.8 - E22.9	Other and unspecified hyperfunction of pituitary gland
E22.1	Hyperprolactinemia
E23.0	Hypopituitarism
E23.6	Other disorders of pituitary gland
E25.0 - E25.9	Adrenogenital disorders
E28.1	Androgen excess
E28.2	Polycystic ovarian syndrome
E28.310 - E28.319	Premature menopause [poor ovarian reserve, spontaneous primary ovarian insufficiency]
E28.39	Other primary ovarian failure [poor ovarian reserve, spontaneous primary ovarian insufficiency]
E29.1	Testicular hypofunction
E89.0	Postprocedural hypothyroidism
E89.40 - E89.41	Postprocedural ovarian failure
I00 - I99.9	Diseases of the circulatory system
I86.1	Scrotal varices
L68.0	Hirsutism
N43.0 - N43.42	Hydrocele and spermatocele
N44.00 - N44.8	Noninflammatory disorders of testis
N46.01 - N46.9	Male infertility
N49.0 - N49.9	Inflammatory disorders of male genital disorders, not elsewhere classified
N50.0 - N50.9	Other and unspecified disorders of male genital organs
N51	Disorders of male genital organs in diseases classified elsewhere
N52.01 - N52.9	Male erectile dysfunction
N53.19	Other ejaculatory dysfunction
N64.3	Galactorrhea not associated with childbirth
N70.01 - N70.93	Salpingitis and oophoritis
N73.0 - N73.9	Other female pelvic inflammatory diseases
N80.00 - N80.9, N80.A0 - N80.D9	Endometriosis

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

Code	Code Description
N83.00 - N83.9	Noninflammatory disorders of ovary, fallopian tube and broad ligament
N84.0	Polyp of corpus uteri
N84.8	Polyp of female genital tract, unspecified [fallopian tube]
N85.6	Intrauterine synechiae
N91.0 - N91.2	Amenorrhea
N92.4	Excessive bleeding in the premenopausal period
N92.5 - N92.6	Other and unspecified irregular menstruation
N94.2	Vaginismus
N95.0 - N95.9	Menopausal and other perimenopausal disorders
N97.0 - N97.9	Female infertility
N98.1	Hyperstimulation of ovaries
N99.83	Residual ovary syndrome
Q50.01 - Q50.6	Congenital malformations of ovaries, fallopian tubes and broad ligaments
Q51.0, Q51.21 - Q51.28 Q51.5 - Q51.7 Q51.820 - Q51.9	Congenital malformations of uterus and cervix
Q52.0 - Q52.9	Other congenital malformations of female genitalia
Q53.00 - Q53.9	Undescended and ectopic testicle
Q55.0 - Q55.21 Q55.29 - Q55.4 Q55.7 - Q55.8	Other congenital malformations of male genital organs
Q96.9	Turner's syndrome, unspecified
Q98.0 - Q98.4	Klinefelter syndrome
R36.1	Hematospermia
R39.83	Unilateral non-palpable testicle
R86.0 - R86.9	Abnormal findings in specimens from male genital organs
R93.811 - R93.9	Abnormal findings on diagnostic imaging of other specified body structures [follow-up on hysterosalpingography abnormalities]
S14.0XXA - S14.9XXS	Injury of nerves and spinal cord at neck level
T50.905A - T50.905S	Adverse effect of unspecified drugs, medicaments and biological substances
T66.xxxA - T66.xxxS	Radiation sickness, unspecified, initial encounter
Z11.3	Encounter for screening for infections with a predominantly sexual mode of transmission [Chlamydia trachomatis screening]
Z14.01 - Z14.02	Hemophilia A carrier
Z14.1	Cystic fibrosis carrier
Z14.8	Genetic carrier of other disease [high-risk of transmitting a genetic disorder from the female partner to the offspring]

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

Code	Code Description
Z20.828	Contact with and (suspected) exposure to other viral communicable diseases [partners of persons infected with hepatitis B]
Z21	Asymptomatic human immunodeficiency virus [HIV] infection status
Z23	Encounter for immunization [rubella] [women susceptible to rubella]
Z31.41	Encounter for fertility testing
Z31.83	Encounter for assisted reproductive fertility procedure cycle
Z31.89	Encounter for other procreative management
Z52.810 - Z52.819	Egg (Oocyte) donor
Z78.0	Asymptomatic menopausal state
Z90.710 - Z90.712	Acquired absence of cervix and uterus
Z90.721 - Z90.722	Acquired absence of ovaries [for treatment of disease]
Z90.79	Acquired absence of other genital organ(s) [removal of testes for treatment of disease]

ICD-10 codes not covered for indications listed in the CPB (not all-inclusive):

N92.4 N95.0 - N95.9	Menopausal and other perimenopausal disorders
Z31.0	Encounter for reversal of previous sterilization
Z78.0	Asymptomatic menopausal state
Z79.890	Hormone replacement therapy
Z90.79	Acquired absence of other genital organ(s)
Z98.51 - Z98.52	Sterilization status

Background

"Infertility" has been defined by the American Society for Reproductive Medicine (ASRM, 2023) as a "disease, condition, or status characterized by any of the following:

- The inability to achieve a successful pregnancy based on a patient's medical, sexual, and reproductive history, age, physical findings, diagnostic testing, or any combination of those factors.
- The need for medical intervention, including, but not limited to, the use of donor gametes or donor embryos in order to achieve a successful pregnancy either as an individual or with a partner.
- In patients having regular, unprotected intercourse and without any known etiology for either partner suggestive of impaired reproductive ability, evaluation should be initiated at 12 months when the female partner is under 35 years of age and at 6 months when the female partner is 35 years of age or older.

Nothing in this definition shall be used to deny or delay treatment to any individual, regardless of relationship status or sexual orientation."

The term primary infertility is applied to a couple who has never achieved a pregnancy; secondary infertility implies that at least one previous conception has taken place. This disease may be present in one or both sexual partners and may be reversible.

Diagnostic investigation of infertility includes complete physical examinations and certain testing for both partners. Infertility treatment may involve a series of procedures in an attempt to correct the cause of

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

infertility.

Recurrent pregnancy loss is distinct from infertility is defined by two or more pregnancy losses. For purposes of determining when evaluation and treatment for infertility or recurrent pregnancy loss are appropriate, pregnancy is defined as a clinical pregnancy documented by ultrasonography or histopathologic examination (ASRM, 2012).

No fertility treatment other than oocyte donation has been shown to be effective for women over 40 years of age with compromised ovarian reserve. Elevated follicle-stimulating hormone (FSH) and estradiol levels are independent predictors of poor prognosis in older women. Common criteria for normal ovarian reserve are an early follicular phase FSH level of less than 10 mIU/ml and an estradiol level of less than 80 pg/ml (ASRM, 2002). Higher cut-off values for FSH have been reported (as high as 20 to 25 mIU/ml for FSH) because of the use of different FSH assay reference standards. Women with diminished ovarian reserve experience decreased responses to ovulation induction, require higher doses of gonadotropin, have higher in-vitro fertilization (IVF) cycle cancellation rates, and experience lower pregnancy rates through IVF.

Aetna covers ovarian stimulation medications and techniques only for women who have a biologic capacity to effectively respond to ovarian stimulation. Serum FSH is a marker of ovarian responsiveness. Ovarian responsiveness is determined by measurement of an unmedicated day 3 FSH obtained within the prior 6 months if the woman is older than age 35 or in the prior 12 months if the individual is age 35 or younger. In women greater than age 40, any single FSH greater than 19mIU/mL, regardless of subsequent test results that may be lower than 19mIU/mL, are indicative of ovarian insufficiency. In women less than age 40, ovarian responsiveness is demonstrated by any unmedicated day 3 FSH of less than 19mIU/ml. Younger women with a day 3 FSH less than 19mIU/ml have the capacity to respond to ovarian stimulation, even if they have had other day 3 FSH measurements greater than 19 mIU/mL.

Guidelines from the Society for Reproductive Medicine and Society for Assisted Reproductive Technology (Pfeifer et a., 2013) recommend that oocyte cryopreservation with appropriate counseling is recommended in patients facing infertility due to chemotherapy or other gonOadotoxic therapies. The guidelines state that more widespread clinic-specific data on the safety and efficacy of oocyte cryopreservation in donor populations are needed before universal donor oocyte banking can be recommended. The guidelines state that there are not yet sufficient data to recommend oocyte cryopreservation for the sole purpose of circumventing reproductive aging in healthy women. The guidelines state that more data are needed before this technology should be used routinely in lieu of embryo cryopreservation.

The American College of Obstetricians and Gynecologists (ACOG) practice bulletin on bariatric surgery and pregnancy (2009) stated that bariatric surgery should not be considered a treatment for infertility.

Although anti-mullerian hormone (AMH) levels appears to be associated with declining ovarian function, there is no consensus on the appropriate threshold value. An assessment by the Institute for Clinical and Health Policy (Pichon-Riviere, et al., 2009) found no clear evidence on the usefulness of AMH in the assisted reproduction program clinical practice setting. The assessment found less evidence for the utility of AMH in other clinical practice settings. Guidelines from the American Society for Reproductive Medicine (2012) concluded "There is mounting evidence to support the use of AMH as a screening test for poor ovarian response, but more data are needed. There is emerging evidence to suggest that a low AMH level (e.g., undetectable AMH) has high specificity as a screen for poor ovarian response but insufficient evidence to suggest its use to screen for failure to conceive."

More recently, the ASRM (2015) concluded: "AMH is a promising screening test and is likely to be more useful in the general ICVF population or in women at high risk for DOR than in women at low risk for DOR. Low AMH cutpoints are fairly specific for poor ovarian reserve, but not for pregnancy. Future studies of AHM as a screening test should incorporate larger numbers of subjects in a high-risk or general risk IVF population. The use of AMH as a routine screening tool for DOR in a low-risk population is not recommended."

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

The American College of Obstetricians and Gynecologists (ACOG, 2015) concluded: "For general obstetrician-gynecologists, the most appropriate ovarian reserve screening tests to use in practice are basal follicle-stimulating hormone (FSH) plus estradiol levels or antimullerian hormone (commonly known as AHM) levels. An antral follicle count, commonly known as AFC, also may be useful if there is an indication to perform transvaginal ultrasonography." The guidelines state that antimullerian hormone level testing is a useful test in women at high risk of diminished ovarian reserve and in women undergoing IVF but has limited benefits in someone at low risk of diminished ovarian reserve.

Current Endocrine Society guidelines on polycystic ovarian syndrome (PCOS) (Lego, et al., 2013) have no recommendation for antimullerian hormone. The guidelines note that "it is possible" that antimullerian hormone may serve as a noninvasive screening or diagnostic test for PCOS in the adolescent population, although there are no well-defined cutoffs. In discussing PCOS in the perimenopausal and menopausal population, the guidelines note that AMH levels decrease with normal aging in women with and without PCOS, but the guidelines make no recommendation for AMH testing in this population.

Steiner (2009) stated that serum and urinary markers of ovarian reserve – follicular phase inhibin B, FSH, and anti-mullerian hormone (AMH) levels – are physiologically associated with ovarian aging, decline with chronologic age, and appear to predict later stages of reproductive aging including the menopause transition and menopause. In infertile women, they can be used to predict low oocyte yield and treatment failure in women undergoing IVF. These markers seem to be affected by common ovarian toxicants, such as smoking, which advance the age at menopause. Although available for commercial use, home test kits have not been shown to predict fertility or infertility in the general population. Clinical use of these markers is limited by the variety of assays, lack of definitive thresholds, and their intercycle variability in older women. Results should be conveyed with caution when highly discrepant with age, in the obese, and in women with irregular menstrual cycles. The author stated that further research is needed to assess their predictive value for determining fertility in the general population.

Nelson et al (2009) stated that individualization of controlled ovarian stimulation (COS) for assisted conception is complicated by variable ovarian response to FSH. These researchers hypothesized that AMH may facilitate treatment strategies for women undergoing COS, to optimize safety and clinical pregnancy rates. A prospective cohort study of 538 patients in 2 centers with differential COS strategies based on a centralized AMH measurement was performed. Anti-Mullerian hormone was associated with oocyte yield after ovarian stimulation in both centers, and a "reduced" AMH (1 to less than 5 pmol/L) was associated with a reduced clinical pregnancy rate. Women with a "normal" AMH (5 to less than 15 pmol/L) treated with a long GnRH-agonist protocol (both centers) showed a low incidence of excess response (0 %) and poor response (0 %). In women with "high" AMH (greater than 15 pmol/L), the antagonist protocol eliminated the need for complete cryo-preservation of embryos due to excess response ($p < 0.001$) and showed a higher fresh cycle clinical pregnancy rate than agonist cycles odds ratio (OR) 4.40 (95 % confidence interval [CI]: 1.95 to 9.93), $p < 0.001$]. The authors concluded that the use of circulating AMH to individualize treatment strategies for COS may result in reduced clinical risk, optimized treatment burden and maintained pregnancy rates, and is worthy of prospective randomized examination.

Nardo et al (2009) evaluated the clinical value of basal AMH measurements compared with other available determinants, apart from chronologic age, in the prediction of ovarian response to gonadotrophin stimulation. Women undergoing their first cycle of controlled ovarian hyperstimulation (COH) for IVF were subject of this study. Basal levels of FSH and AMH as well as antral follicle count (AFC) were measured in 165 subjects. All patients were followed prospectively and their cycle outcomes recorded. Main outcome measures included predictive value of FSH, AMH, and AFC for extremes of ovarian response to stimulation. Out of the 165 women, 134 were defined as normal responders, 15 as poor responders, and 16 as high responders. Subjects in the poor response group were significantly older then those in the other 2 groups. Anti-Müllerian hormone levels and AFC were markedly raised in the high responders and decreased in the poor responders. Compared with FSH and AFC, AMH performed better in the prediction of excessive response to ovarian stimulation-AMH area under receiver operating characteristic curve (ROC(AUC)) 0.81, FSH ROC(AUC) 0.66, AFC ROC(AUC) 0.69. For poor response, AMH (ROC(AUC) 0.88) was a significantly better predictor than FSH (ROC(AUC) 0.63) but not AFC (ROC(AUC) 0.81). Anti-Mullerian hormone prediction of ovarian response was independent of age and polycystic ovarian syndrome (PCOS).

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

Anti-Mullerian hormone cutoffs of greater than 3.75 ng/ml and less than 1.0 ng/ml would have modest sensitivity and specificity in predicting the extremes of response. The authors concluded that circulating AMH has the ability to predict excessive and poor response to stimulation with exogenous gonadotrophins. Overall, this biomarker is superior to basal FSH and AFC, and has the potential to be incorporated in to work-up protocols to predict patient's ovarian response to treatment and to individualize strategies aiming at reducing the cancellation rate and the iatrogenic complications of COH.

Su and associates (2010) examined if AMH and inhibin B were impacted by breast cancer treatment by comparing cancer survivors to age-matched control women and determined the association between these hormones and post-chemotherapy menstrual pattern. Breast cancer patients (n = 127) with American Joint Committee on Cancer stage I to III disease who were pre-menopausal at diagnosis were enrolled post-chemotherapy and observed. The primary end point was chemotherapy-related amenorrhea (CRA) (greater than or equal to 12 months of amenorrhea following chemotherapy). Matched pair analyses compared AMH, inhibin B, and FSH levels between cancer and age-matched control subjects. Associations between hormones, CRA status, and change in CRA status over time were assessed. The median age of the patients at chemotherapy was 43.2 years (range of 26.7 to 57.8 years). At enrollment, median follow-up since chemotherapy was 2.1 years, and 55 % of subjects had CRA. Compared with age-matched controls, cancer subjects had significantly lower AMH ($p = 0.004$) and inhibin B ($p < 0.001$) and higher FSH ($p < 0.001$). Inhibin B ($p = 0.001$) and AMH ($p = 0.002$) were found to be significantly associated with risk of CRA, even after controlling for FSH. Anti-mullerian hormone was significantly lower ($p = 0.03$) and FSH was significantly higher ($p = 0.04$) in menstruating subjects who developed subsequent CRA. The authors concluded that AMH and inhibin B are 2 additional measures of post-chemotherapy ovarian function in late reproductive-aged breast cancer survivors. They stated that with further research and validation, these hormones may supplement limited current tools for assessing and predicting post-chemotherapy ovarian function.

In a Cochrane review, Duffy et al (2010) the effectiveness of adjuvant growth hormone (GH) in IVF protocols. These investigators searched the Cochrane Menstrual Disorders and Subfertility Groups trials register (June 2009), the Cochrane Central Register of Controlled Trials (Cochrane Library Issue 2, 2009), MEDLINE (1966 to June 2009), EMBASE (1988 to June 2009) and Biological Abstracts (1969 to June 2009). All randomized controlled trials were included if they addressed the research question and provided outcome data for intervention and control participants. Assessment of trial risk of bias and extraction of relevant data was performed independently by 2 reviewers. A total of 10 studies (440 subfertile couples) were included. Results of the meta-analysis demonstrated no difference in outcome measures and adverse events in the routine use of adjuvant GH in IVF protocols. However, meta-analysis demonstrated a statistically significant difference in both live birth rates and pregnancy rates favoring the use of adjuvant GH in IVF protocols in women who are considered poor responders without increasing adverse events, OR 5.39, 95 % CI: 1.89 to 15.35 and OR 3.28, 95 % CI: 1.74 to 6.20 respectively. The authors concluded that although the use of GH in poor responders has been found to show a significant improvement in live birth rates, they were unable to identify which sub-group of poor responders would benefit the most from adjuvant GH. The result needs to be interpreted with caution, the included trials were few in number and small sample size. Thus, before recommending GH adjuvant in IVF, further research is needed to fully define its role.

Guidelines from the American Society for Reproductive Medicine (2012) concluded that "inhibin B is not a reliable measure of ovarian reserve" and that "the routine use of inhibin B as a measure of ovarian reserve is not recommended."

In a meta-analysis, Toulis et al (2010) evaluated the diagnostic accuracy of inhibin B and AMH as markers of persistent spermatogenesis in men with non-obstructive azoospermia (NOA). A search was conducted in the electronic databases MEDLINE, EMBASE and Cochrane Central Register of Controlled Trials from inception through June 2009. A total of 36 different studies reported data on the predictive value of 1 or more index markers (serum inhibin B: 32 studies, seminal inhibin B: 5 studies, serum AMH: 2 studies, seminal AMH: 4 studies) and were included in the systematic review. Nine studies, which had serum inhibin B as index marker, met the predefined criteria and were included in the meta-analysis. Serum inhibin B showed a sensitivity of 0.65 (95 % CI: 0.56 to 0.74) and a specificity of 0.83 (CI: 0.64 to 0.93) for the

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

prediction of the presence of sperm in testicular sperm extraction (TESE). When the pre-test probability of 41 % was incorporated in a Fagan's nomogram, resulted in a positive post-test probability of 73 % and a negative post-test probability of 23 % for the presence of sperm in TESE. The authors concluded that serum inhibin B can not serve as a stand-alone marker of persistent spermatogenesis in men with NOA. Although limited, evidence on serum AMH and serum/seminal AMH do not support their diagnostic value in men with NOA.

Steiner et al (2011) generated estimates of the association between markers of ovarian aging and natural fertility in a community sample at risk for ovarian aging. Women aged 30 to 44 years with no history of infertility who had been trying to conceive for less than 3 months provided early-follicular phase serum and urine (n = 100). Subsequently, these women kept a diary to record menstrual bleeding and intercourse and conducted standardized pregnancy testing for up to 6 months. Serum was analyzed for estradiol, FSH, AMH, and inhibin B. Urine was analyzed for FSH and estrone 3-glucuronide. Diary data on menstrual cycle day and patterns of intercourse were used to calculate day-specific fecundability ratios. Sixty-three percent of participants conceived within 6 months. After adjusting for age, 18 women (18 %) with serum AMH levels of 0.7 ng/ml or less had significantly reduced fecundability given intercourse on a fertile day compared with women with higher AMH levels (fecundability ratio 0.38; 95 % CI: 0.08 to 0.91). The day-specific fecundability for women with early-follicular phase serum FSH values greater than 10 mIU/ml compared with women with lower FSH levels was also reduced, although nonsignificantly (11 % of women affected; fecundability ratio 0.44; 95 % CI: 0.08 to 1.10). The association with urinary FSH was weaker (27 % women affected; fecundability ratio 0.61; 95 % CI: 0.26 to 1.26), and the associations for the other markers were weaker still. The authors concluded that early-follicular phase AMH appears to be associated with natural fertility in the general population. Moreover, they stated that larger studies are needed to confirm these findings and to explore the way the different endocrine markers interact as potential joint predictors of fertility.

In a meta-analysis, Polyzos and associates (2010) examined the effect of double versus single intra-uterine insemination (IUI) per treatment cycle in women with unexplained infertility. Main outcome measure was clinical pregnancy rates per couple. Electronic searches of the Cochrane Central Trials Registry and Medline without year and language restriction through March 2009 were performed; hand searching of the abstract books of the European Society of Human Reproduction and Embryology and American Society for Reproductive Medicine annual meetings (2001 to 2008) was carried out. A total of 6 randomized trials, involving 829 women, were included in the analysis. Fifty-four (13.6 %) clinical pregnancies were recorded for treatment with double IUI and 62 (14.4 %) for treatment with single IUI. There was no significant difference between the single and double IUI groups in the probability for clinical pregnancy (OR, 0.92; 95 % CI: 0.58 to 1.45; p = 0.715). The authors concluded that double IUI offers no clear benefit in the overall clinical pregnancy rate in couples with unexplained infertility.

Guercini et al (2005) reported that in chronic prostatitis there are many causes that may provoke a therapeutical failure of a systemic antibiotic treatment. At the moment a consensus has not been reached on the effectiveness of the many therapeutical options that are available with not one of these approaches being effective in all patients. In the authors' view the main causes of treatment failure are the well-known hurdle to antibiotic diffusion inside the glandular parenchyma associated with the so-called intra-prostatic bacterial biofilms and the possible presence of local auto-immune reactions. Given this background, these researchers tested ultrasound-guided intra-prostate infiltration of a cocktail of antibiotics and betamethasone, for therapeutical options. A total of 320 patients, referred for treatment because of symptoms indicative of chronic prostatitis, were enrolled in this study. The inclusion criteria were the severity of the symptoms and the failure of repeated cycles of antibiotics in the previous 12 months. At the initial consultation patients completed the NIH Prostatitis Symptoms Index (NIH-CPSI). All underwent:

- digital rectal examination (DRE),
- transrectal prostatic ultrasound scan (TRUS),
- uroflowmetry,
- cultures of first voiding and after prostatic massage urine and cultures of sperm for saprophytic and pathogen germs, yeasts and protozoa,

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

- DNA amplification with polymerase chain reaction (PCR) on urine and sperm, for Chlamydia trachomatis, Mycoplasmas (Ureaplasma urealyticum and Mycoplasma hominis), Gonococcus, HPV and HCV.

Patients on the basis of laboratory results received a cocktail of antibiotics associated with betamethasone. The cocktail was administered as prostate infiltration. Administration was repeated after 7 and 14 days. Final assessment of the effectiveness of therapy included not only the NIH-CPSI scores but also the patient's subjective judgement expressed as a "percentage overall improvement". The percentage judgements were arbitrarily divided into 4 classes:

- 0 to 30 %: no improvement (Class I);
- 30 to 50 %: satisfactory improvement (Class II);
- 50 to 80 %: good improvement (Class III); and
- 80 to 100 %: cured (Class IV).

Statistical analysis of the results showed 68 % of patients were included in the Class IV and 13 % were non-responders (Class I). The authors concluded that this is one of the more valid therapeutical approaches to chronic bacterial or abacterial prostatitis; but it also required more studies.

McGrath et al (2009) stated that cycle-dependent fluctuations in natural killer (NK) cell populations in endometrium and circulation may differ, contributing to unexplained infertility. They conducted a study whereby NK cell phenotypes were determined by flow cytometry in endometrial biopsies and matched blood samples. While circulating and endometrial T cell populations remained constant throughout the menstrual cycle in fertile and infertile women, circulating NK cells in infertile women increased during the secretory phase. However, increased expression of CD94, CD158b (secretory phase), and CD158a (proliferative phase) by endometrial NK cells from infertile women was observed. These changes were not reflected in the circulation. In infertile women, changes in circulating NK cell percentages were found exclusively during the secretory phase and not in endometrium; cycle-related changes in NK receptor expression were observed only in infertile endometrium. While having exciting implications for understanding NK cell function in fertility, these data emphasized the difficulty in attaching diagnostic or prognostic significance to NK cell analyses in individual patients.

Winger et al (2011) examined if quantification of peripheral blood Treg cell levels could be used as an indicator of miscarriage risk in newly pregnant women with a history of immunologic reproductive failure. A total of 54 pregnant women with a history of immunologic infertility and/or pregnancy loss were retrospectively evaluated (mean age of 36.7 +/- 4.9 years, 2.8 +/- 2.5 previous miscarriages; 1.5 +/- 1.9 previous IVF failures). Twenty-three of these women experienced another first trimester miscarriage, and 31 of these women continued their current pregnancies past 12 weeks ("pregnancy success"). The following immunologic parameters were assessed in the first trimester: NK cell 50:1 cytotoxicity, CD56(+) 16(+) CD3(-) (NK), CD56(+) CD3(+) (NKT), TNFα/IL-10, IFNγ/IL-10, CD4(+) CD25(-) Foxp3(+), total CD4(+) Foxp3(+) (CD4(+) CD25(+) Foxp3 plus CD25(-) Foxp3(+)), and CD4(+) CD25(+) Foxp3(+) levels. Patients with successful ongoing pregnancies experienced a mean (CD4(+) CD25(+) Foxp3(+)) "Treg" level of 0.72 +/- 0.52 %, while those that miscarried in the first trimester experienced a mean Treg level of 0.37 +/- 0.29 % ($p = 0.005$). Markers not significantly different between the loss and success groups were NK 50:1 cytotoxicity ($p = 0.63$), CD56(+) 16(+) 3(+) NK cells ($p = 0.63$), CD56(+) 3(+) NKT ($p = 0.30$), TNFα(+) IL-10(+) ($p = 0.13$), IFNg(+) IL-10(+) ($p = 0.63$), and CD4(+) 25(-) Foxp3(+) cells ($p = 0.10$), although total CD4(+) Foxp3(+) levels remained significant ($p = 0.02$) and CD4(+) 25(+) Foxp3(+) showed the most significant difference ($p = 0.005$). Mean day of blood draw was 49.2 +/- 36.1 days pregnant (median of 39.0 days). In addition, patients with a low Treg level (less than 0.7 %) in the first trimester experienced a significantly lower ongoing pregnancy rate than those with a higher Treg level (greater than 0.7 %) in the first trimester [44 % (15/34) versus 80 % (16/20); $p = 0.01$]. Of the 18 successful pregnancies with sequential Treg results, 85 % (11/13) showed a T-regulatory-cell-level increase (mean Treg change 0.33 +/- 0.32), while only 40 % (2/5) of the failed pregnancies showed a Treg increase (mean Treg change -0.08 +/- 0.28; $p = 0.02$). The authors concluded that from these data, they proposed that CD4(+) CD25(+) Foxp3(+) T regulatory cells may serve as a superior pregnancy marker for assessing miscarriage risk in newly pregnant women. Moreover, they stated that larger follow-up studies are needed for confirmation.

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

In a prospective, randomized controlled trial, Ben-Meir et al (2010) examined if supplementation with hCG throughout the secretory phase of hormonally modulated cycles of frozen-thawed embryos might positively affect the outcome of such cycles. Patients were randomly divided into 2 groups by the last digit of their identification number. Group A received the authors' standard protocol for endometrial preparation, whereas group B patients were given an additional 250 microg of recombinant hCG on day of progesterone (P) initiation, the day of embryo transfer, and 6 days later. Throughout the cycle, and to compare between the groups, serial ultrasound examinations and hormonal tests of E(2) and P serum levels were obtained. Main outcome measures were implantation and clinical pregnancy rates (PR). A total of 165 patients were enrolled in this study – 78 in the control group and 87 in the hCG-treated group. Progesterone levels and endometrial thickness were similar throughout the cycle in both groups. The E(2) level was significantly higher in group B on the day of embryo transfer and 6 days later. The PRs did not differ between the 2 groups (28.2 % and 32.2 % for groups A and B, respectively). Similarly, the implantation rates were comparable between the groups (12.7 % and 14.9 %, respectively). The authors concluded that no advantage was found concerning PR and implantation rate by supplementing the secretory phase with hCG in patients undergoing transfer of frozen-thawed embryo in hormonally modulated cycles.

In a systematic review and meta-analysis, Momeni et al (2011) evaluated the relationship between endometrial thickness on the day of hCG administration and pregnancy outcome in in-vitro fertilization cycles. These investigators identified 484 articles using Cochrane library, PubMed, Web of Science, and Embase searches with various key words including endometrial thickness, pregnancy, assisted reproductive technology, endometrial pattern, and in-vitro fertilization. A total of 14 studies with data on endometrial thickness and outcome were selected, representing 4,922 cycles (2,204 pregnant and 2,718 non-pregnant). The meta-analysis with a random effects model was performed using comprehensive meta-analysis software. These researchers calculated the standardized mean difference (SMD), odds ratio (OR), and 95 % confidence intervals (CIs). There was a significant difference in the mean endometrial thickness between pregnant and non-pregnant groups ($p < 0.001$), with a SMD of 0.4 mm (95 % CI: 0.22 to 0.58). The OR for pregnancy was 1.40 (95 % CI: 1.24 to 1.58). The authors concluded that the mean endometrial thickness was significantly higher in pregnant women compared to non-pregnant. The mean difference between 2 groups was less than 1 mm, which may not be clinically meaningful. Moreover, they stated that although there may be a relationship between endometrial thickness and pregnancy, implantation potential is probably more complex than a single ultrasound measurement can determine.

van der Linden et al (2011) determined the relative safety and effectiveness of methods of luteal phase support in subfertile women undergoing assisted reproductive technology (ART). These investigators searched the Cochrane Menstrual Disorders and Subfertility Group (MDSG) Specialised Register, Cochrane Central Register of Controlled Trials (CENTRAL), MEDLINE, EMBASE, PsycINFO, CINAHL, Database of Abstracts of Reviews of Effects (DARE), LILACS, conference abstracts on the ISI Web of Knowledge, OpenSigle for grey literature from Europe, and ongoing clinical trials registered online. The final search was in February 2011. Randomized controlled trials of luteal phase support in ART investigating progesterone, hCG or GnRH agonist supplementation in IVF or intra-cytoplasmic sperm injection (ICSI) cycles. Quasi-randomized trials and trials using frozen transfers or donor oocyte cycles were excluded. These researchers extracted data per women and 3 review authors independently assessed risk of bias. They contacted the original authors when data were missing or the risk of bias was unclear; and they entered all data in 6 different comparisons. These investigators calculated the Peto odds ratio (Peto OR) for each comparison. A total of 69 studies with 16,327 women were included. The authors assessed most of the studies as having an unclear risk of bias, which we interpreted as a high-risk of bias. Because of the great number of different comparisons, the average number of included studies in a single comparison was only 1.5 for live birth and 6.1 for clinical pregnancy. Five studies (746 women) compared hCG versus placebo or no treatment. There was no evidence of a difference between hCG and placebo or no treatment except for ongoing pregnancy: Peto OR 1.75 (95 % CI: 1.09 to 2.81), suggesting a benefit from hCG. There was a significantly higher risk of ovarian hyper-stimulation syndrome (OHSS) when hCG was used (Peto OR 3.62, 95 % CI: 1.85 to 7.06). There were 8 studies (875 women) in the second comparison, progesterone versus placebo or no treatment. The results suggested a significant effect in favor of progesterone for the live birth rate (Peto OR 2.95, 95 % CI: 1.02 to 8.56) based on one study. For clinical pregnancy (CPR) the results also suggested a significant result in favor of progesterone (Peto OR 1.83, 95 % CI: 1.29 to 2.61) based on seven studies. For the other outcomes the results indicated no difference in effect. The third comparison (15 studies, 2,117 women)

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

investigated progesterone versus hCG regimens. The hCG regimens were subgrouped into comparisons of progesterone versus hCG and progesterone versus progesterone + hCG. The results did not indicate a difference of effect between the interventions, except for OHSS. Subgroup analysis of progesterone versus progesterone + hCG showed a significant benefit from progesterone (Peto OR 0.45, 95 % CI: 0.26 to 0.79). The fourth comparison (9 studies, 1,571 women) compared progesterone versus progesterone + estrogen. Outcomes were subgrouped by route of administration. The results for clinical pregnancy rate in the subgroup progesterone versus progesterone + transdermal oestrogen suggested a significant benefit from progesterone + estrogen. There was no evidence of a difference in effect for other outcomes. Six studies (1,646 women) investigated progesterone versus progesterone + GnRH agonist. These researchers subgrouped the studies for single-dose GnRH agonist and multiple-dose GnRH agonist. For the live birth, clinical pregnancy and ongoing pregnancy rate the results suggested a significant effect in favor of progesterone + GnRH agonist. The Peto OR for the live birth rate was 2.44 (95 % CI: 1.62 to 3.67), for the clinical pregnancy rate was 1.36 (95 % CI: 1.11 to 1.66) and for the ongoing pregnancy rate was 1.31 (95 % CI: 1.03 to 1.67). The results for miscarriage and multiple pregnancies did not indicate a difference of effect. The last comparison (32 studies, 9,839 women) investigated different progesterone regimens: Intra-muscular (IM) versus oral administration, IM versus vaginal or rectal administration, vaginal or rectal versus oral administration, low-dose vaginal versus high-dose vaginal progesterone administration, short protocol versus long protocol and micronized progesterone versus synthetic progesterone. The main results of this comparison did not indicate a difference of effect except in some subgroup analyses. For the outcome clinical pregnancy, subgroup analysis of micronized progesterone versus synthetic progesterone showed a significant benefit from synthetic progesterone (Peto OR 0.79, 95 % CI: 0.65 to 0.96). For the outcome multiple pregnancies, the subgroup analysis of IM progesterone versus oral progesterone suggested a significant benefit from oral progesterone (Peto OR 4.39, 95 % CI: 1.28 to 15.01). The authors concluded that this review showed a significant effect in favor of progesterone for luteal phase support, favoring synthetic progesterone over micronized progesterone. Overall, the addition of other substances such as estrogen or hCG did not seem to improve outcomes. They also found no evidence favoring a specific route or duration of administration of progesterone. These investigators found that hCG, or hCG plus progesterone, was associated with a higher risk of OHSS. The use of hCG should therefore be avoided. There were significant results showing a benefit from addition of GnRH agonist to progesterone for the outcomes of live birth, clinical pregnancy and ongoing pregnancy. For now, progesterone seems to be the best option as luteal phase support, with better pregnancy results when synthetic progesterone is used.

Morley et al (2013) stated that recurrent miscarriage (RM) is defined as the loss of 3 or more consecutive pregnancies. Further research is required to understand the causes of RM, which remain unknown for many couples. Human chorionic gonadotropin is vital for maintaining the corpus luteum, but may have additional roles during implantation which support its use as a therapeutic agent for RM. In a Cochrane review, these investigators determined the efficacy of hCG in preventing further miscarriage in women with a history of unexplained RM. They searched the Cochrane Pregnancy and Childbirth Group's Trials Register (September 30, 2012) and reference lists of retrieved studies. Randomized controlled trials investigating the efficacy of hCG versus placebo or no treatment in preventing RM were included for analysis. Quasi-randomized trials were included. Cluster-randomized trials and trials with a cross-over design were excluded. Two review authors independently assessed trials for inclusion and assessed the methodological quality of each study. Date were extracted by 2 review authors and checked for accuracy. These investigators included 5 studies (involving 596 women). Meta-analysis suggested a statistically significant reduction in miscarriage rate using hCG. The number of women needed to treat to prevent subsequent pregnancy loss was 7. However, when 2 studies of weaker methodological quality were removed, there was no longer a statistically significant benefit (risk ratio 0.74; 95 % CI: 0.44 to 1.23). There were no documented adverse effects of using hCG. The authors concluded that the evidence supporting hCG supplementation to prevent RM remains equivocal. A well-designed randomized controlled trial of adequate power and methodological quality is required to determine whether hCG is beneficial in RM.

Also, an UpToDate review on “Overview of treatment of female infertility” (Kuohung and Hornstein, 2014) does not mention the use of human chorionic gonadotropin as a management tool.

Current guidelines recommend hCG in men only for pituitary hypogonadism to address infertility issues. It is not recommended for long-term use outside of infertility treatment. The European Association of

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

Urology's guidelines on "Male hypogonadism" (Dohle et al, 2012) noted that "In patients with secondary hypogonadism and fertility issues, and in selected cases of primary hypogonadism, hCG treatment can be chosen to support endogenous testosterone production for the period of infertility treatment. The dosage has to be adjusted individually to prevent suppression of FSH serum levels. hCG treatment has higher costs than testosterone treatment. There is insufficient information about the therapeutic and adverse effects of long-term hCG treatment. This type of treatment can therefore not be recommended for male hypogonadism, except in patients in whom fertility treatment is an issue".

In a Cochrane review, Siristatidis et al (2013) compared outcomes associated with in-vitro maturation (IVM) followed by IVF or ICSI versus conventional IVF or ICSI, among women with PCOS undergoing assisted reproductive technologies (ART). These searched the Menstrual Disorders and Subfertility Group (MDSG) Specialised Register of controlled trials to May 2013 for any relevant trials identified from the title, abstract, or keyword sections. This was followed by a search of the electronic database MEDLINE, EMBASE, LILACS and CINAHL, without language restriction. They also performed a manual search of the references of all retrieved articles; sought unpublished papers and abstracts submitted to international conferences, searched the clinicaltrials.gov and WHO portal registries for submitted protocols of clinical trials, and contacted experts. In addition, these researchers examined the National Institute of Clinical Excellence (NICE) fertility assessment and treatment guidelines and hand-searched reference lists of relevant articles (from 1970 to May 2013). All randomized controlled trials (RCTs) on the intention to perform IVM before IVF or ICSI compared with conventional IVF or ICSI for subfertile women with PCOS. Three review authors independently assessed eligibility and quality of trials. Primary outcome measure was live birth rate per randomized woman. There were no RCTs suitable for inclusion in the review, although there are currently 3 ongoing trials that have not yet reported results. The authors concluded that although promising data on the IVM technique have been published, unfortunately there is still no evidence from RCTs upon which to base any practice recommendations regarding IVM before IVF or ICSI for women with PCOS.

Furthermore, an UpToDate review on "Fertility preservation in patients undergoing gonadotoxic treatment or gonadal resection" (Sonmezer and Oktay, 2014) states that "When embryo cryopreservation is not feasible, cryopreservation of oocytes matured in vivo is a reasonable option. In vitro maturation of oocytes is an investigational procedure; implantation and ongoing pregnancy rates are lower than with conventional in vitro fertilization (IVF) using in vivo matured oocytes".

ICSI has become standard of care for fertilization of frozen oocytes in in vitro cycles despite a lack of controlled studies (see, e.g., Kazem, et al., 1995; Gook, et al., 2005; Li, et al., 2005). Gook and Edgar (2007) explained that: "In contrast to the low normal fertilization rates observed in cryopreserved mouse oocytes, higher normal fertilization rates (∼50%) were observed following insemination of human oocytes cryopreserved using the DMSO [citing Al-Hasani et al., 1987; Siebzehnruebl et al., 1989; Hunter et al., 1991; Bernard et al., 1992] and the PROH procedures [citing Al-Hasani et al., 1987; Gook et al., 1994; Serafini et al., 1995]. Further evidence that cryopreservation using the PROH procedure had no adverse affect on fertilization was demonstrated by the observation of equivalent fertilization rates (∼50%) following insemination and ICSI [citing Gook et al., 1995; Li et al., 2005]. In contrast, Kazem et al. (1995) reported a lower rate with insemination (3%) relative to ICSI (43%). Despite the fact that there is no evidence from controlled comparisons of insemination techniques to suggest that ICSI is required to fertilize human cryopreserved oocytes, it has been adopted as the method of choice in subsequent clinical studies."

In contrast to the low normal fertilization rates observed in cryopreserved mouse oocytes, higher normal fertilization rates (∼50%) were observed following insemination of human oocytes cryopreserved using the DMSO (Al-Hasani et al., 1987; Siebzehnruebl et al., 1989; Hunter et al., 1991; Bernard et al., 1992) and the PROH procedures (Al-Hasani et al., 1987; Gook et al., 1994; Serafini et al., 1995). Further evidence that cryopreservation using the PROH procedure had no adverse affect on fertilization was demonstrated by the observation of equivalent fertilization rates (∼50%) following insemination and ICSI (Gook et al., 1995b; Li et al., 2005). In contrast, Kazem et al. (1995) reported a lower rate with insemination (3%) relative to ICSI (43%). Despite the fact that there is no evidence from controlled comparisons of insemination techniques to suggest that ICSI is required to fertilize human cryopreserved oocytes, it has been adopted as the method of choice in subsequent clinical studies.

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

Chen et al (2013) stated that reactive oxygen species (ROS) are an array of molecules including oxygen-centered radicals, which are endowed with 1 or more unpaired electrons and non-radical oxygen derivatives such as hydrogen peroxide, which behave, to a large extent, like a double-edged sword in human sperm biology. These investigators reviewed the current knowledge of ROS in sperm physiology and pathology, as well as related therapies in spermatozoal dysfunction. They searched for keywords from PUBMED, including reactive oxygen species, oxidative stress, sperm function, and antioxidant therapy. Low levels of ROS exert critical function in normal sperm physiology, such as fertilizing ability (acrosome reaction, hyper-activation, capacitation, and chemotaxis) and sperm motility; while increased ROS generation and/or decreased antioxidant capacity leads to the imbalance between oxidation and reduction in living systems, which is called sperm oxidative stress. This condition was widely considered to be a significant contributory factor to sperm DNA damage/apoptosis, lipid peroxidation, and reduced motility, which in turn, increased risk of male factor infertility/subfertility and birth defects. Under the current status quo, numerous subsequent studies have concentrated on antioxidant therapy. Although utility of such a therapeutic strategy significantly improved sperm function and motility in a myriad of experimental and clinical reports, the overall effectiveness still remains controversial mainly due to non-standardized assay to measure the level of ROS and sperm DNA damage, various antioxidant supplementation strategies, and inadequate fertilization and pregnancy data after clinical treatment. Therefore, standardized assessment and evaluation of ROS and total antioxidant capacity in semen should be established to keep ROS in a physiological level and prevent over-treatment of antioxidants toward reductive stress, which should be kept in mind, especially in assisted reproductive procedure. The authors noted that the significance of large sample size populations, double-blind randomized, placebo-controlled clinical trials of antioxidant therapies is emphasized in this review to achieve optimal ingredients and dosage of antioxidants for patients with reactive oxygen-induced male fertility/subfertility.

Also, an UpToDate review on “Evaluation of male infertility” (Swerdloff and Wang, 2014) states that “Generation of reactive oxygen species may be a cause of sperm dysfunction and a predictor of fertilization in vitro. Reactive oxygen species lead to lipid peroxidation of the sperm membrane and are also deleterious to sperm motility. This is still regarded as a research test and is not often used for diagnosis of a specific sperm defect”.

Cryopreservation of immature oocytes and in vitro maturation are considered experimental procedures. The term in vitro maturation refers to the maturation in culture of immature oocytes after their recovery from follicles that may or may not have been exposed to exogenous FSH but were not exposed to either exogenous LH or hCG prior to retrieval to induce meiotic resumption. Guidelines from the American Society for Reproductive Medicine (2013) state that in vitro maturation should only be performed as an experimental procedure in specialized centers for carefully selected patients evaluating both efficacy and safety. The guidelines state that the intitial results of in vitro maturation suggest the potential for clinical application. However, at this time, patients must be made aware that the implantation and pregnancy rates are significantly lower than with standard IVF, limiting more universal utilization.

Chighizola and de Jesus (2014) noted that since the late 1980s some publications have proposed that antiphospholipid antibodies (aPL) may have some relationship with infertility, considering reported deleterious effects that aPL exert on trophoblast proliferation and growth. Although not included in current classification criteria for antiphospholipid syndrome, many physicians investigated for aPL in patients with a history of infertility, including antibodies not listed in classification criteria, and most of those patients will receive anticoagulant therapy if any of those antibodies have a result considered positive. These investigators performed a review of literature searching for studies that investigated the association of aPL and infertility and if aPL positivity alters IVF outcome. The definition of infertility, routine work-up to exclude other causes of infertility, definition of IVF failure as inclusion criteria and control populations were heterogeneous among studies. Most of them enrolled women over 40 years of age, and exclusion of other confounding factors was also inconsistent. Of 29 studies that assessed aPL positivity rates in infertile women, the majority had small sample sizes, implying a lack of power, and 13 (44.8 %) reported higher frequency of aPL in infertile patients compared to controls, but most of them investigated a panel of non-criteria aPL tests, whose clinical significance is highly controversial. Only 2 studies investigated all 3 criteria tests, and medium-high titer of anticardiolipin cut-off conforming to international guidelines was used in 1 study. Considering IVF outcome, there was also disparity in this definition: few studies assessed

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

the live birth rate, others the implantation rate. Of 14 publications that addressed the relationship between aPL and IVF outcome, only 2 described a detrimental effect of these autoantibodies. The authors concluded that available data do not support an association between aPL and infertility, and aPL positivity does not seem to influence IVF outcome. They stated that well-designed clinical studies recruiting women with a clear diagnosis of infertility and a high-risk aPL profile should be performed to test whether clinically relevant aPL do-or not-exert an effect on human fertility.

Furthermore, an UpToDate review on "Evaluation of female infertility" (Kuohung and Hornstein, 2015b) states that "Testing for antibodies – Routine testing for antiphospholipid, antisperm, antinuclear, and antithyroid antibodies is not supported by existing data. Although an association between antiphospholipid antibodies and recurrent pregnancy loss has been established, the other autoimmune factors remain under investigation as markers of fertility treatment failure".

In a Cochrane review, McDonnell et al (2014) examined the effectiveness and safety of functional ovarian cyst aspiration prior to ovarian stimulation versus a conservative approach in women with an ovarian cyst who were undergoing IVF or ICSI. These investigators searched the Menstrual Disorders and Subfertility Group (MDSG) Specialised Register, Cochrane Central Register of Controlled Trials (CENTRAL), MEDLINE, EMBASE, PsycINFO, CINAHL, ClinicalTrials.gov, Google Scholar and PubMed. The evidence was current to April 2014 and no language restrictions were applied. These researchers included all RCTs comparing functional ovarian cyst aspiration versus conservative management of ovarian cysts that have been seen on transvaginal ultrasound (TVS) prior to COH for IVF or ICSI. Ovarian cysts were defined as simple, functional ovarian cysts greater than 20 mm in diameter. Oocyte donors and women undergoing donor oocyte cycles were excluded. Study selection, data extraction and risk of bias assessments were conducted independently by 2 review authors. The primary outcome measures were live birth rate and adverse events. The overall quality of the evidence for each comparison was rated using Grades of Recommendation, Assessment, Development and Evaluation (GRADE) Working Group methods. A total of 3 studies were eligible for inclusion (n = 339), all of which used agonist protocols. Neither live birth rate nor adverse events were reported by any of the included studies. There was no conclusive evidence of a difference between the group who underwent ovarian cyst aspiration and the conservatively managed group in the clinical pregnancy rate (OR 1.40, 95 % CI: 0.67 to 2.94, 3 studies, 339 women, I(2) = 0 %, low-quality evidence). This suggested that if the clinical pregnancy rate in women with conservative management was assumed to be 5 %, the chance following cyst aspiration would be between 4 % and 14 %. There was no evidence of a difference between the groups in the mean number of follicles recruited (0.55 follicles, 95 % CI: -0.48 to 1.59, 2 studies, 159 women, I(2) = 0 %, low-quality evidence) or mean number of oocytes collected (0.41 oocytes, 95 % CI: -0.04 to 0.85, 3 studies, 339 women, I(2) = 0 %, low-quality evidence). Findings for the cancellation rate (2 studies) were inconsistent but neither study reported a benefit for the aspiration group. The main limitations of the evidence were imprecision, inconsistency, questionable applicability, and poor reporting of study methods. The authors concluded that there is insufficient evidence to determine whether drainage of functional ovarian cysts prior to controlled ovarian hyperstimulation influences live birth rate, clinical pregnancy rate, number of follicles recruited, or oocytes collected in women with a functional ovarian cyst. They stated that the findings of this review do not provide supportive evidence for this approach, particularly in view of the requirement for anesthesia, extra cost, psychological stress and risk of surgical complications.

Gleicher et al (2014) noted that a few years ago the ASRM, the European Society for Human Reproduction and Embryology (ESHRE) and the British Fertility Society declared preimplantation genetic screening (PGS#1) ineffective in improving IVF pregnancy rates and in reducing miscarriage rates. These investigators reviewed a presumably upgraded form of the procedure (PGS#2) that has recently been re-introduced. PGS#2 in comparison to PGS#1 is characterized by: (i) trophectoderm biopsy on day 5/6 embryos in place of day-3 embryo biopsy; and (ii) fluorescence in-situ hybridization (FISH) of limited chromosome numbers is replaced by techniques, allowing aneuploidy assessments of all 24 chromosome pairs. Reviewing the literature, the authors were unable to identify properly conducted prospective clinical trials in which IVF outcomes were assessed based on "intent-to-treat". Whether PGS#2 improves IVF outcomes can, therefore, not be determined. Re-assessments of data, alleged to support the effectiveness of PGS#2, indeed, suggested the opposite. Like with PGS#1, the introduction of PGS#2 into unrestricted IVF practice again appears premature, and threatens to repeat the PGS#1 experience, when thousands of women experienced reductions

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

in IVF pregnancy chances, while expecting improvements. The authors concluded that PGS#2 is an unproven and still experimental procedure, which, until evidence suggests otherwise, should only be offered under study conditions, and with appropriate informed consents.

Lee et al (2015) examined if preimplantation genetic diagnosis for aneuploidy (PGD-A) with analysis of all chromosomes during ART is clinically and cost effective? These investigators performed a systematic review of the literature for full text English language articles using MEDLINE, EMBASE, SCOPUS, Cochrane Library databases, NHS Economic Evaluation Database and EconLit. The Downs and Black scoring check-list was used to assess the quality of studies. Clinical effectiveness was measured in terms of pregnancy, live birth and miscarriage rates. A total of 19 articles meeting the inclusion criteria, comprising 3 RCTs in young and good prognosis patients and 16 observation studies were identified; 5 of the observational studies included a control group of patients where embryos were selected based on morphological criteria (matched cohort studies). Of the 5 studies that included a control group and reported implantation rates, 4 studies (including 2 RCTs) demonstrated improved implantation rates in the PGD-A group. Of the 8 studies that included a control group, 6 studies (including 2 RCTs) reported significantly higher pregnancy rates in the PGD-A group, and in the remaining 2 studies, equivalent pregnancies rates were reported despite fewer embryos being transferred in the PGD-A group. The 3 RCTs demonstrated benefit in young and good prognosis patients in terms of clinical pregnancy rates and the use of single embryo transfer. However, studies relating to patients of advanced maternal age, recurrent miscarriage and implantation failure were restricted to matched cohort studies, limiting the ability to draw meaningful conclusions. The authors concluded that given the uncertain role of PGD-A techniques, high-quality experimental studies using intention-to-treat analysis and cumulative live birth rates including the comparative outcomes from remaining cryopreserved embryos are needed to evaluate the overall role of PGD-A in the clinical setting. It is only in this way that the true contribution of PGD-A to ART can be understood.

There is some evidence that in women with high T-helper 1/T-helper 2 (Th1/Th2) ratios, there is an increased incidence of pregnancy loss and infertility. Thus, this test has been used by infertility specialists. However, there are no studies demonstrating the clinical utility of these measurements. A review by Ly et al (2010) stated: “Th1 dominance may well be a result of the miscarriage rather than a cause, and much more basic knowledge is needed about the complex cytokine networks in pregnancy and the correlation between cytokine production in peripheral mononuclear cells and decidual lymphocytes before tests measuring cytokines can be introduced in clinical practice”.

Ozkan et al (2014) noted that implantation necessitates complex interactions among the developing embryo, decidualizing endometrium, and developing maternal immune tolerance and/or alterations in cellular and humoral immune responses. Over-stimulation of Th1 or Th2 cytokines in systemic and local environments, alterations of the prevalence of interleukin-17 (IL-17) and regulatory T cell (Treg) cytokines have also been suggested to contribute to the pathogenesis of implantation failure. These researchers investigated the plasma levels of IL-4, IL-6, IL-10, tumor necrosis factor-alpha (TNFα), gamma interferon (IFNγ), transforming growth factor-beta (TGFβ), IL-17, IL-35, and suppressors of cytokine signaling 3 (SOCS3) in infertile and fertile women. This case-control study was conducted with 80 women suffering from unexplained infertility and 40 fertile women. Peripheral venous blood samples were drawn on day 21 of the menstrual cycle. The extracted plasma samples were assayed by an enzyme linked immunosorbent assay (ELISA). Statistical analysis was performed using SPSS version 16.0. The main findings were as follows: despite the significantly high IL-17 and IL-35 plasma levels of infertile women, IL-35/IL-17 ratio was significantly lower in the infertile group compared with that in the fertile group; SOCS3 plasma levels showed an inverse relation with plasma levels of all cytokines except IL-35; increased plasma IL-17 levels (greater than 3.42 pg/ml) have a negative impact on fertility; TNFα/IL-10, IFNγ/IL-10, IFNγ/IL-6, and IFNγ/IL-4 ratios were significantly higher in infertile group compared with those in the fertile group. The authors concluded that it is not possible to show the major immunological factor(s) of unexplained infertility, but these findings pointed out that the decreased suppressor activity of the immune system may play a role in implantation failure.

Furthermore, UpToDate reviews on “Overview of infertility” (Kuohung and Hornstein, 2015a) and “Evaluation of female infertility” (Kuohung and Hornstein, 2015b) do not mention the use of Th1/Th2 ratio or intracellular cytokine assay as a management tool.

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

Follitropins and Menotropins

For follitropins and menotropins, see CPB 1082 - Follitropins and Menotropins.

Human Chorionic Gonadotropin

For human chorionic gonadotropin (hCG), see CPB 1081 - Human Chorionic Gonadotropin (hCG).

Early Embryo Viability Assessment (Eeva) Test

In a prospective, multi-center, cohort study, Conaghan et al (2013) evaluated the first computer-automated platform for time-lapse image analysis and blastocyst prediction and determined how the screening information may assist embryologists in day 3 (D3) embryo selection. A total of 160 women aged 18 years or older undergoing fresh IVF treatment with basal antral follicle count greater than or equal to 8, basal FSH less than 10 IU/ml, and greater than or equal to 8 normally fertilized oocytes were included in this study. A non-invasive test combining time-lapse image analysis with the cell-tracking software, Eeva (Early Embryo Viability Assessment), was used to measure early embryo development and generate usable blastocyst predictions by D3. Main outcome measure was improvement in the ability of experienced embryologists to select which embryos are likely to develop to usable blastocysts using D3 morphology alone, compared with morphology plus Eeva. Experienced embryologists using Eeva in combination with D3 morphology significantly improved their ability to identify embryos that would reach the usable blastocyst stage (specificity for each of 3 embryologists using morphology versus morphology plus Eeva: 59.7 % versus 86.3 %, 41.9 % versus 84.0 %, 79.5 % versus 86.6 %). Adjunctive use of morphology plus Eeva improved embryo selection by enabling embryologists to better discriminate which embryos would be unlikely to develop to blastocyst and was particularly beneficial for improving selection among good-morphology embryos. Adjunctive use of morphology plus Eeva also reduced inter-individual variability in embryo selection. The authors concluded that previous studies have shown improved implantation rates for blastocyst transfer compared with cleavage-stage transfer; addition of Eeva to the current embryo grading process may improve the success rates of cleavage-stage ETs.

VerMilyea et al (2014) noted that computer-automated time-lapse analysis has been shown to improve embryo selection by providing quantitative and objective information to supplement traditional morphology. In a blinded, multi-center study, these researchers examined relationship between such computer-derived outputs (High, Medium, Low scores), embryo implantation and clinical pregnancy. Data were collected from 6 clinics, including 205 patients whose embryos were imaged by the Eeva(TM) System. The Eeva scores were blinded and not considered during embryo selection. Embryos with high and medium scores had significantly higher implantation rates than those with low scores (37 % and 35 % versus 15 %; $p < 0.0001$; $p = 0.0004$). Similar trends in implantation rates were observed in different IVF centers each using their own protocols. Further analysis revealed that patients with at least 1 high embryo transferred had significantly higher clinical pregnancy rates than those with only low embryos transferred (51 % versus 34 %; $p = 0.02$), although patients' clinical characteristics across groups were comparable. The authors concluded that these data, together with previous research and clinical studies, confirmed that computer-automated Eeva scores provided valuable information, which may improve the clinical outcome of IVF procedures and ultimately facilitate the trend of single embryo selection.

In summary, there is currently insufficient evidence to support the use of the EEVA test for improving embryo selection.

CAG-Repeat Polymorphisms in the Polymerase γ Gene (POLG) and Male Infertility

Zhang et al (2015) stated that CAG-repeat in the polymerase γ (POLG) gene encoding polymerase γ for mitochondria is important to spermatogenesis. Compared with a few researchers who raised alteration of CAG-repeat-affected male reproductive ability, others did not find the association between CAG-repeat polymorphisms and male infertility. These researchers performed a comprehensive meta-analysis to determine the association; 13 case-control studies were screened out using keywords search. From these

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

studies, characteristics were extracted for conducting meta-analysis. Odds ratio (OR) and 95 % CI were used to describe the results; the results indicated that CAG-repeat allele was not a risk factor to male infertility (pooled OR = 1.03, 95 % CI: 0.79 to 1.34, p = 0.828). Four different genetic comparisons also demonstrated a negative result: heterozygote comparison (not 10/10 versus 10/10; pooled OR = 0.99, 95 % CI: 0.77 to 1.27, p = 0.948), homozygote comparison (not 10/not 10 versus 10/10; pooled OR = 1.08, 95 % CI: 0.56 to 2.06, p = 0.816), the recessive genetic comparison (not 10/not 10 versus not 10/10 + 10/10; pooled OR = 1.07, 95 % CI: 0.58 to 1.95, p = 0.829) and the dominant genetic comparison (not 10/not 10 + not 10/10 versus 10/10; pooled OR = 0.97, 95 % CI: 0.72 to 1.29, p = 0.804). The authors concluded that based on current researches, this meta-analysis demonstrated no apparent association between POLG-CAG-repeat and male infertility. Similarly, CAG-repeat was not a sensitive site to male infertility.

DuoStim IVF Protocol

DuoStim IVF cycles are IVF cycles for women with poor response to prior IVF cycles or diminished ovarian reserve. DuoStim has two steps. Step one includes stimulation, aspiration, fertilization and freeze of the embryos. Step two occurs after aspiration in step one and involves stimulation of any follicles that were too small to aspirate in step one. This second crop of follicles go to aspiration and fertilization.

Massin (2017) noted that the advent of embryo and oocyte vitrification today gives reproductive specialists an opportunity to consider new strategies for improving the practice and results of in-vitro fertilization (IVF) attempts. As the freezing of entire cohorts does not compromise, and may even improve, the results of IVF attempts, it is possible to break away from the standard sequence of stimulation-retrieval-transfer. The constraints associated with ovarian stimulation in relation to the potential harmful effects of the hormonal environment on endometrial receptivity can be avoided. This review examined the new stimulation protocols where progesterone is used to block the LH surge. Thanks to 'freeze all' strategies, the increase in progesterone could actually be no longer a cause for concern. There are 2 ways of using progesterone, whether it be endogenous, as in luteal phase stimulation, or exogenous, as in the use of progesterone in the follicular phase i.e., progestin-primed ovarian stimulation. These investigators performed a literature search (until September 2016) on Medline. The following text words were utilized to generate the list of citations: progestin primed ovarian stimulation, luteal phase stimulation, luteal stimulation, DuoStim, double stimulation, random start. Articles and their references were then examined in order to identify other potential studies. All of the articles were reported in this review. The use of progesterone during ovarian stimulation is effective in blocking the LH surge, whether endogenous or exogenous, and it does not affect the number of oocytes collected or the quality of the embryos obtained. Its main constraint is that it requires total freezing and delayed transfer. A variety of stimulation protocols can be derived from these 2 methods, and their implications were discussed, from fertility preservation to ovarian response profiles to organization for the patients and clinics. These new regimens enable more flexibility and are of emerging interest in daily practice. However, their medical and economic significance remains to be demonstrated. The authors concluded that the use of luteal phase or follicular phase protocols with progestins could rapidly develop in the context of oocyte donation and fertility preservation not related to oncology. Their place could develop even more in the general population of patients in IVF programs. The strategy of total freezing continues to develop, thanks to technical improvements, in particular vitrification and PGS on blastocysts, and thanks to studies showing improvements in embryo implantation when the transfer take place far removed from the hormonal changes caused by ovarian stimulation.

Vaiarelli et al (2018) stated that the management and treatment of patients with poor ovarian response is still a controversial issue in IVF. Increasing evidences demonstrate that the number of oocytes retrieved after a controlled ovarian stimulation (COS) greatly influences the clinical outcome in terms of cumulative live-birth per started cycle. For this reason, any COS should aim to optimize the number of oocytes according to the ovarian reserve of the patient. These investigators provided an overview of new strategies proposed to manage poor responders according to the novel POSEIDON classification. Gonadotrophins cannot compensate for the absence of follicles in the ovary, therefore, COS in poor responders may benefit from the exploitation of multiple follicular waves within a single ovarian cycle, for instance, through luteal phase stimulation or double stimulation (follicular plus luteal) in the same ovarian cycle (DuoStim) protocols. The authors concluded that many strategies have been proposed to manage poor responder patients, however, a

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

consensus upon which is the most beneficial has not been yet reached; and DuoStim is the most promising approach to increase the number of oocytes collected in a single ovarian cycle; however, more embryological and clinical data is needed, as well as an analysis of its cost-effectiveness.

Cimadomo et al (2018) noted that 3 theories of follicle recruitment have been postulated to date: the "continuous recruitment" theory, the "single recruitment episode" theory, and the "wave" theory. Yet, a clear characterization of this crucial biological process for human reproduction is missing. Recent advances implemented in IVF, such as blastocyst culture, aneuploidy testing and vitrification, have encouraged clinicians to maximize the exploitation of the ovarian reserve through tailored stimulation protocols, which is crucial especially for poor prognosis patients aiming to conceive after IVF. Luteal phase stimulations (LPS) has been already successfully adopted to treat poor prognosis or oncological patients through DuoStim, LPS-only or random-start ovarian stimulation approaches. Nevertheless, little, and mainly retrospective, evidence has been produced to support the safety of LPS in general. Feasibility of the LPS approach would severely question the classic "single recruitment episode" theory of follicular development. In a case-control study, these researchers determined if the mean numbers of blastocysts obtained from sibling cohorts of oocytes recruited after follicular phase stimulation (FPS) and LPS in the same ovarian cycle are similar. This trial was carried out with paired follicular phase- and luteal phase-derived cohorts of oocytes collected after stimulations in the same ovarian cycle (DuoStim) at 2 private IVF clinics between October 2015 and December 2017. This study included 188 poor prognosis patients undergoing DuoStim with pre-implantation genetic testing for aneuploidies (PGT-A); FPS and LPS were performed with the same daily dose of recombinant-gonadotrophins in an antagonist protocol. Blastocyst culture, trophectoderm biopsy, vitrification and frozen-warmed euploid single blastocyst transfers were performed. The primary outcome was the mean number of blastocysts obtained per oocyte retrieval from paired-FPS- and LPS-derived cohorts (required sample size = 165 patients; power = 90 %). Mean blastulation and euploidy rates were monitored, along with the number of oocytes, euploid blastocysts and clinical outcomes. Significantly fewer blastocysts were obtained after FPS than LPS (1.2 ± 1.1 versus 1.6 ± 1.6, $p < 0.01$), due to fewer oocytes collected (3.6 ± 2.1 versus 4.3 ± 2.8, $p < 0.01$) and a similar mean blastocyst rates per retrieval (33.1 % ± 30.3 % versus 37.4 % ± 30.8 %, p = NS). The number of oocytes collected were correlated ($R = 0.5$, $p < 0.01$), while the blastocyst rates were uncorrelated among paired-FPS- and LPS-derived cohorts. Overall, a significantly lower chance of producing blastocyst(s) was reported after FPS than after LPS: 67.6 % (n = 127/188, 95 % confidence interval [CI]: 60.3 to 74.1) versus 77.1 % (n = 145/188, 95 % CI: 70.3 to 82.8; $p = 0.05$). The mean euploidy rates per retrieval were similar between FPS- and LPS-derived cohorts of oocytes (13.6 % ± 22.8 % versus 16.3 % ± 23.4 %, p = NS). Thus, on average fewer euploid blastocysts (0.5 ± 0.8 versus 0.7 ± 1.0, $p = 0.02$) resulted from FPS. Similar ongoing-pregnancy/delivery rates were reported, to-date, after FPS- and LPS-derived euploid single blastocyst transfers: 42.4 % (n = 28/66, 95 % CI: 30.5 to 55.2) versus 53.8 % (n = 35/65, 95 % CI: 41.1 to 66.1; p = NS). The authors concluded that this study provided evidence that the follicles recruited during the anovulatory phase of the ovarian cycle may be rescued through LPS. Of note, LPS-derived cohorts of oocytes were also larger than paired-FPS-derived cohorts and the oocytes showed comparable competence. These data supported the putative benefits of LPS in poor prognosis and oncological patients. Furthermore, they encourage additional clinical and basic research studies on this topic, which may revolutionize the basics of human folliculogenesis, as well as the future concept of approaches to ovarian stimulation in IVF.

The authors stated that more studies need to be conducted in the future to confirm the safety of LPS, especially in terms of ovarian and follicular environment, as well as the clinical, peri-natal and post-natal outcomes. The findings of this study showed preliminary data suggesting a similar ongoing implantation/delivery rate (greater than 22 weeks) between FPS- and LPS-derived euploid blastocysts, that need to be extended in the future, to populations other than poor prognosis patients and using approaches other than DuoStim together with a constant monitoring of the related peri-natal and post-natal outcomes.

Vaiarelli et al (2018) stated that a panel of experts known as the POSEIDON group has recently re-defined the spectrum of poor responder patients and introduced the concept of sub-optimal response. Since an ideal management for these patients is still missing, they highlighted the importance of tailoring the ovarian stimulation based on the chance of each woman to obtain an euploid blastocyst. Interestingly, a novel pattern of follicle recruitment has been defined: multiple waves may arise during a single ovarian cycle. This evidence opened important clinical implications for the treatment of poor responders. For instance, double

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

stimulation in the follicular (FPS) and luteal phase (LPS) of the same ovarian cycle (DuoStim) is an intriguing option to perform 2 oocyte retrievals in the shortest possible time. These investigators reported their 2-year experience of DuoStim application in 4 private IVF centers. To-date, 310 poor prognosis patients completed a DuoStim protocol and underwent IVF with blastocyst-stage pre-implantation-genetic-testing. LPS resulted into a higher mean number of oocytes collected than FPS; however, their competence (i.e., fertilization, blastocyst, euploidy rates, and clinical outcomes after euploid single-embryo-transfer) was comparable. Importantly, the rate of patients obtaining at least 1 euploid blastocyst increased from 42.3 % (n = 131/310) after FPS to 65.5 % (n = 203/310) with the contribution of LPS. A summary of the putative advantages and disadvantages of DuoStim was reported here through a Strengths-Weaknesses-Opportunities-Threats analysis. The strengths of this approach made it very promising. Moreover, the authors concluded that DuoStim still needs a more extensive and wider validation to testify its safety. Interesting future perspectives to investigate its clinical efficacy/efficiency would entail

I. a RCT comparing double-FPS versus DuoStim;
II. the application of DuoStim in cancer patients for fertility preservation;
III. as well as in prospective analyses focused on patients clustered according to either the Bologna criteria or the Poseidon stratification.

I. a RCT comparing double-FPS versus DuoStim;
II. the application of DuoStim in cancer patients for fertility preservation;
III. as well as in prospective analyses focused on patients clustered according to either the Bologna criteria or the Poseidon stratification.

They stated that until such evidence would be produced, DuoStim should be clinically applied only to a population of patients of poor prognosis and/or to whom time represents a critical issue.

The authors stated that the weaknesses of DuoStim are: a higher number of stimulations appeared to be canceled in the LP than in the FP; no RCT or cost-effectiveness analysis has been performed to-date investigating the use of DuoStim; a freeze-all approach is mandatory; it has been applied only to poor prognosis patients. The opportunities are: a decrease in the time and increase in the chance to obtain at least 1 competent embryo in a single menstrual cycle; the DuoStim protocol might be better-tolerated from the patients than consecutive FPS cycles; the drop-out rate might be reduced; the knowledge regarding the mechanisms of follicular recruitment and ovarian physiology might be increased. The threats are: an analysis of the cost-effectiveness is yet eagerly needed; the total dose of gonadotrophins to be administrated is substantial; few biological, gynecological, obstetrical, and neonatal evidence of safety have been produced to date. The strengths of this approach make it very promising. However, more studies are needed in the future to limit its weaknesses, shed light on its putative threats, and realize its opportunities.

There is a clinical trial on “DuoStim in Cases of PGT: Comparison of Embryo Quantity and Embryonic Quality Using MitoScore” that is currently recruiting participants (last updated September 3, 2018).

Furthermore, an UpToDate review on “In vitro fertilization” (Paulson, 2019) does not mention DuoStim / double stimulation as a management tool.

Embryo Transfers

To reduce multiple gestations and promote singleton pregnancies, the 2021 American Society for Reproductive Medicine and Society for Assisted Reproductive Technology (ASRM/SART) guidance provides age- and prognosis-based recommendations for embryo transfer in assisted reproductive technology (ART) cycles. Patients with a favorable prognosis (defined by factors such as young age, availability of high-quality or euploid embryos, prior live birth, and vitrified blastocysts) should receive fewer embryos:

- Transfer of a euploid embryo should be limited to one, regardless of patient age;
- Patients less than 35 years of age should be strongly encouraged to receive a single-embryo transfer, regardless of the embryo stage;

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

- Patients between 35 to 37 years of age, strong consideration should be made for a single-embryo transfer;
- Patients between 38 to 40 years of age, no more than 3 untested cleavage-stage embryos or 2 blastocysts should be transferred;
- Patients between 41 to 42 years of age should plan to receive no more than 4 untested cleavage-stage embryos or 3 blastocysts.

The ASRM/SART (2021) committee outlines other scenarios where deviation from standard embryo transfer limits may be considered: (i) patients in any age group with an unfavorable prognosis may receive an additional embryo, with counseling on the increased risk of multiple gestation; (ii) favorable prognosis patients who fail to conceive after multiple cycles with high-quality embryos may consider transferring an additional embryo; (iii) patients with medical conditions that could be worsened by multiple pregnancy should be limited to single embryo transfer; and (iv) in rare cases where embryo transfer exceeds recommended limits, both counseling and justification must be documented in the medical record. In women 43 years of age or older, there are insufficient data to recommend a limit on the number of embryos to transfer when the patient uses her own oocytes. Caution should be exercised as the risk associated with multiple pregnancy increases dramatically with advancing maternal age.

In donor-oocyte cycles, the donor's age determines embryo transfer limits. If the donor is less than 38 years and other favorable criteria are met, single embryo transfer should be planned. In gestational carrier (GC) cycles, single embryo transfer is strongly recommended due to the health risks of multiple gestations. For frozen embryo transfer (FET) cycles, embryo transfer decisions should be on the basis of the age of the woman when the embryos were cryopreserved and include the presence of high-quality vitrified embryos, euploid embryos, first FET cycle, or previous live birth after a prior transfer with sibling embryo(s). Embryo transfer numbers should not exceed the recommended limit on the number of fresh embryos transferred for each age group.

Single-embryo transfer is strongly recommended in all gestational carrier (GC) cycles due to the significant health risks associated with multiple gestations for the carrier; at a minimum, embryo transfer should follow age-based limits determined by the age of the oocyte source (either the intended parent or donor). In frozen embryo transfer (FET) cycles, favorable prognostic factors include the age of the woman at the time of embryo cryopreservation, presence of high-quality vitrified or euploid embryos, undergoing a first FET cycle, or having had a prior live birth from sibling embryos. Regardless of these factors, the number of embryos transferred in FET cycles should not exceed the age-specific limits established for fresh embryo transfers.

Endometrial Receptivity Analysis (Igenomix)

The Endometrial Receptivity Analysis (ERA; Igenomix) is a test designed to evaluate endometrial receptivity. It is the 1st diagnostic test that determines each woman's unique personalized ET timing; thus, synchronizing the ET with the individualized window of implantation. The ERA uses RNA obtained from an endometrial tissue sample. The RNA analyzed and then classified by Igenomix's ERA predictor as receptive or non-receptive, depending on the expression profile of the RNA. The ERA test couples next-generation sequencing (NGS) to a computational predictor to identify transcriptomic signatures for each endometrial stage: proliferative (PRO), pre-receptive (PRE), receptive (R) and post-receptive (POST). If the endometrium is non-receptive, the test enables clinicians to find a personalized window of implantation for each patient in the majority of cases.

In an open-label, multi-center RCT, Simon and colleagues (2020) examined if clinical performance of personalized ET (PET) guided by ERA would differ from frozen ET (FET) or fresh ET in infertile patients undergoing IVF. This trial included 458 patients aged 37 years or younger undergoing IVF with blastocyst transfer at 1st appointment; they were randomized to PET guided by ERA, FET or fresh ET in 16 reproductive clinics. Clinical outcomes by intention-to-treat (ITT) analysis were comparable, but cumulative pregnancy rate (CPR) was significantly higher in the PET (93.6 %) compared with FET (79.7 %) ($p = 0.0005$) and fresh ET groups (80.7 %) ($p = 0.0013$). Analysis per protocol demonstrated that livebirth rates (LBRs) at 1st ET were 56.2 % in PET versus 42.4 % in FET ($p = 0.09$), and 45.7 % in fresh ET groups (p =

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

0.17). Cumulative LBRs (CLBRs) after 12 months were 71.2 % in PET versus 55.4 % in FET (p = 0.04), and 48.9 % in fresh ET (p = 0.003). Pregnancy rates at the 1st ET in PET, FET and fresh ET arms were 72.5 % versus 54.3 % (p = 0.01) and 58.5 % (p = 0.05), respectively. Implantation rates at 1st ET were 57.3 % versus 43.2 % (p = 0.03), and 38.6 % (p = 0.004), respectively. Obstetrical outcomes, type of delivery and neonatal outcomes were similar in all groups. The authors concluded that, to the best of their knowledge, this was the 1st RCT aiming to provide proof-of-principle evidence for the potential of using a personalized diagnosis of the endometrial factor in the work-up of the infertile couple at the 1st appointment. While the ITT analysis showed no beneficial effect of the ERA test except for a statistically significant CPR compared with FET and fresh ET, the per protocol analysis demonstrated a significant improvement in pregnancy rates at the 1st and cumulative rates up to 12 months, and implantation rates at the 1st attempt, indicating the potential of the ERA test to diagnose the endometrial factor in the work-up of the infertile couple. Moreover, these researchers stated that these findings need to be confirmed in a larger randomized clinical trial.

The authors stated that the main drawback of this study was the unexpected 50 % patient drop-out rate versus 30 % initially planned. This situation has rendered the study under-powered to detect statistical significance by ITT analysis in the PET arm versus FET and fresh ET, except for higher CLBR. This sample size affected the per protocol analysis; although the analysis detected a 13.8 and 10.5-percentage points increase in LBR in the PET group versus FET and fresh ET at the 1st ET, the difference was not significant because the study was powered to detect statistical differences for a 15-percentage point increase in the primary and secondary outcomes. Second, in the historical setting in which this RCT was designed, ERA was performed using microarray technology with early algorithms, and in some cases 2 endometrial biopsies were needed for diagnosis. At the time of publication, ERA is being carried out by NGS combined with refined algorithms informed by the analysis and clinical follow-up of more than 50,000 endometrial samples worldwide; additionally, only 1 endometrial biopsy is needed. Because of the indicated constraints of the study, these researchers are performing a new ERA 2.0 RCT, in which the current sequencing technology and refined algorithms will be included together with proper power of the study.

In a prospective, cohort study, Riestenberg and associates (2021) compared the LBR between patients who undergo personalized ET (pET) after ERA versus FET with standard timing in 1st single euploid FET cycles. These researchers also reported the rate of displacement of the window of implantation (WOI) in an infertile population without a history of implantation failure. Subjects were patients who underwent their 1st single euploid programmed FET; they underwent 1st autologous single euploid programmed FET between January 2018 and April 2019. Subjects underwent endometrial biopsy with ERA followed by pET as indicated. Main outcome measures included LBR as well as rate of receptive and non-receptive ERA. A total of 228 single euploid FET cycles were included in this analysis. Of those, 147 (64.5 %) were ERA/pET cycles, and 81 (35.5 %) were standard timing FET cycles. Endometrial receptivity array was receptive in 60/147 (40.8 %) and non-receptive in 87/147 (59.2 %) patients. Non-receptive ERAs were pre-receptive in 93.1 % of cases. The LBR did not differ between patients who underwent FET with standard timing and patients who underwent ERA/pET, 45/81 (56.6 %) and 83/147 (56.5 %), respectively. The authors concluded that the findings of this study did not support the routine use of ERA in an unselected patient population undergoing 1st autologous single euploid programmed ET.

In a retrospective, cohort study, Bergin and co-workers (2021) examined the impact of the ERA on LBRs in FET cycles. Autologous FET cycles between January 1, 2014, and June 30, 2019 were reviewed. Multiple co-variates that impacted outcomes were used for propensity score matching; 133 ERA patients were matched to 353 non-ERA patients. Patients were assigned to the ERA group if they had an ERA during treatment and underwent at least 1"personalized" FET (pFET) based on the ERA recommendations. Main outcome measures included LBRs per cycle in the FET cycle after ERA compared with that of matched non-ERA patients. The LBRs for the ERA group, 49.62 %, and the matched non-ERA group, 54.96 %, (OR 0.8074; 95 % CI: 0.5424 to 1.2018) were not significantly different, nor was a difference observed in sub-analyses based on prior number of FETs or receptivity status. The authors concluded that ERA identified a patient's putative WOI with the objective of improving synchrony with the embryo; thus, attaining higher LBRs. This study used propensity score matching to control for multiple co-variates in a heterogenous group of patients to compare LBRs. There was no difference in the LBRs in patients who underwent ERA compared with that of those who did not.

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

In a retrospective study, Eisman and associates (2021) examined the use of the ERA in women with prior failed ET. This trial included patients who underwent an ERA test with a subsequent FET. Women were classified based on their indication for an ERA test: greater than or equal to 1 prior failed ET (cases), or as a prophylactic measure (controls). A subset analysis of women with greater than or equal to 3 prior failed ETs was carried out. Pregnancy outcomes of the subsequent cycle were examined, including conception, clinical pregnancy, and ongoing pregnancy/livebirth. A total of 222 women were included, 131 (59 %) women with greater than or equal to 1 prior failed ET and 91 (41 %) controls. Among the 131 women with greater than or equal to 1 prior failed ET, 20 women (9 %) had greater than or equal to 3 prior failed ETs. The proportion of non-receptive ERA tests in the 3 groups were the following: 45 % (greater than or equal to 1 prior failed ET), 40 % (greater than or equal to 3 prior failed ETs), and 52 % (controls). The results did not differ between cases and controls. The pregnancy outcomes did not differ between women with greater than or equal to 1 prior failed ET and controls. In women with greater than or equal to 3 prior failed ETs, there was a lower ongoing pregnancy/LBR (28 % versus 54 %, p = 0.046). The authors concluded that women with greater than or equal to 1 prior failed ET and greater than or equal to 3 prior failed ETs had a similar prevalence of non-receptive endometrium compared to controls. Women with greater than or equal to 3 prior failed ETs had a lower ongoing pregnancy/LBR despite a personalized FET, suggesting that there are additional factors in implantation failure beyond an adjustment in progesterone exposure.

Rafeal (2021) discussed the limitations of the ERA methodology to increase implantation. Such limitations varied from the assumed inconsistency of the endometrial biopsy, the variable number of genes found to be dysregulated in endometrium samples without the embryonal-induced effect, the failure to account for the simultaneous serum progesterone level, and the expected low percentage of patients who may need this add-on procedure, to the difficulties in synchronizing the endometrium with hormone replacements in successive cycles and the inherent peri-natal risks associated with routine cryo-preservation of embryos. Without a gold standard to compare, the claim that the WOI might be off by ± 12 hours only requires a good argument for the advantage it provides to human procreation, knowing that embryos can linger for days before actual embedding starts and that the window is actually a few days. Moreover, the intra-patient variations in the test need to be addressed. The author stated that clinicians tend to grasp any new idea that could potentially improve results, also to demonstrate that they are competitive and updated; however, even after many years of practice, most add-ons have been deemed not proven. New tests, such as the ERA, until shaped and proven, should be offered only under research protocols that separate compounding factors, considering all the reservations, and keeping in mind that freezing embryos is not risk free. As witnessed before, a single RCT or even a meta-analysis should not be accepted as a final proof towards its overall utility as a new solution or its wide-spread use. It is the duty of the clinical societies and peer-reviewed journals to follow-up on the evidence, and filter new ideas and technological procedures, to avert physicians from repeating false directions or mistakes for longer than necessary. The author concluded that like all other add-ons, it is doubtful that the ERA test use would significantly enhance implantation success rates.

Evaluation of CYP1A1 rs4646903 T > C Genetic Variations for Risk of Male Infertility

Cao and colleagues (2019) stated that number of studies have been carried out to examine the relationship between the CYP1A1 rs4646903 polymorphism and male infertility risk, but the sample size was small and the results were conflicting. These researchers conducted a meta-analysis to examine these associations. They performed a systematic search to identify all relevant studies from Medline, Web of science, Embase, China biology medical literature database (CBM), China National Knowledge Infrastructure (CNKI), WanFang and Weipu (VIP) databases up to June 30, 2018. The ORs with 95 % CIs were calculated to assess the strength of associations. All of the statistical analyses were conducted using Revman 5.3 and Stata 14.0. A total of 10 studies entailing 3,028 cases and 3,258 controls were included in this analysis. Overall, significant association was observed between the CYP1A1 rs4646903 polymorphism and male infertility (C versus T: OR=1.42, 95 % CI: 1.14 to 1.76; CC versus TT: OR =2.13, 95 % CI:1.36 to 3.34; CC versus CT+TT: OR=1.96, 95 % CI: 1.30 to 2.95; CC+CT versus TT: OR=1.51, 95 % CI:1.16 to 1.97). In subgroup analysis by ethnic group, a statistically significant association was observed in Asians (C versus T: OR=1.59, 95 % CI:1.22 to 2.08), but not in non-Asians (C versus T: OR=1.01, 95 % CI: 0.79 to 1.30). Furthermore, none of the individual studies significantly affected the association between CYP1A1 rs4646903 polymorphism and male infertility, according to sensitivity analysis. The authors concluded that the findings

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

of this meta-analysis supported that the CYP1A1 rs4646903 polymorphism might contribute to individual susceptibility to male infertility in Asians. Moreover, these researchers stated that large sample size, well-designed, and population-based studies are needed to validate the association between CYP1A1 gene variant rs4646903 polymorphism and male infertility risk.

The authors stated that this study had several drawbacks. First, only 10 studies were incorporated in the meta-analysis, the sample size of included published articles was small. Second, sub-group analyses such as by infertility type and source of control group were not performed, due to the lack of information. Third, the effects of gene-gene and gene-environment interactions on male infertility susceptibility were not estimated, as the studies enrolled lacked of information. Finally, these findings were based on unadjusted estimates, due to the lack of data of smoking, age, and other environmental exposure factors.

Evaluation of FAS/FASL Genetic Variations for Risk of Male Infertility

Asgari and colleagues (2019) noted that studies suggested that FAS/FASL polymorphisms are associated with male infertility; however, their results are still inconclusive. In a systematic review and meta-analysis, these researchers examined the overall association of FAS/FASL polymorphisms and risk of male infertility. They carried out a search on the databases of Science Direct, PubMed and Google Scholar. For performing the meta-analysis, pooled OR values with 95 % CI were applied in order to analyze the strength of association between the FAS/FASL polymorphisms and risk of male infertility. A total of 7 relevant studies published up to September 2018 were considered. FASL-844C/T genotype results of 559 patients and 623 healthy individuals were included in this study. For FAS-670A/G genotype effect, 751 patients and 821 healthy individuals were examined. Results showed that all analysis models including dominant, recessive and allelic models of FASL-844C/T and FAS-670A/G polymorphism had no significant effect on infertility in men ($p > 0.05$ and $p > 0.05$, respectively). According to sensitivity analysis, these findings were stable. The authors demonstrated that FAS/FASL polymorphisms might not be an effective factor on male reproductive health. These researchers stated that for precise determination of FAS/FASL polymorphisms effects on male infertility, large-scale, case-control studies are needed.

Evaluation of Telomere Length for Female and Male Infertility

Vasilopoulos and colleagues (2019) noted that telomere length (TL) has long been associated with aging, as telomeres serve as protective caps of chromosomes, and are thus deeply involved in the preservation of genome integrity and are vital to cellular functions. Traditionally, a strong link connects aging and infertility in both sexes, with an earlier onset in females. Over the last 10 years, telomeres have attracted increasing attention due to the role they play in fertility. In this review, these investigators examined the potential positive or negative association between relative TL and different factors of female and male infertility. They carried out a systematic search of the PubMed database. A total of 206 studies were identified; 45 met the criteria of validity and relevance. Following an analysis and a comparison of the study outcomes, several clear trends were observed. The majority of female infertility factors were associated with a shorter TL, with the exception of endometriosis, premature ovarian failure and clear cell carcinoma that were associated with a longer TL and PCOS, which revealed conflicting results among several studies, leading to ambiguous conclusions. Male infertility factors were associated with a shorter TL. The authors concluded that although the findings of this review could provide an outline of general trends in the association of TL with infertility factors, further epidemiological and original research studies are needed to examine the basis of these varying lengths of telomeres. Moreover, these researchers stated that many questions must first be addressed before the use of TL as a marker for identification of reproductive capacity can be employed in a clinical setting. They stated that further studies are needed to understand the bases of TL associations with biological aging and reproductive capacity, and to determine how this knowledge can be used in medical applications.

Evaluation of Vaginal Microbiota

In a systematic review and meta-analysis, Hong and colleagues (2020) examined the association between vaginal microbiota and infertility. These investigators searched a range of electronic databases for appropriate articles, including PubMed, Web of Science, Embase, Chinese National Knowledge

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

Infrastructure (CNKI), and Wanfang, from inception to September 8, 2019. Identified articles were then screened using strict inclusion and exclusion criteria. By referring to Tamarelle's method, these researchers divided vaginal microbiota into 2 categories: low-Lactobacillus vaginal microbiota (LL-VMB) and high-Lactobacillus vaginal microbiota (HL-VMB). Patients were defined as HL-VMB if they had a Nugent score of 0 to 3, a negative Amesel/Spiegel's test, or if the vaginal community status was dominated by either L. crispatus, L. iners, L. gasseri and L. jensenii via 16S rRNA sequencing. Otherwise, cases were regarded as LL-VMB. Statistical analyses were carried out with STATA 13.0 statistical software. Effect estimates are presented as ORs with 95 % CIs. A total of 15 articles were included in the final analysis. The HL-VMB was negatively related to infertility; a fixed model showed that the pooled OR was 0.83 (95 % CI: 0.77 to 0.90). There was no significant publication bias, as determined by Begg's test (p = 0.488) and Egger's test (p = 0.652). Using a random effect model, the pooled OR for intermediate bacterial vaginitis (BV) and infertility was 1.39 (95 % CI: 1.10 to 1.76) and the pooled OR for positive BV was 1.72 (95 % CI: 1.10 to 2.69). Subgroup and sensitivity analyses further demonstrated that the associations identified were stable; however, the acquired evidence was insufficient to make inferences with regards to the mechanisms underlying these relationships. The authors concluded that this systematic review and meta-analysis identified a negative correlation between HL-VMB and female infertility. However, due to a variety of limitations, the evidence acquired did not allow these researchers to identify the specific mechanisms underlying this association. They stated that further high-quality studies are needed to verify the causal relationship and examine the molecular mechanisms involved.

Furthermore, an UpToDate review on “Evaluation of female infertility” (Kuohung and Hornstein, 2021) does not mention evaluation of vaginal microbiota as a management option.

Genetic Testing of CFTR Mutations for a Man with Congenital Absence of the Vas Deferens

Yu and associates (2012) stated that numerous studies have reported cystic fibrosis transmembrane conductance regulator (CFTR) mutations in congenital bilateral absence of the vas deferens (CBAVD) patients, but their results are not completely consistent. These investigators performed a systemic review and meta-analysis with emphasis on clarifying further the genetic association of CFTR mutations with CBAVD. They searched the Medline database until March, 2011 for eligible articles reporting CFTR mutations in CBAVD. Relevant data from each included study were abstracted by 2 independent reviewers. The overall frequency of CFTR mutations in CBAVD and OR for common specific alleles were pooled under random-effect or fixed-effect model as appropriate. Subgroup analysis was performed by ethnicity, and potential heterogeneity and bias were both assessed. Among CBAVD patients, 78 % had at least 1 CFTR mutation, 46 % having 2 and 28 % only 1. Moreover, the common heterozygous F508del/5T and F508del/R117H were observed in 17 % and 4 % of CBAVD cases, respectively, and the allele frequency in CBAVD was 17 % for F508del, 25 % for 5T and 3 % for R117H. Subgroup analysis indicated an increased frequency of cases with 2 mutations in Caucasian patients than in non-Caucasian (68 % versus 50 %, p = 0.012), but no differences for cases with at least 1 mutation (88 % versus 77 %, p = 0.163) or with only 1 mutation (17 % versus 25 %, p = 0.115). Caucasian patients had higher F508del frequency, but lower 5T frequency, than non-Caucasian (22 % versus 8 %, p = 0.001; 20 % versus 31 %, p = 0.009). Summary OR was 9.25 for 5T [95 % CI: 7.07 to 12.11, p = 0.000], with moderate heterogeneity (I(2)= 49.20 %, p= 0.019) and evident bias (Egger's test, p = 0.005), and it was 19.43 for 5T/(TG)12_13 (95 % CI: 10.48 to 30.03, p = 0.000) without any evidence of heterogeneity (I(2)= 0.1 %, p = 0.391) and bias (Egger's test, p = 0.160). The OR for 5T/(TG)12_13 was significantly higher than that for 5T allele (p = 0.000). The authors concluded that these findings demonstrated a high frequency of CFTR mutations in CBAVD patients, and these exhibited evident ethnic differences. In addition, 5T allele and 5T/(TG)12_13 may contribute to the increased risk for CBAVD, with the 5T penetrance probably being modulated by adjacent (TG)12_13.

The European Association of Urology’s guidelines on “Male infertility” (Jungwirth et al, 2012) stated that “Men with congenital bilateral absence of the vas deferens [CBAVD] often have mild clinical stigmata of cystic fibrosis (CF) (e.g., history of chest infections). Children born after intracytoplasmic sperm injection (ICSI), where the father has CBAVD and is either heterozygous or homozygous, must be followed-up. When a man has CBAVD, it is important to test him and his partner for CF mutations. If the female partner is

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

found to be a carrier of cystic fibrosis transmembrane conductance regulator (CFTR), the couple must consider very carefully whether to proceed with ICSI using the husband's sperm, as the risk of having a baby with CF will be 25 % if the man is heterozygous and 50 % if the man is homozygous. If the female partner is negative for known mutations, the risk of being a carrier of unknown mutations is about 0.4 %".

Sharma and co-workers (2014) noted that CF is usually considered a rare disease in the Indian population; 2 studies have reported on the frequency of CFTR gene mutations in Indian males with CAVD. However, data on the spectrum of CFTR gene mutations are still lacking. These researchers identified the spectrum of CFTR gene mutations and investigated an association of CF genetic modifiers in the penetrance of CAVD in infertile Indian men. A total of 60 consecutive infertile males with a diagnosis of CAVD were subjected to CFTR gene analysis that revealed 13 different CFTR gene mutations and 1 intronic variant that led to aberrant splicing. p.Phe508del (n = 16) and p.Arg117His (n = 4) were among the most common severe forms of CFTR mutations identified. The IVS8-T5 allele, which is considered as a mild form of CFTR mutation, was found with an allelic frequency of 28.3 %; 8 novel mutations were also identified in the CFTR gene from this patient cohort. It was noteworthy that the spectrum of CFTR gene mutation was heterogeneous, with exon 4 and exon 11 as hot spot regions. Moreover, these investigators also found an association of the CF genetic modifiers, viz., transforming growth factor (TGF)-β1 and endothelial receptor type-A (EDNRA) genes with the CAVD phenotype. The findings were of considerable clinical significance because men suffering from infertility due to CAVD can decide to use artificial reproduction technology. The children of men with CAVD are at risk of carrying CFTR mutations; therefore, genetic counseling is a crucial step for such patients. With special reference to developing countries, such as India, where whole gene sequencing is not feasible, the outcome of this study will make the screening procedure for CFTR gene simpler and more cost-effective as these researchers have identified hot spot regions of the CFTR gene that are more prone to mutation in Indian males with CAVD. Moreover, this was the first study from the Indian population to investigate the association of CF genetic modifiers with penetrance of the CAVD phenotype. They stated that the observed association of the genetic modifiers TGF-β1 and EDNRA in the penetrance of CAVD further supports their involvement in genesis of the vas deferens.

Yang and colleagues (2015) discussed the findings and significance of the detection of the CFTR gene mutation in azoospermia patients with congenital unilateral absence of the vas deferens (CUAVD). These researchers collected peripheral blood samples from 6 azoospermia patients with CUAVD for detection of the CFTR gene mutations and single nucleotide polymorphisms (SNPs). They analyzed the genome sequences of the CFTR gene in comparison with the website of the UCSC Genome Browser on Human December 2013 Assembly. Missense mutation of c. 592G > C in exon 6 was found in 1 of the 6 azoospermia patients with CUAVD and splicing mutation of c. 1210-12T[5] was observed in the non-coding region before exon 10 in 2 of the patients, both with the V470 haplotype in exon 11. The authors concluded that mutations of the CFTR gene can be detected in azoospermia patients with CUAVD and the detection of the CFTR gene mutation is necessary for these patients.

Furthermore, an UpToDate review on "Evaluation of male infertility" (Swerdloff and Wang, 2017) states that "Absence of the vas deferens on physical examination, together with low seminal fluid volume and acidic pH, suggest congenital absence of vas deferens. Low or absent semen fructose will help to confirm the diagnosis of this disease, because the seminal vesicles are usually also absent. These patients should be tested for the CFTR gene mutations and, if positive in either the man or the female partner, genetic counseling is necessary before IVF and ICSI".

Germ Cell Transplantation or Cultured Testicular Stem Cells for the Treatment of Male Infertility

An UpToDate review on "Treatment of male infertility" (Wang and Swerdloff, 2017) states that "Mammalian (mouse) germ cells undergo self-renewal, can be maintained in vitro for several hours, can initiate organized, normal spermatogenesis when transplanted to mice depleted of germ cells due to genetic mutation or after chemotherapy, and can result in normal progeny after successful mating with females. Successful germ cell transplants can be achieved from mouse to mouse, rat to rat, and rat to immune-compromised mouse. Recently, successful ectopic xenografts of testis from a number of species including primates into mice have

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

allowed studies of drugs and toxicants on spermatogenesis without having to administer the agent to the species. These observations suggest that germ cell transplantation or cultured testicular stem cells may become a treatment for male infertility and for genetic diseases in men that can be corrected and eradicated in germ cell lines. This possibility raises serious ethical, social, and moral issues".

Hyperbaric Oxygen Therapy

Metelev et al (2015) examined the potential of hyperbaric oxygen (HBO) for reduction of sperm DNA fragmentation level and reactive oxygen species (ROS) in semen. The study included 90 men with idiopathic infertility. Patients of the treatment group (n = 60) underwent HBO before IVF. In the control group (n = 30) IVF was carried out without prior course of HBO. Sperm DNA fragmentation analysis was carried out using the TUNEL assay, the level of ROS in the ejaculate was measured by chemiluminescence. Hyperbaric oxygen therapy resulted in a significant decrease in the mean level of sperm DNA fragmentation from 33.2 ± 7.5 to 11.9 ± 5.9 %, and the median ROS in sperm from 0.89 to 0.39 mV/s ($p < 0.05$). In the control group these changes were not statistically significant. Pregnancy after IVF occurred in 63.3 % (38/60) of sexual partners of the treatment group men and in 36.7 % (11/30) of the control group ($p < 0.05$). The authors concluded that the high efficiency of HBO in overcoming the adverse effects of oxidative stress on sperm parameters suggested that it is a promising method for the treatment of men with idiopathic infertility.

Hysterosalpingo-Foam Sonography (HyFoSy)

Hysterosalpingogram (HSG), which uses dye, fluoroscopy and x-ray techniques, is the standard of care test used for evaluation of tubal patency due to its therapeutic and diagnostic benefits. Hysterosalpingo-contrast sonography (HyCoSy), which uses ultrasound and microbubble or agitated saline contrast, is considered a reasonable alternative (Kuohung and Hornstein, 2024).

Transvaginal hysterosalpingo-foam sonography (HyFoSy) is a novel ultrasound method that uses intrauterine foam contrast (e.g., ExEm Foam) to evaluate the patency of the fallopian tubes in women who are having difficulty getting pregnant.

ExEm Foam (ExEm Inc.) is an FDA-approved ultrasound contrast agent indicated for sonohysterosalpingography to assess fallopian tube patency in women with known or suspected infertility. ExEm Foam is a reconstituted air polymer-type A foam that is infused via a catheter that is inserted into the cervix. The foam creates an echogenic effect in the fallopian tubes and peritoneal cavity which can be visualized with transvaginal ultrasound.

In a randomized controlled trial, Dreyer et al (2014) evaluated whether HyFoSy is a less painful first line tubal patency test than serial HSG. The trial included 40 subfertile women, ages 18 to 41 years, with an indication for tubal patency testing as part of the fertility workup according to the Dutch Nederlandse Vereniging voor Obsteterie & Gynaecologie-guidelines. The women were randomized to either HyFoSy or serial HSG. Visual Analogue Scale (VAS) pain scores were used during tubal patency testing. The authors found that the median VAS score for pain perception during the HyFoSy procedure was 1.7 cm (interquartile range: 2.1) compared with 3.7 cm (interquartile range: 4.2) during HSG. The HyFoSy procedure also had a statistically significantly shorter procedure time compared with HSG, with a median of 5.0 minutes (interquartile range: 3.0) for HyFoSy versus 12.5 minutes (interquartile range: 16.0) for HSG. The authors concluded that the HyFoSy procedure is a less painful and less time-consuming tubal patency test compared with HSG.

In a prospective observational study, Ludwin et al (2017) reviewed the accuracy of HyFoSy in comparison to HyCoSy with air/saline and to laparoscopy with dye in 132 infertile women consecutively enrolled in the study from 2013 to 2015. 2D-Air/saline-HyCoSy, 2D/3D-HyFoSy and 2D/3D-HDF-HyFoSy and laparoscopy were performed independently. The authors found that 2D-Air/saline-HyCoSy, 2D/3D-HyFoSy and 2D/3D-HDF-HyFoSy indicated that 46 (17.8%), 27 (10.4%) and 24 (9.2%) of the 259 tubes were occluded, respectively; additionally, inconclusive results were obtained for 8 (3%), 5 (1.9%) and 3 (1.2%) tubes, respectively. The reference method revealed 18 (6.9%) occluded Fallopian tubes. 2D-Air/saline-

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

HyCoSy had a high negative predictive value (NPV) (99.5%) that was similar to that of 2D/3D-HyFoSy (99%) and 2D/3D-HDF-HyFoSy (99.6%) (P > 0.05), but had a very low positive predictive value (PPV) (30.4%). The use of 2D/3D-HyFoSy, especially 2D/3D-HDF-HyFoSy, which had a significantly higher PPV (48% and 71%, P < 0.05 and P < 0.01; respectively), resulted in fewer false positive and inconclusive findings than the use of 2D-air/saline-HyCoSy. The LR- and LR+ was 0.14 and 14.8, respectively, for 2D/3D-HyFoSy, 0.06 and 32.1, respectively, for 2D/3D-HDF-HyFoSy, and 0.08 and 6.9, respectively, for 2D-air/saline-HyCoSy. The number of inconclusive or positive results per patient was significantly fewer with 2D/3D-HyFoSy (odds ratio, OR = 0.5, CI = 0.3-0.95, P < 0.05) and 2D/3D-HDF-HyFoSy (OR = 0.4, 95% CI = 0.2-0.8, P < 0.01) than with 2D-air/saline-HyCoSy. The authors acknowledged limitations to their study. An unselected infertile population with a low prevalence of tubal occlusion is suitable for estimating the diagnostic accuracy of imaging tests only as a screening tool. The authors concluded that 2D-Air/saline-HyCoSy, which has a high NPV, is suitable as an initial test and basic screening method, but 2D/3D-HDF-HyFoSy, which has a significantly higher PPV, can be used as a standard to verify any questionable or positive results obtained with 2D HyCoSy. This strategy may significantly reduce the need for laparoscopy as a reference standard.

In a prospective randomized study, Piccioni et al (2017) compared sonohysterosalpingography (sono-HSG) with foam instillation (HyFoSy) versus saline solution (HyCoSy) in the evaluation of tubal patency in 37 infertile women. The women were randomized into two groups: HyFoSy (group I) and HyCoSy (group II). The patients of both groups underwent laparoscopy with dye test. The authors assessed the diagnostic performance (sensitivity, specificity, and overall accuracy) of HyFoSy and HyCoSy, compared with laparoscopy and dye test, in the assessment of tubal patency. The authors found that sono-HSG findings in tubal patency assessment obtained in the HyFoSy group were concordant with laparoscopic results in 94.4% of cases, with a sensitivity of 87.5% and a specificity of 100%, whereas in the HyCoSy group, concordance occurred in only 57.8% of examinations, with a sensitivity of 50% and a specificity of 66.6%. The authors concluded that HyFoSy allows a more accurate diagnosis of tubal patency compared with HyCoSy; however, limitations in this study include small sample size.

In a pilot study, Riganelli et al (2018) aimed to compare 2D and 3D-sonohysterosalpingography (2D-3D-HyFoSy) with previous diagnostic laparoscopy in the diagnosis of tubal patency, and compare each procedure in terms of procedure's time, perceived pain and complication rate. The authors prospectively recruited 50 infertile women, previously submitted to laparoscopy and randomly allocated into 2D-HyFoSy (group I; n=25) and 3D-HyFoSy (group II; n=25). They analyzed the results in term of sensitivity, specificity, positive predictive value and negative predictive value in tubal patency evaluation of both procedures in comparison with laparoscopy. The authors state that 2D-HyFoSy findings obtained in group I were concordant with laparoscopy in 81% of cases, with a sensitivity of 80% and a specificity of 92%. In group II, a correspondence was present in 88% of examinations, with a sensitivity and specificity of 98% and 91.4% respectively. 3D-HyFoSy was found to be faster and less painful than 2D (P<0.001). The authors concluded that in the diagnosis of tubal occlusion, in the high-risk population, it seems advisable to use the 3D-HyFoSy as the first-level examination, while, in low-risk patients, if the tubes appear obstructed in 2D-HyFoSy, the 3D-HyFoSy should be indicated before submitting patients to operative laparoscopy.

An expert panel on women's imaging discussed the 2020 American College of Radiology (ACR) appropriateness criteria in female infertility. The authors state that "Hysterosalpingo-foam sonography in combination with 2-D or 3-D imaging has demonstrated improved accuracy of 93.7% and better concordance with laparoscopy compared with 2-D or 3-D air or saline HyCoSy for assessment of tubal patency. The addition of high-definition flow (bidirectional Doppler feature) achieved even higher accuracy at 96.9%, comparable with the reference method of laparoscopic chromopertubation with dye. However, high-definition flow still needs to be validated by other groups". However, the citations used for support included Ludwin et al (2017), a prospective observational study comparing HyFoSy and HyCoSy in 132 women, and Piccioni et al (2017), a prospective randomized study comparing foam and saline in 37 infertile women.

In a multicenter, prospective, randomized, non-inferiority study, van Welie et al (2022) evaluated whether HyFoSy could replace HSG as a first-choice tubal patency test. The study included infertile women between 18 and 41 years old who were scheduled for tubal patency testing during their fertility work-up in the Netherlands. Women with anovulatory cycles not responding to ovulation induction, endometriosis, severe

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

male infertility or a known iodine contrast allergy were excluded. The primary outcome for the comparison of the HyFoSy- and HSG-based strategies was ongoing pregnancy leading to live birth within 12 months after inclusion in an intention-to-treat analysis. Participating women underwent both HyFoSy and HSG in randomized order. In case of discordant results, women were randomly allocated to either a management strategy based on HyFoSy or one based on HSG. Between May 2015 and January 2019, 1026 women underwent HyFoSy and HSG. HyFoSy was inconclusive in 97 of them (9.5%), HSG was inconclusive in 30 (2.9%) and both were inconclusive in 9 (0.9%). In 747 women (73%) conclusive tests results were concordant. Of the 143/1026 (14%) with discordant results, 105 were randomized to clinical management based on the results of either HyFoSy or HSG. In this group, 22 of the 54 women (41%) allocated to management based on HyFoSy and 25 of 51 women (49%) allocated to management based on HSG had an ongoing pregnancy leading to live birth (Difference -8%; 95% CI: -27% to 10%). In total, clinical management based on the results of HyFoSy was estimated to lead to a live birth in 474 of 1026 women (46%) versus 486 of 1026 (47%) for management based on HSG (Difference -1.2%; 95% CI: -3.4% to 1.5%). Given the pre-defined margin of -2%, statistically significant non-inferiority of HyFoSy relative to HSG could not be demonstrated (P = 0.27). The mean pain score for HyFoSy on the 1-10 Visual Analogue Scale (VAS) was 3.1 (SD 2.2) and the mean VAS pain score for HSG was 5.4 (SD 2.5; P for difference < 0.001). The authors acknowledged limitations in the study. Since all women underwent both tubal patency tests, no conclusions on a direct therapeutic effect of tubal flushing could be drawn. A majority of the women (85%) had a low risk for tubal pathology. The authors concluded that HyFoSy or HSG produce similar tubal pathology findings in a majority of infertile couples and, where they differ, a difference in findings does not lead to substantial difference in pregnancy outcome, while HyFoSy is associated with significantly less pain. However, before final conclusions on clinical management can be drawn, a head-to-head comparison between HyFoSy and HSG with oil-based contrast may be needed.

In an UpToDate review on "Female infertility: Evaluation", Kuohung and Hornstein (2024) do not discuss use of HyFoSy, or intrauterine foam contrast with ultrasound, for evaluation of patency of the fallopian tubes.

Intralipid Infusion for the Treatment of Female Infertility

In a double-blind, randomized controlled trial (RCT), Dakhly et al (2016) examined the efficacy of intralipid supplementation in women with recurrent spontaneous abortion (RSA) and elevated natural killer (NK) cell activity undergoing in-vitro fertilization/intracytoplasmic sperm injection (IVF/ICSI). This study was carried out between February 10, 2013, and April 30, 2015 at a center in Egypt. Women with unexplained secondary infertility, RSA, and elevated levels of NK cells (greater than 12 %) were enrolled and randomly assigned to receive intralipid (2-ml diluted at 20 % in 250 ml saline) or saline (250 ml) infusion on the day of oocyte retrieval using random numbers and sealed envelopes. Patients and attending physicians were masked to group assignment. The infusions were repeated within 1 week of a positive pregnancy test and then every 2weeks until the end of the 1st trimester. The primary outcome was chemical pregnancy 14 days after embryo transfer. Analyses were by intention-to-treat (ITT). A total of 296 women were enrolled. Chemical pregnancy was recorded for 84 (58.3 %) of 144 women in the intralipid group and 76 (50.0 %) of 152 in the control group (p = 0.129). The authors concluded that intralipid supplementation did not increase frequency of chemical pregnancy; however, findings related to ongoing pregnancy and livebirth should be further examined.

In a systematic review and meta-analysis, Achilli et al (2018) examined the evidence on the role of immunotherapy in IVF and in the management of recurrent pregnancy loss (RPL). These researchers carried out a literature search using Medline, PubMed, CINAHL, and Embase until May 2017. Only RCTs were included, and a meta-analysis was performed where appropriate. Subjects were women undergoing IVF treatment with or without a history of recurrent implantation failure and women with idiopathic RPL. Assessment of the efficacy of commonly used immunomodulators such as IV use of (i) immunoglobulin, (ii) lymphocyte immunotherapy and (iii) intralipid; intra-uterine infusion of (iv) granulocyte colony-stimulating factor (G-CSF) and (v) peripheral blood mononuclear cells; subcutaneous administration of (vi) tumor necrosis factor (TNF)-alpha inhibitors, (vii) leukemia inhibitory factor; and oral administration of (viii) glucocorticoids. The primary outcomes were livebirth rate and miscarriage rate; secondary outcome was clinical pregnancy rate. Of the 7,226 publications identified, 53 were selected during the initial screening; 30

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

satisfied the selection criteria and were included in this review. The authors concluded that the available medical literature showed controversial results regarding the role of immunotherapy when used for improving reproductive outcomes. This study did not show a role for immunotherapy in improving the livebirth rate in women undergoing IVF treatment or in the prevention of idiopathic RPL. Currently, immunotherapy should be used in the context of research and should not be used in routine clinical practice to improve reproductive outcomes.

In a RCT, Al-Zebeidi et al (2020) examined the effect of empiric intralipid infusion therapy on pregnancy outcomes for patients with unexplained recurrent implantation failure (RIF) undergoing ICSI. A total of 142 patients with a history of unexplained RIF (3 or more cycles) were included in this study. Patients were randomized into 2 groups, study group (n = 71) and control group (n = 71). The study group received intralipid 20 % infusion on the day of embryo transfer (ET) and a 2nd dose on the day of pregnancy test. The clinical pregnancy rate in the study group was 36.6 % (n = 26) compared to 28.2 % (n = 20) in the control group (odds ratio [OR] 1.47, confidence interval [CI]: 0.72 to 2.98, p = 0.282). The livebirth rate in the study group was 18.3 % (n = 13) and 14.1 % (n = 10) in the control group (OR 1.37, CI: 0.55 to 3.36, p = 0.49). No side effects of intralipid therapy were reported in the study period. There was improvement in the pregnancy rate among women with unexplained RIF who received empiric intralipid infusion therapy; however, this improvement did not reach statistical significance.

Furthermore, an UpToDate review on “Treatments for female infertility” (Kuohung and Hornstein, 2022) does not mention intralipid infusion as a management / therapeutic option.

Intrauterine Administration of Human Chorionic Gonadotropin (hCG) for Subfertile Women Undergoing Assisted Reproduction

Craciunas and colleagues (2018) examined if intra-uterine (intra-cavity) administration of hCG (IC-hCG) around the time of ET improves clinical outcomes in sub-fertile women undergoing assisted reproduction. These investigators performed searches on January 9, 2018 using Cochrane methods. They looked for RCTs evaluating IC-hCG around the time of ET, irrespective of language and country of origin. Two review authors independently selected studies, assessed risk of bias, extracted data from studies, and attempted to contact study authors when data were missing. They performed statistical analysis using Review Manager 5. These researchers assessed evidence quality using GRADE methods. Primary outcomes were live-birth and miscarriage; secondary outcomes were clinical pregnancy rate and complications. A total of 17 RCTs examined the effects of IC-hCG administration for 4,751 sub-fertile women undergoing assisted reproduction; IC-hCG was administered in variable doses at different times before the ET; hCG was obtained from the urine of pregnant women or from cell cultures using recombinant DNA technology. Most studies (12/17) were at high risk of bias in at least 1 of the 7 domains assessed. Common problems were unclear reporting of study methods and lack of blinding. The main limitations for evidence quality were high risk of bias and serious imprecision. For analyses of live-birth and clinical pregnancy, there was considerable heterogeneity ($I^2 > 75$ %) and therefore these researchers presented subgroups for dosage and stage of ET. Exploration for sources of heterogeneity revealed 2 key pre-specified variables as important determinants: stage of ET (cleavage versus blastocyst stage) and dose of IC-hCG (less than 500 international units (IU) versus greater than or equal to 500 IU). They performed meta-analyses within subgroups defined by stage of embryo and dose of IC-hCG. Live-birth rates among women having cleavage-stage ET with an IC-hCG dose of less than 500 IU compared to women having cleavage-stage ET without IC-hCG showed no benefit of the intervention and would be consistent with no substantive difference or disadvantage of indeterminate magnitude (risk ratio (RR) 0.76, 95 % CI: 0.58 to 1.01; 1 RCT; 280 participants; $I^2 = 0$ %; very low-quality evidence). In a clinic with a live-birth rate of 49 % per cycle, use of IC-hCG of less than 500 IU would be associated with a live-birth rate ranging from 28 % to 50 %. Results showed an increase in live-birth rate in the subgroup of women undergoing cleavage-stage ET with an IC-hCG dose of greater than or equal to 500 IU compared to women having cleavage-stage ET without IC-hCG (RR 1.57, 95 % CI: 1.32 to 1.87; 3 RCTs; 914 participants; $I^2 = 0$ %; moderate-quality evidence). At a clinic with a live-birth rate of 27 % per cycle, use of IC-hCG of greater than or equal to 500 IU would be associated with a live-birth rate ranging from 36 % to 51 %. Results showed no substantive differences in live-birth among women having blastocyst-stage ET with an IC-hCG dose of greater than or equal to 500 IU compared to women having blastocyst-stage ET

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

without IC-hCG (RR 0.92, 95 % CI: 0.80 to 1.04; 2 RCTs; 1,666 participants; $I^2 = 0$ %; moderate-quality evidence). At a clinic with a live-birth rate of 36 % per cycle, use of IC-hCG of greater than or equal to 500 IU would be associated with a live-birth rate ranging from 29 % to 38 %. Evidence for clinical pregnancy among women having cleavage-stage ET with an IC-hCG dose of less than 500 IU showed no benefit of the intervention and would be consistent with no substantive difference or disadvantage of indeterminate magnitude (RR 0.88, 95 % CI: 0.70 to 1.10; 1 RCT; 280 participants; $I^2 = 0$ %; very low-quality evidence). Results showed an increase in clinical pregnancy rate in the subgroup of women having cleavage-stage ET with an IC-hCG dose of greater than or equal to 500 IU compared to women having cleavage-stage ET without IC-hCG (RR 1.49, 95 % CI: 1.32 to 1.68; 12 RCTs; 2,186 participants; $I^2 = 18$ %; moderate-quality evidence). Results showed no substantive differences in clinical pregnancy among women having blastocyst-stage ET with an IC-hCG dose of greater than or equal to 500 IU (RR 0.99, 95 % CI: 0.85 to 1.15; 4 RCTs; 2,091 participants; $I^2 = 42$ %; moderate-quality evidence) compared to women having blastocyst-stage ET with no IC-hCG. No RCTs examined blastocyst-stage ET with an IC-hCG dose of less than 500 IU. These researchers were uncertain whether miscarriage was influenced by intra-uterine hCG administration (RR 1.04, 95 % CI: 0.81 to 1.35; 11 RCTs; 3,927 participants; $I^2 = 0$ %; very low-quality evidence). Reported complications were ectopic pregnancy (4r RCTs; 1,073 participants; 4 events overall), heterotopic pregnancy (1 RCT; 495 participants; 1 event), intra-uterine death (3 RCTs; 1,078 participants; 22 events), and triplets (1 RCT; 48 participants; 3 events). Events were few, and very low-quality evidence was insufficient to permit conclusions to be drawn. The authors concluded that there was moderate quality evidence that women undergoing cleavage-stage transfer using an IC-hCG dose of greater than or equal to 500 IU had an improved live-birth rate. There was insufficient evidence for IC-hCG treatment for blastocyst transfer. There should be further trials with live-birth as the primary outcome to identify the groups of women who would benefit the most from this intervention. There was no evidence that miscarriage was reduced following IC-hCG administration, irrespective of embryo stage at transfer or dose of IC-hCG. Events were too few to allow conclusions to be drawn with regard to other complications.

Intrauterine Injection/Infusion of Platelet Rich Plasma (PRP) for the Treatment of Female Infertility

Maleki-Hajiagha and colleagues (2020) stated that previous studies have provided conflicting results regarding the use of platelet-rich plasma (PRP) in women undergoing IVF or ICSI. In a systematic review and meta-analysis, these researchers examined the effect of the intrauterine infusion of PRP on the outcome of ET in women undergoing IVF/ICSI. They searched databases, including PubMed, Embase, Scopus, Web of Science, and the Cochrane Database of Clinical Trials (CENTRAL). Meta-analysis using a random-effects model was carried out to calculate the pooled estimates. A total of 7 studies involving 625 patients (311 cases and 314 controls) were included. The probability of chemical pregnancy (n = 3, RR: 1.79, 95 % CI: 1.29 to 2.50; $p < 0.001$, I2 = 0 %), clinical pregnancy (n = 7, RR: 1.79, 95 % CI: 1.37 to 2.32; $p < 0.001$, I2 = 16 %), and implantation rate (n = 3, RR: 1.97, 95 % CI: 1.40 to 2.79; $p < 0.001$, I2 = 0 %) was significantly higher in women who received PRP compared with control. There was no difference between women who received PRP compared with control group regarding miscarriage (RR: 0.72, 95 % CI: 0.27 to 1.93; $p = 0.51$, I2 = 0 %). Following the intervention, endometrial thickness increased in women who received PRP compared to control group (SMD: 1.79, 95 % CI: 1.13 to 2.44; $p < 0.001$, I2 = 64 %). The authors concluded that the findings of this systematic review suggested that PRP is an alternative treatment strategy in patients with thin endometrium and recurrent implantation failure (RIF). Moreover, these researchers stated that further prospective, large, and high-quality RCTs are needed to identify the subpopulation that would most benefit from PRP.

The Practice Committee of the American Society for Reproductive Medicine's guideline on "Evidence-based treatments for couples with unexplained infertility" (2020) did not mention PRP as a management option.

Lin and associates (2021) noted that clinical studies for the effectiveness of PRP in improving uterine endometrial function remain controversial. Most studies suggested that PRP treatment resulted in positive reproductive results in patients with thin endometrium, RIF, chronic endometritis (CE) and Asherman syndrome (AS); however, the best dosage and timing of PRP application remains unknown. Few studies examined the underlying mechanisms of the PRP treatment. PRP might improve the endometrial receptivity

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

via the improvement of cell proliferation, vascularization, anti-inflammatory properties and the reduction in the degree of fibrosis, with the help of the concentrated peptides, growth factors and cytokines in PRP. Moreover, these researchers stated that for its potential clinical application, further explication of the mechanism is needed. They noted that although increasing evidence suggests the possible value of PRP treatment on the endometrium of infertile women, more carefully designed studies, especially RCTs, on larger scales are needed. Furthermore, the standard PRP preparation procedure and the strict indication of the PRP treatment should be published; studies and follow-ups are needed on the long-term health and complications of the resulting child(ren).

Mouanness and co-workers (2021) stated that endometrial receptivity and thickness play an important role in achieving a pregnancy. Intrauterine autologous PRP infusion has been used in infertile women with RIF and thin endometrial lining thickness (EMT). These investigators carried out a literature search in PubMed for studies including in-vitro, animal, and human studies as well as in abstracts presented at national conferences. Animal studies demonstrated a decrease in the expression of inflammatory markers and fibrosis, and increased endometrial proliferation rate, increased expression of proliferative genes, and increased pregnancy rates. The in-vitro studies showed that PRP was associated with increased stromal and mesenchymal cell proliferation, increased expression of regenerative enzymes, and enhancement in cell migration. In infertile women undergoing ART, 1 randomized clinical trial showed that PRP intrauterine infusion improved EMT, implantation rate, and CPR in patients with thin EMT, while 3 other trials involving subjects with RIF showed conflicting results related to CPR. Case series and cohort studies showed conflicting results pertaining to CPR. The authors concluded that data to-date suggested that PRP may be beneficial in improving endometrial thickness and endometrial receptivity; however, they stated that further large, prospective and high-quality trials are needed to examine its effect and to identify the population of patients that would benefit the most.

Intrauterine Insemination (IUI)

Statistics show that the cumulative incidence of IUI pregnancies tends to level off after four IUI cycles for heterosexual couples, whereas the cumulative incidence of IUI pregnancies continues to increase significantly with more than four cycles for same-sex female couples and unpartnered females (Capinello, et al., 2021; Johal, et al., 2021; Linara-Demakakou, et al., 2020; Morshedi, et al., 2003; Quetrell, et al., 2023: Soria, et al., 2012).

In a randomized controlled trial (RCT; the FASTT Trial), Reindollar et al (2010) examined the value of gonadotropin/intra-uterine insemination (FSH/IUI) therapy for infertile women aged 21 to 39 years. Participants were couples with unexplained infertility. Couples were randomized to receive either conventional treatment (n = 247) with 3 cycles of clomiphene citrate (CC)/IUI, 3 cycles of FSH/IUI, and up to 6 cycles of IVF or an accelerated treatment (n = 256) that omitted the 3 cycles of FSH/IUI. Main outcome measure(s) included the time it took to establish a pregnancy that led to a live-birth and cost-effectiveness, defined as the ratio of the sum of all health insurance charges between randomization and delivery divided by the number of couples delivering at least 1 live-born baby. An increased rate of pregnancy was observed in the accelerated arm (hazard ratio [HR], 1.25; 95 % confidence interval [CI]: 1.00 to 1.56) compared with the conventional arm. Median time to pregnancy was 8 and 11 months in the accelerated and conventional arms, respectively. Per cycle pregnancy rates for CC/IUI, FSH/IUI, and IVF were 7.6 %, 9.8 %, and 30.7 %, respectively. Average charges per delivery were $9,800 lower (95 % CI: $25,100 lower to $3,900 higher) in the accelerated arm compared to conventional treatment. The observed incremental difference was a savings of $2,624 per couple for accelerated treatment and 0.06 more deliveries. The authors concluded that a randomized clinical trial demonstrated that FSH/IUI treatment was of no added value.

The authors stated that beginning treatment with CC/IUI will result in pregnancy in nearly 25 % of the couples, with minimal risk of multiple births and at a low cost. As clinical practice moves to limiting the number of embryos transferred during IVF procedures in young women to 1 or 2 embryos, eliminating gonadotropin/IUI from the stepwise infertility paradigm will result in pregnancies with the lowest possible risk for multiple births. Compared with conventional infertility treatment and when the woman is younger than 40 years, an accelerated approach to IVF that starts with CC/IUI, but eliminates gonadotropin/IUI, results in a shorter time to pregnancy, with fewer treatment cycles, and at a suggested cost savings.

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

In a secondary analysis of 2 RCTs (the FASTT Trial and the FORT-T Trial), Kaser et al (2014) examined if Day-3 FSH and E2 levels at the upper limits of normal affect live-birth rates and treatment trajectory in a conventional versus "fast track" treatment program for IVF. Infertile women aged 21 to 42 years were randomized to conventional or accelerated treatment with controlled ovarian hyperstimulation (COH)-IUI and/or IVF (n = 603 patients contributing 2,717 total cycles). Patients were stratified according to basal FSH and E2: FSH less than 10 mIU/ml and E2 less than 40 pg/ml (group 1A), FSH less than 10 mIU/ml and E2 40 pg/ml or higher (group 1B), FSH, 10 to 15 mIU/ml and E2 less than 40 pg/ml (group 2A), and FSH, 10 to 15 mIU/ml and E2 40 pg/ml or higher (group 2B). Main outcome measure(s) included number of cancelled cycles, dis-enrollment for poor response, and cumulative live-birth rates per couple. Women in groups 2A and 2B were more likely to have cancelled cycles and be dis-enrolled for poor response. While no live-births occurred in group 2B during COH-IUI (0/19 couples, 0/58 cycles), IVF still afforded these patients a reasonable chance of success (6/18 couples, 6/40 cycles, 33.3 % live-birth rate per couple). The specificity and positive predictive value (PPV) of basal FSH of 10 to 15 mIU/ml and E2 40 pg/ml or higher for no live-birth during COH-IUI treatment were both 100 %. The authors concluded that women who initiated infertility treatment with FSH of 10 to 15 mIU/ml and E2 40 pg/ml or higher on Day-3 testing were unlikely to achieve live-birth after COH-IUI treatment.

Bordewijk et al (2019) stated that in women with normo-gonadotropic anovulation who ovulate but do not conceive after 6 cycles with clomiphene citrate (CC), medication is usually switched to gonadotrophins, with or without IUI. The cost-effectiveness of these changes in policy is unknown. These researchers examined if 6 cycles of ovulation induction with gonadotrophins are more cost-effective than 6 cycles of ovulation induction with CC with or without IUI in normo-gonadotropic anovulatory women not pregnant after 6ovulatory cycles with CC. They carried out an economic evaluation of ovulation induction with gonadotrophins compared with CC with or without IUI in a 2-by-2 factorial multi-center RCT in normo-gonadotropic anovulatory women not pregnant after 6 ovulatory cycles with CC. Between December 2008 and December 2015, women were allocated to 6 cycles with gonadotrophins plus IUI, 6 cycles with gonadotrophins plus intercourse, 6 cycles with CC plus IUI or 6 cycles with CC plus intercourse. The primary outcome was conception leading to a live-birth achieved within 8 months of randomization. These investigators carried out a cost-effectiveness analysis on direct medical costs. They calculated the direct medical costs of ovulation induction with gonadotrophins versus CC and of IUI versus intercourse in 6 subsequent cycles. They included costs of medication, cycle monitoring, interventions, and pregnancy leading to live-birth. Resource use was collected from the case report forms and unit costs were derived from various sources. These researchers calculated incremental cost-effectiveness ratios (ICER) for gonadotrophins compared to CC and for IUI compared to intercourse. They used non-parametric bootstrap re-sampling to examine the effect of uncertainty in their estimates. The analysis was carried out according to the intention-to-treat (ITT) principle. The authors allocated 666 women in total to gonadotrophins and IUI (n = 166), gonadotrophins and intercourse (n = 165), CC and IUI (n = 163), or CC and intercourse (n = 172). Mean direct medical costs per woman receiving gonadotrophins or CC were €4,495 versus €3,006 (cost difference of €1,475 (95 % CI: €1,457 to €1,493)). Live-birth rates were 52 % in women allocated to gonadotrophins and 41 % in those allocated to CC (relative risk (RR) 1.24; 95 % CI: 1.05 to 1.46). The ICER was €15,258 (95 % CI: €8,721 to €63,654) per additional live-birth with gonadotrophins. Mean direct medical costs per woman allocated to IUI or intercourse were €4,497 versus €3,005 (cost difference of €1,510 (95 % CI: €1,492 to €1,529)). Live-birth rates were 49% in women allocated to IUI and 43 % in those allocated to intercourse (RR = 1.14; 95 % CI: 0.97 to 1.35). The ICER was €24,361 (95 % CI: €-11,290 to €85,172) per additional live-birth with IUI.

The authors concluded that gonadotrophins were more effective, but more expensive than CC; thus, the use of gonadotrophins in women with normo-gonadotropic anovulation who have not conceived after 6 ovulatory CC cycles depends on society's willingness to pay for an additional child. In view of the uncertainty around the cost-effectiveness estimate of IUI, these data were insufficient to make recommendations on the use of IUI in these women. In countries where ovulation induction regimens are reimbursed, policy makers and health care professionals may use these findings in their guidelines.

Physiological, Hyaluronan-selected Intracytoplasmic Sperm Injection (PICSI)

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

Physiological, hyaluronan-selected intracytoplasmic sperm injection (PICSI) is a variation of intracytoplasmic sperm injection (ICSI) that uses hyaluronic acid (HA) to select sperm. The selection of sperm for ICSI by the hyaluronic acid binding method is based on the ability of competent mature sperm to attach themselves to HA through specific receptors found on the sperm plasma membrane. Sperm with diminished maturity, increased levels of chromosomal aberrations, or failed spermatogenetic membrane remodeling will not bind to HA. The selection of sperm bound to HA for ICSI may facilitate the selection of individual mature sperm with low levels of chromosomal aneuploidies and thus be associated with lower rates of miscarriage after ICSI (Rosen, 2024).

In a systematic review and meta-analysis, Lepine et al (2019) evaluated the effectiveness and safety of advanced sperm selection techniques on assisted reproductive technologies (ART) outcomes. The analysis included eight randomized controlled trials (RCTs) (4147 women) comparing advanced sperm selection techniques versus standard IVF, ICSI, or another technique. The authors report that the quality of evidence ranged from very low to low. Primary outcomes measured were livebirth and miscarriage per woman randomly assigned. Secondary outcome measures included clinical pregnancy per woman randomly assigned. Secondary adverse events measured included miscarriage per clinical pregnancy and fetal abnormality. The authors found that miscarriage risk was lower for patients undergoing ICSI with hyaluronic acid-selected sperm (PICSI) versus ICSI alone (3 to 6 percent for PICSI versus 7 percent with ICSI alone, risk ratio [RR] 0.61, 95% CI 0.45-0.83, 3005 women); however, live birth rates were similar between the groups (RR 1.09, 95% CI 0.97-1.23, 2903 women). The authors concluded that the evidence suggests that sperm selected by hyaluronic acid binding may have little or no effect on live birth or clinical pregnancy but may reduce miscarriage. The authors report being uncertain of the effect of Zeta sperm selection on live birth, clinical pregnancy, and miscarriage due principally to the very low quality of the evidence for this intervention. They are also uncertain of the effect of the other selection techniques on live birth, miscarriage, or pregnancy. Further high-quality studies are required to evaluate whether any of these advanced sperm selection techniques can be recommended for use in routine practice.

Miller and colleagues (2019) noted that sperm selection strategies aimed at improving success rates of ICSI include binding to hyaluronic acid (herein termed hyaluronan). Hyaluronan-selected sperm have reduced levels of DNA damage and aneuploidy. Use of hyaluronan-based sperm selection for ICSI (so-called physiological ICSI [PICSI]) has been reported to reduce the proportion of pregnancies that end in miscarriage. However, the effect of PICSI on live-birth rates is uncertain. In a parallel, 2-group, randomized trial (HABSelect), these investigators examined the efficacy of PICSI versus standard ICSI for improving live-birth rates among couples undergoing fertility treatment. This study included couples undergoing an ICSI procedure with fresh embryo transfer at 16 assisted conception units in the United Kingdom. Eligible women (aged 18 to 43 years) had a BMI of 19 to 35 kg/m2 and a FSH concentration of 3.0 to 20.0 mIU/ml or, if no FSH measurement was available, an AMH concentration of at least 1.5 pmol/L. Eligible men (aged 18 to 55 years) had not had a vasovasostomy or been treated for cancer in the 24 months before recruitment and were able, after at least 3 days of sexual abstinence, to produce freshly ejaculated sperm for the treatment cycle. Couples were randomly assigned (1:1) with an online system to receive either PICSI or a standard ICSI procedure. The primary outcome was full-term (greater than or equal to 37 weeks' gestational age [GA]) live-birth, which was assessed in all eligible couples who completed follow-up. Between February 1, 2014 and August 31, 2016, a total of 2,772 couples were randomly assigned to receive PICSI (n = 1,387) or ICSI (n = 1,385), of whom 2,752 (1,381 in the PICSI group and 1,371 in the ICSI group) were included in the primary analysis. The term live-birth rate did not differ significantly between PICSI (27.4 % [379/1,381]) and ICSI (25.2% [346/1,371]) groups (OR 1.12, 95 % CI: 0.95 to 1.34; p = 0.18). There were 56 serious adverse events (AEs) in total, including 31 in the PICSI group and 25 in the ICSI group; most were congenital abnormalities and none was attributed to treatment. The authors concluded that compared with ICSI, PICSI did not significantly improve term live-birth rates. Thus, these researchers stated that the wider use of PICSI is not recommended at present.

Martin and Woodland (2021) state that several studies reported improved fertility treatment outcomes when using PICSI compared with conventional ICSI; however, the majority of studies are underpowered. Recently, a large, multicenter, randomized controlled trial, known as the Hyaluronic Acid Binding Sperm Selection (HABSelect) trial, found a significant reduction in miscarriage rates with PICSI, but no significant effect on live birth rate. There are still many avenues through which PICSI may provide an advantage, subject to

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

confirmation by future research, such as improved long-term health of offspring. Other advanced sperm selection techniques include intracytoplasmic morphologically selected sperm injection, magnetic-activated cell sorting, and Zeta potential sperm selection; however, the most recent Cochrane review concluded that there is currently insufficient evidence to ascertain whether these techniques improve clinical outcomes, such as live birth rates.

Sickle Cell Disease and Infertility Risks

Sickle cell disease (SCD) is a chronic, inherited blood disorder characterized by the production of abnormal hemoglobin S, which causes red blood cells to become rigid and sickle-shaped. This condition can result info a vaso-occlusive pain crises, anemia, and end-organ injury.

In a narrative review, Pecker and Cameron (2024) state that SCD and its treatments present infertility risks. The purpose of their review is to support clinicians in providing counselling about fertility and infertility risks to girls and women with SCD and their families. The authors state ovaries are considered an SCD end-organ and that there is growing evidence to support the claim that in SCD, the reproductive lifespan is reduced. They report that this is best substantiated for those with hemoglobin SS or hemoglobin Sβ0-thalassemia. The authors state that a treatment paradigm that acknowledges the risks of untreated SCD, the benefits of treating SCD, and the potential for compromised fertility in the future is warranted. Moreover, robust data addressing significant questions about menstruation, contraception, pregnancy, infertility, menopause and how these issues are affected by genotype and SCD therapies would allow clinicians to claim more solid ground and clarify recommendations. An updated treatment paradigm that coordinates optimizing SCD treatment with fertility care is indicated.

SCD-related fertility issues impact men and women alike. "The standard protocol for managing infertility remains identifying the source of the infertility—meaning whether the issue lies with the male or female in a relationship, or both. Just because the risk of infertility is higher in SCD does not preclude infertility from being caused by other factors" (Lai, 2022).

Stem Cell Therapy for the Treatment of Female Infertility

In a systematic review, Ahmadian and colleagues (2020) examined the evidence on stem cell therapy for ovarian disorders. These researchers evaluated different published studies on stem cell-based therapy for the treatment of various types of ovarian insufficiency and disorders such as premature ovarian insufficiency (POI) in the affected female population in animal or human clinical studies. They monitored 5 databases, including PubMed, Cochrane, Embase, Scopus, and ProQuest. A comprehensive online search was carried out using including criteria targeting application of stem cells in animal models for menopause. Two independent reviewers carefully evaluated titles and abstracts of studies. The stem cell type, source, dosage, route of administration were high-lighted in various POI animals models. Non-relevant and review articles were excluded. A total of 648 published studies were identified during the initial comprehensive search process from which 41 were selected according to designed criteria. Based on this analysis, stem cells could accelerate ovarian tissues rejuvenation, regulate systemic sex-related hormones levels and eventually increase fertility rate. The authors concluded that the evidence suggested that stem cell-based therapies could be considered as an alternative modality to deal with women undergoing POI.

An UpToDate review on “Treatments for female infertility” (Kuohung and Hornstein, 2020) does not mention stem cell therapy as a therapeutic option.

Uterine Transplant (Donor Hysterectomy) for the Treatment of Infertility

Daolio and colleagues (2020) noted that uterine transplantation (UTx) associated with IVF restores fertility in women affected by absolute uterine factor infertility (AUFI). Pregnancies achieved both in women undergoing any solid organ transplantation and following IVF are associated with an increased risk of maternal and neonatal complications. In a systematic review, these investigators examined this risk in UTx-IVF treated women focusing on the safety and efficacy features of the treatment. A total of 22 studies and 3

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

press releases reporting on 52 UTx-IVF treatments were identified. Regarding the safety of treatment, 38/52 (73.1 %) of surgical procedures led to the restoration of uterine function in recipients, 12/52 (23.1 %) of recipients experienced post-operative complications requiring hysterectomy, and 2/52 (3.8 %) of procedures failed before uterine recipients' surgery due to intra-operative complications. Regarding the efficacy of treatment, results focused on transplanted patients showing full recovery of organ functioning: 16/38 (42.1 %) of patients achieved a pregnancy, including 2 women who gave births twice. UTx-IVF pregnancies led to 16 deliveries and all newborns were healthy; 6 out of 16 (37.5 %) UTx pregnancies faced major complications during gestation. Pre-term births (PTBs) occurred in 10/16 (62.5 %) UTx deliveries. The authors concluded that these findings showed that the risk of gestational and delivery complications deserves important consideration in AUFI women receiving UTx-IVF treatments; however, these observations were preliminary and need to be revised after larger series of data are published. Moreover, these researchers stated that although the accuracy of UTx-IVF safety and efficacy outcomes could not be evaluated definitively until UTx centers provide additional, meaningful data from larger series of UTx cases treated homogeneously, they critically evaluated the following aspects. UTx surgery in recipients is mainly complicated by vascular impairments and uterine infections with a negative impact on graft survival and maintenance. The incidence of post-operative complications leading to hysterectomy in UTx recipients was estimated to be 23 %, but this value could be under-estimated.

The authors stated that this review had several drawbacks. Published studies represented only a small percentage of the cases actually performed by the time of writing, and this could have influenced the accuracy of the estimations of safety and efficacy in both UTx recipients and children, which could be potentially under-estimated. As a matter of fact, more than 60 UTx procedures have been performed worldwide, and 18 babies have been delivered, but 50 % of these cases has not yet been scientifically published. The majority of studies did not describe post-operative, maternal and neonatal outcomes simultaneously, probably due to the different timings of occurrence concerning UTx surgery, gestational period and delivery. Data were often published at distance and reported in more than 1 study; or referred to as personal communications. Furthermore, many studies did not capture the learning curve of the surgical teams at different centers worldwide, and it is hoped that future research will provide more standardized UTx surgical procedures.

Ricci and co-workers (2021) stated that UTx is an evolving procedure to allow for childbearing in women with AUFI . These researchers examined the available evidence using a comprehensive PubMed literature search. They carried out a systematic medical sub-headings search strategy with the terms "uterus transplant" and "uterine transplantation". Of the 75 full-text articles reviewed for eligibility, 68 were included in the qualitative synthesis. Of these, 9 were included in the meta-analysis on living donor uterine transplant, 5 on deceased donor uterine transplant, and 6 case reports of single uterine transplants. The authors concluded that UTx is a nascent field undergoing a rapid rate of evolution as programs mature their data and increase the number of procedures performed.

Malasevskaia and Al-Awadhi (2021) noted that UTx restores fertility in women with AUFI, and allows the opportunity to conceive, experience gestation, and acquire motherhood. The number of cases being carried out is increasing dramatically, with detailed outcomes from 65 cases now available. Pregnancies achieved through UTx and following IVF were associated with an increased risk for further mother and newborn babies. This review focused on the safety and efficacy of UTx, which is associated with significant risk, with approximately 25 % of grafts removed because of complications. The authors concluded that UTx is realizable in women with uterine factor infertility; however, it is associated with a significant complication risk. The risk of the procedure and gestational and delivery complications deserve important consideration before receiving such treatments. Nevertheless, these observations are preliminary and should be revised after a larger series of data are published. Moreover, these investigators stated that further studies should concentrate on the safest operation method to minimize the complications on the donor and recipient. Furthermore, further research is needed on the immunosuppressive protocol, which is used before and during the pregnancy period. Long-term observation of the patients (recipients) after hysterectomy is demanded. As they have to have a big intervention, which changes the anatomy of pelvic floor muscles, the direction of ureters, and place of the ovaries, these may predispose to pelvic organ prolapse and urinary tract infections (UTIs) with further kidney damage. In additional, these procedures could affect the vascular supply of pelvic organs and lower limbs.

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

Fronek and colleagues (2021) reported the results of the first 10 UTx procedures carried out at their institution. The program started in April 2016 as a 2-arm study comparing the efficacy of UTx from live donors (LD) and deceased donors (DD). Between April 2016 and April 2018, these researchers carried out 5 DD UTx and 5LD UTx; 2 grafts had to be removed early due to thrombosis; and 1 graft was removed due to chronic rejection and previous herpes simplex infection at month 7. Graft survival was 70 % at 1 year. Recipient survival was 100 % at 2 years; LD survival was 100 % at t3 years; and 3 livebirths have been achieved, 2 from a LD and 1 from a graft from a nulli-parous DD. Vaginal anastomotic stenosis occurred in 63 % (5/8) of grafts. Self-expanding stents have shown preliminary suitability for the treatment of vaginal stenosis; 3 recipients developed severe acute rejection. The authors concluded that the interim results of this study demonstrated mid-term viability in 70 % of grafts. The LD UTx produced 2 livebirths and the DD UTx produced 1 livebirth. Moreover, these researchers stated that nulli-parous donors should be considered for donation; however, their potential needs further clarification. These investigators also observed 4 miscarriages in 2 recipients. This preliminary report rendered further evidence of the feasibility of UTx from both LDs and DDs.

An UpToDate review on "Uterus transplantation for absolute uterine factor infertility: Ethics, patient selection, and consent" (Brannstrom and Diaz-Garcia, 2021a) states that "Uterus transplantation is a complex, multi-step procedure for the treatment of absolute uterine factor infertility (AUFI). AUFI refers to infertility that is fully attributable to the uterus because of absence (congenital or surgical) or abnormalities (anatomic or functional) that prevent embryo implantation or completion of a pregnancy to term … As human uterus transplantation is in the beginning stages, the optimal inclusion and exclusion criteria for both donors and recipients are not yet known. Living donors and recipients undergo extensive testing to ensure medical and psychological appropriateness. The authors' protocol includes consultation by the following services: gynecology, transplantation surgery, psychology, clinical immunology, anesthesiology, internal medicine, and radiology".

Furthermore, an UpToDate review on "Uterus transplantation for absolute uterine factor infertility: Surgery, immunosuppression, and obstetric management" (Brannstrom and Diaz-Garcia, 2021b) states that "Uterus transplantation is an experimental procedure for the treatment of absolute uterine factor infertility (AUFI) … Uterus transplantation is the first type of organ or tissue transplantation that is only temporary and not intended to stay with the host for life. The uterus is removed when childbearing is deemed complete to limit the amount and duration of immunosuppressive medication, and thereby reduce the risk for immunosuppressive side effects such as nephrotoxicity and increased risk of certain malignancies. Major challenges anticipated at the time of the graft removal surgery are adhesions that have formed around the graft's vessels and the non-anatomic location of the vessels".

Jones and associates (2021) stated that following the diagnosis of AUFI, women may experience considerable psychological harm as a result of a loss of reproductive function and the realization of permanent and irreversible infertility. Adoption enables women with AUFI, and their partners, to experience social and legal parenthood, also often providing benefits for the adopted child. Surrogacy offers the opportunity to have genetically related offspring. Outcomes are generally positive in both surrogates and the children born as a result. Uterine transplantation is the only option to restore reproductive anatomy and functionality. These researchers noted that more than 70 UTx cases have now been undertaken and, following at least 18 live-births after successful procedures, UTx is now considered a feasible fertility-restoring treatment for women with AUFI. However, it is associated with considerable surgical and immunosuppressive-related risk and, based on cases performed so far, a greater than 25 % risk of unplanned hysterectomy.

Brannstrom and colleagues (2021) noted that women with AUFI, because of uterine absence, or the presence of a non-functional uterus, were regarded as being untreatable until 2014 when the 1st birth following UTx took place in Sweden. This proof-of-concept occurred in a woman with MRKH syndrome with congenital uterine absence, who received a uterus from a 61-year-old LD. Since then, several births after UTx have occurred in Sweden and subsequently in other countries, including both LD and DD transplants. A great majority of the recipients were women with MRKH syndrome. The safety and effectiveness of UTx can be determined only when a complete study cohort of transplanted women have reached the definitive endpoint of graft hysterectomy. The different outcomes of transplanted women included graft failure, as well as graft

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

survival with failure to achieve live-birth, or live-birth(s). Published data from a completed trial are not yet available. The results that these researchers had to rely on were reports of completed surgeries and interim outcomes that may be as early as a few months after surgery and up to several years after UTx. These investigators provided an update on all published clinical UTx data and major results, including live-births up to mid-2021. The interim results of a number of UTx studies have been published. LD UTx procedures have been reported from 4 European countries (Sweden, the Czech Republic, Germany, and Spain), 4 Asian nations (Saudi Arabia, India, China, and Lebanon), as well as some from the U.S. DD UTx procedures have been reported from Turkey, the Czech Republic, the U.S., and Brazil. To the authors' knowledge, there also exist unpublished UTx cases from some of the countries mentioned above and from at least 4 other countries (Serbia, France, Mexico, and Italy). They estimated that at least 80 UTx procedures have been carried out, resulting in more than 40 births. This study included only data from published, peer-reviewed, research papers. The results of 62 UTx cases showed an overall surgical success rate, as defined by a technically successful transplantation with a subsequent regular menstrual pattern, of 76 %. The success rates for LD and DD UTx procedures were 78 % and 64 %, respectively. The rate of serious post-surgical complications requiring invasive or radiological intervention was 18 % for LDs and 19 % for recipients. The cumulative live-birth rate in successful UTx procedures was estimated to be above 80 %; 24 births after UTx have been reported and the results showed a high rate of PTB, with an associated high proportion of respiratory distress syndrome. The authors concluded that UTx has proven to be a successful treatment for AUFI at several centers around the world. The modest success rate and the fairly high complication rate among LDs, indicated that further research and development under strict governance are needed before this option should be widely offered.

Furthermore, an UpToDate review on "Uterus transplantation for absolute uterine factor infertility: Surgery, immunosuppression, and obstetric management" (Brannstrom, and Diaz-Garcia, 2022) states that "Uterus transplantation is an experimental procedure for the treatment of absolute uterine factor infertility (AUFI). AUFI refers to infertility that is completely attributable to uterine absence (congenital or surgical) or an abnormality (anatomic or functional) that prevents embryo implantation or completion of pregnancy to term".

Vaginal Sildenafil for the Treatment of Female Infertility

Check et al (2004) examined if sildenafil improves endometrial thickness better than vaginal estradiol (E2) in women with a history of thin endometria. Women failing to attain an 8 mm endometrial thickness on either the oocyte retrieval cycle or their 1st frozen embryo transfer (ET) despite an oral graduated E2 regimen were treated again with graduated oral E2 and were also randomly assigned to vaginal sildenafil or vaginal E2 therapy. Endometrial thickness was compared between the groups. Neither vaginal E2 nor sildenafil significantly improved endometrial thickness or blood flow in the subsequent frozen ET-cycle. The authors concluded that these data failed to corroborate previous claims that 25-mg sildenafil 4 times daily intra-vaginally can improve endometrial thickness.

Zinger et al (2006) stated that vaginal sildenafil citrate has been shown to be useful in increasing endometrial thickness and achieving pregnancy in women with varied uterine disorders. However, it failed to demonstrate favorable results in the setting of Asherman's syndrome, a disease characterized by the presence of uterine synechiae. These investigators have successfully applied this treatment in 2 women noted to have inadequate endometrium after surgical resection of uterine synechiae. Both patients had a history of a post-partum uterine curettage with subsequent secondary infertility. Asherman's syndrome was surgically demonstrated and treated in both patients. Post-operatively, both patients were noted to have a thin endometrium and failed to conceive despite fertility treatment. Subsequently, these women achieved pregnancy in the 1st treatment cycle with vaginal sildenafil citrate. Using trans-vaginal ultrasound, endometrial thickness was noted to improve when sildenafil citrate was administered. It is suspected that this medication causes selective vasodilatation, resulting in improved endometrial development.

Malinova et al (2013) noted that evaluation of endometrial receptivity remains a challenge in clinical practice. Ultrasound evaluation of endometrial thickness and texture and measurement of uterine artery blood flow has been used for endometrial assessment. These researchers investigated the role of combination of sildenafil citrate and serophene on endometrial thickness, endometrial volume, endometrial FI and VFI on

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

angiohistogram, RI and PI to a. uterine on the day of hCG, in prediction of IUI outcome in infertile women. A total of 42 patients were selected randomly who had anovulatory infertility. In the sildenafil citrate plus serophene group (Group I), patients got 25 mg sildenafil citrate (Silden) vaginally and serophene 100 to 150 mg orally, and in serophene group (Group II), 100 to 150 mg of serophene was given orally. Mean endometrial thickness and endometrial volume was 11.8 +/- 2.6 v/s 10.2 +/- 2.8 and 5.2 +/- 1.4 v/s 3.6 +/- 1.8, respectively in group I and in group II ($p < 0.05$). There was significant decrease in PI and RI to a. uterina in group I. The authors concluded that combination of sildenafil citrate and serophene is an effective agent as a first-line of treatment for ovulation induction. This was a relatively small study, and its findings were confounded by the combined use of sildenafil and serophene.

Soliman and colleagues (2017) developed and characterized in-situ thermos-sensitive gels for the vaginal administration of sildenafil as a potential treatment of endometrial thinning occurring as a result of using clomiphene citrate for ovulation induction in women with type II eugonadotrophic anovulation. While sildenafil has shown promising results in the treatment of infertility in women, the lack of vaginal pharmaceutical preparation and the side effects associated with oral sildenafil limit its clinical effectiveness. Sildenafil citrate in-situ forming gels were prepared using different grades of Pluronic (PF-68 and PF-127). Muco-adhesive polymers as sodium alginate and hydroxyethyl cellulose were added to the gels in different concentrations and the effect on gel properties was studied. The formulations were evaluated in terms of viscosity, gelation temperature (Tsol-gel), muco-adhesion properties, and in-vitro drug release characteristics. Selected formulations were evaluated in women with clomiphene citrate failure due to thin endometrium (Clinicaltrial.gov identifier NCT02766725). The Tsol-gel decreased with increasing PF-127 concentration and it was modulated by addition of PF-68 to be within the acceptable range of 28 to 37 °C. Increasing Pluronic concentration increased gel viscosity and muco-adhesive force but decreased drug release rate. Clinical results showed that the in-situ sildenafil vaginal gel significantly increased endometrial thickness and uterine blood flow with no reported side effects. Further, these results were achieved at lower frequency and duration of drug administration. The authors concluded that sildenafil thermos-sensitive vaginal gels might result in improved potential of pregnancy in anovulatory patients with clomiphene citrate failure due to thin endometrium. These preliminary findings need to be validated by well-designed studies.

Furthermore, an UpToDate review on “Overview of treatment of female infertility” (Kuohung and Hornstein, 2017) does not mention sildenafil as a management tool.

Vasodilators for Women Undergoing Fertility Treatment

Gutarra-Vilchez et al (2014) noted that since 1978, when Patrick Steptoe and Robert Edwards achieved the birth of the first test tube baby, ARTs have been refined and improved. However, the rate of successful pregnancies brought to term has barely increased. Thus, closer evaluation of the interventions is needed along with working towards improving uterus receptivity. Vasodilators have been proposed to increase endometrial receptivity, thicken the endometrium and favor uterine relaxation, all of which could improve uterine receptivity and enhance the chances for successful assisted pregnancies. In a Cochrane review, these investigators evaluated the safety and effectiveness of vasodilators in women undergoing fertility treatment. The authors concluded that

I. evidence was insufficient to show that vasodilators increased the live-birth rate in women undergoing fertility treatment;
II. low-quality evidence suggested that vasodilators may increase clinical pregnancy rates in comparison with placebo or no treatment, and
III. evidence was insufficient to show whether any particular vasodilator, administered alone or in combination with other active medications, was superior, and evidence was insufficient to allow the review authors to reach any conclusions regarding adverse effects.

I. evidence was insufficient to show that vasodilators increased the live-birth rate in women undergoing fertility treatment;
II. low-quality evidence suggested that vasodilators may increase clinical pregnancy rates in comparison with placebo or no treatment, and

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

III. evidence was insufficient to show whether any particular vasodilator, administered alone or in combination with other active medications, was superior, and evidence was insufficient to allow the review authors to reach any conclusions regarding adverse effects.

They stated that adequately powered studies are needed so that each treatment can be evaluated more accurately.

In a Cochrane review, Gutarra-Vilchez and colleagues (2018) examined the safety and effectiveness of vasodilators in women undergoing fertility treatment. These investigators searched the following electronic databases, trial registers, and websites: the Cochrane Gynaecology and Fertility Group (CGF) Specialized Register of controlled trials, the Cochrane Central Register of Controlled Trials, via the Cochrane Register of Studies Online (CRSO), Medline, Embase, PsycINFO, the Cumulative Index to Nursing and Allied Health Literature (CINAHL), Web of Knowledge, the Open System for Information on Grey Literature in Europe (OpenSIGLE), the Latin American and Caribbean Health Science Information Database (LILACS), clinical trial registries, and the reference lists of relevant articles. They conducted the search in October 2017 and applied no language restrictions; RCTs comparing vasodilators alone or in combination with other treatments versus placebo or no treatment or versus other agents in women undergoing fertility treatment were selected for analysis. Four review authors independently selected studies, assessed risk of bias, extracted data, and calculated RRs. They combined study data using a fixed-effect model and assessed evidence quality using GRADE methods. The primary outcomes were live-birth or ongoing pregnancy and vasodilator side effects. Secondary outcomes included clinical pregnancy, endometrial thickness, multiple pregnancy, miscarriage, and ectopic pregnancy. These investigators included 15 studies with a total of 1,326 women. All included studies compared a vasodilator versus placebo or no treatment. They judged most of these studies as having unclear risk of bias. Overall, the quality of evidence was low-to-moderate for most outcomes. The main limitations were imprecision due to low numbers of events and participants and risk of bias due to unclear methods of randomization. Vasodilators probably made little or no difference in rates of live-birth compared with placebo or no treatment (RR 1.18, 95 % CI 0.83 to 1.69; 3 RCTs; n = 350; $I^2 = 0$ %; moderate-quality evidence) but probably increase overall rates of side effects including headache and tachycardia (RR 2.35, 95 % CI: 1.51 to 3.66; 4 RCTs; n = 418; $I^2 = 0$ %; moderate-quality evidence). Evidence suggested that if 236 per 1,000 women achieved live-birth with placebo or no treatment, then between 196 and 398 per 1,000 will do so with the use of vasodilators. Compared with placebo or no treatment, vasodilators may slightly improve clinical pregnancy rates (RR 1.45, 95 % CI: 1.19 to 1.77; 11 RCTs; n = 1,054; $I^2 = 6$ %; low-quality evidence). Vasodilators probably made little or no difference in rates of multiple gestation (RR 1.15, 95 % CI: 0.55 to 2.42; 3 RCTs; n = 370; $I^2 = 0$ %; low-quality evidence), miscarriage (RR 0.83, 95 % CI 0.37 to 1.86; 3 RCTs; n = 350; $I^2 = 0$ %; low-quality evidence), or ectopic pregnancy (RR 1.48, 95 % CI: 0.25 to 8.69; 2 RCTs; n = 250; $I^2 = 5$ %; low-quality evidence). All studies found benefit for endometrial thickening, but reported effects varied ($I^2 = 92$ %) and ranged from a MD of 0.80 higher (95 % CI: 0.18 to 1.42) to 3.57 higher (95 % CI: 3.01 to 4.13) with very low-quality evidence, so these researchers were uncertain how to interpret these results. The authors concluded that evidence was insufficient to show whether vasodilators increase the live-birth rate in women undergoing fertility treatment. However, low-quality evidence suggested that vasodilators may slightly increase clinical pregnancy rates. Moderate-quality evidence showed that vasodilators increased overall side effects in comparison with placebo or no treatment. These investigators stated that adequately powered studies are needed so that each treatment can be evaluated more accurately.

EMMA (Endometrial Microbiome Metagenomic Analysis) and ALICE (Analysis of Infectious Chronic Endometritis)

Endometrial Microbiome Metagenomic Analysis (EMMA) (Igenomix) is a test to evaluate the microbial composition of the endometrium in women undergoing assisted reproductive treatments or experiencing recurrent implantation failure or pregnancy loss. EMMA uses advanced molecular techniques to identify and quantify bacteria present in the endometrium, distinguishing between beneficial species such as Lactobacillus, which support a healthy uterine environment, and pathogenic bacteria that may contribute to chronic endometritis and infertility. By providing a detailed profile of the endometrial microbiome, EMMA is intended to assist clinicians to tailor treatments, such as targeted antibiotics or probiotics, to restore microbial

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

balance and improve conditions for embryo implantation. The test is often performed alongside Analysis of Infectious Chronic Endometritis (ALICE), which specifically screens for chronic endometritis.

Iwami et al. (2025) investigated whether personalized interventions guided by EMMA and ALICE could improve pregnancy outcomes and reduce time to conception in women with recurrent implantation failure or recurrent pregnancy loss. Conducted across 14 IVF centers in Japan, the prospective cohort included 527 women under 42 years old. Endometrial biopsies were analyzed using next-generation sequencing to classify patients into normal, abnormal (dysbiosis), or mild dysbiosis/ultralow biomass groups. Treatment was tailored to microbiota profiles: no intervention for normal cases, antibiotics followed by probiotics for abnormal cases, and probiotics alone for mild or ultralow profiles. Results showed that while overall ongoing pregnancy rates were similar across groups, Kaplan–Meier analysis revealed that women with dysbiosis who received targeted therapy achieved pregnancy significantly earlier than untreated groups, particularly within six to twelve months. Subgroup analysis indicated that patients with dysbiosis but negative ALICE results responded best to treatment, and age emerged as the only significant predictor of cumulative pregnancy rates, with older women benefiting most from microbiota correction. The study reported no severe adverse events, only minor antibiotic-related rashes. Limitations included potential contamination during sampling, inability of sequencing to distinguish live from dead bacteria, exclusion of fungi and viruses, and uncertainty about the pathogenic role of some bacteria found in healthy women.

Glossary of Terms

Table: Glossary of Terms

Term	Definition
Agglutination	clumping spermatozoa in the ejaculate
Amenorrhea	condition in which a woman does not have menstrual periods
Aneuploidy	abnormal number of chromosomes in a cell. The majority of embryos with aneuploidies are not compatible with life.
Anovulation	condition in which a woman does not ovulate or ovulates rarely
Anti-sperm antibody test	a test used to determine if antibodies on the surface of sperm are interfering with the ability of sperm to move, penetrate the cervical mucus, or fertilize and egg
Artificial insemination	general term for the procedure in which sperm are inserted directly into a woman's cervix, fallopian tubes, or uterus
Aspermia	lack of external ejaculation
Assisted hatching	ART procedure in which the zona pellucida of an embryo is either thinned or perforated by chemical, mechanical or laser methods
Assisted reproduction technology (ART)	interventions that include the in vitro handling of both human oocytes and sperm or of embryos for the purpose of reproduction. This includes, but is not limited to, IVF and embryo transfer (ET), intracytoplasmic sperm injection (ICSI), embryo biopsy, preimplantation genetic testing (PGT), assisted hatching, gamete intrafallopian transfer (GIFT), zygote

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

	intrafallopian transfer (ZIFT), gamete and embryo cryopreservation, semen, oocyte and embryo donation, and gestational carrier cycles
Azoospermia	absence of spermatozoa in the ejaculate
Blastocyst	stage of preimplantation embryo development that occurs around day 5–6 after insemination or ICSI
Cleavage stage embryos	the early stage of embryo development where the fertilized egg is rapidly dividing into smaller cells, but the overall size of the embryo remains relatively constant. This stage typically occurs on days 2-3 after fertilization
Cryopreservation	process of slow freezing or vitrification to preserve biological material (e.g., gametes, zygotes, cleavage-stage embryos, blastocysts or gonadal tissue) at extreme low temperature for future use
Cryptorchidism	testis not in scrotal position within the neonatal period and, up to but not limited to, 1 year post birth. If the testis has not descended into the scrotum, this condition can cause primary testicular failure and increased risk of testicular cancer development
Donor insemination	process of placing laboratory processed sperm or semen from a man into the reproductive tract of a woman who is not his intimate sexual partner, for the purpose of initiating a pregnancy
Embryo	biological organism resulting from the development of the zygote, until eight completed weeks after fertilization, equivalent to 10 weeks of gestational age
Embryo bank	repository of cryopreserved embryos stored for future use
Embryo transfer cycle	an ART cycle in which one or more fresh or frozen/thawed embryos at cleavage or blastocyst stage are transferred into the uterus or Fallopian tube
Follicle stimulating hormone (FSH)	hormone produced in the pituitary gland that causes cells in the ovaries to grow
Frozen-thawed embryo transfer (FET) cycle	an ART procedure in which cycle monitoring is carried out with the intention of transferring to a woman, frozen/thawed or vitrified/warmed embryo(s)/blastocyst(s). Note: A FET cycle is initiated when specific medication is provided or cycle monitoring is started in the female recipient with the intention to transfer an embryo
Gamete intrafallopian transfer (GIFT)	an assisted reproductive technique that involves removing sperm and eggs, mixing them together and placing them into the fallopian tubes
Gonadotropin releasing hormone (Gn-RH)	hormone produced in the hypothalamus that is involved in triggering ovulation
Gonadotropin releasing hormone agonists and antagonists (GnRH agonist)	synthetic hormones that perform the same function as natural Gn-RH

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

Human chorionic gonadotropin (hCG)	hormone that can be used to trigger ovulation
Hypogonadotropic hypogonadism	also known as secondary hypogonadism, is a condition where the hypothalamus or pituitary gland fails to produce enough hormones that signal to the testes or ovaries to produce sex hormones like testosterone or estrogen.
Hysterosalpinogram	X-ray which involves injecting dye through the cervix into the uterus to determine if the fallopian tubes are open and the uterine cavity is normal
Hysteroscopy	procedure in which a thin, telescope-like instrument is inserted through the cervix into the uterus, allowing the doctor to see and photograph the area, and correct problems if needed
Iatrogenic infertility	impairment of fertility caused by surgery, radiation, chemotherapy, or other medical treatment affecting reproductive organs or processes (Huang et al, 2025)
Infertility	generally defined as an inability to get pregnant after a year of unprotected sexual intercourse or due to an impairment of a person's capacity to reproduce either as an individual or with his/her partner
Intracytoplasmic sperm injection (ICSI)	laboratory procedure in which sperm and eggs are retrieved from both partners. A single spermatozoon (sperm) is injected directly into an oocyte cytoplasm (egg), then the fertilized egg is implanted into the woman's uterus
Intra-uterine insemination (IUI)	artificial insemination technique in which sperm are put directly into a woman's uterus at the time they are ovulating to attempt a pregnancy
In vitro fertilization (IVF)	assisted reproductive technique that involves removing sperm and eggs, fertilizing them in a laboratory, then placing a fertilized egg in the uterus
Luteinizing hormone	hormone that triggers ovulation
Luteal phase	second half of the menstrual cycle
Morphology	size and shape of sperm
Motility	ability of sperm to move by themselves
Oligospermia	when a man has too few sperm to fertilize an egg normally
Ovulation	when the ovaries release a mature egg that is ready for fertilization
Oocyte	a developing egg
Oocyte donation	use of oocytes from an egg donor for reproductive purposes
Ovum	a mature egg
Preimplantation genetic testing (PGT)	test performed to analyze the DNA from oocytes (polar bodies) or embryos (cleavage stage or blastocyst) for HLA-typing or for determining genetic abnormalities. These include: PGT for aneuploidies (PGT-A); PGT for monogenic/single gene defects

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

	(PGT-M); and PGT for chromosomal structural rearrangements (PGT-SR)
Zona pellucida	the glycoprotein coat surrounding the oocyte
Zygote	fertilized egg
Zygote intrafallopian transfer (ZIFT)	assisted reproductive technique similar to IVF that involves removing sperm and eggs, combining them outside the body, and inserting fertilized eggs into the fallopian tubes

Source: Adapted from American Society for Reproductive Medicine (ASRM), 2017; WebMD, 2025

Appendix

Laboratory Services

The following numbers of laboratory services per cycle are considered medically necessary.

Table: Laboratory Services per Cycle

	Natural monitoring	Clomid monitoring	Clomid IUI	Inj Mon Cycle	Inj IUI	IVF	GIFT	FET Code	PM
Transvaginal ultrasound	2	6	6	8	10				
Estradiol	2	6	6	8	10	10	10	10	
FSH	2	6	6	8	10	10	10	10	
LH	2	6	6	8	10	10	10	10	
Progesterone	2Footnotes for progesterone*	2Footnotes for progesterone*	2Footnotes for progesterone*	8	10	10	10	10	3
hCG	2	2	2	2	2	2	2	2	3

Key: IUI: intra-uterine insemination; Inj: injection; Mon: monthly; IVF: in-vitro fertilization; GIFT: gamete intra-fallopian transfer; FET: frozen embryo transfer; PM: pregnancy monitoring; FSH: follicle stimulating hormone; LH: luteinizing hormone; hCG: human chorionic gonadotropin.

Footnotes for progesterone***Note**: More than 2 progesterone measurements may be medically necessary for infertile women with irregular and prolonged menstrual cycles. For infertile women with regular menstrual cycles, a mid-luteal serum progesterone measurement (day 21 of a 28-day cycle) is considered medically necessary. For infertile women with irregular menstrual cycles, this test would need to be repeated at the mid-luteal phase and weekly thereafter until the next menstrual cycle starts.

Table: Female Gonadotropin Injectable Vial Management

Medication	Standard	FDA-recommended Dosing

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

	Limit	
Gonadotropins/Menotropins (Initial CycleFootnotes for Initial Cycle*)		
Follitropin alfa (Gonal-f) 450 unit vial	10 vials per 28 days	Dosing must be individualized for each person. Ovulation induction: First cycle starting dose is 75 IU daily with incremental adjustment of up to 37.5 IU after 14 days. If necessary, increase dose by same magnitude every 7 days (in general up to 35 days of treatment). The initial dose in subsequent cycles is individualized based on prior response. Maximum daily dose of 300 IU. ART: Starting dose of 150 IU per day until adequate follicular development (up to 10 days of therapy in most cases). In persons whose endogenous gonadotropin levels are suppressed, starting dose of 225 IU daily. Consider dose adjustment after 5 days based on response (adjust no more frequently than every 3-5 days by no more than 75-150 IU). Maximum daily dose of 450 IU.Source: Merck & Co., 2020
Follitropin alfa (Gonal-f) 1050 unit vial	6 vials per 28 days	
Follitropin alfa (Gonal-f RFF) 300/0.5ml pen injector	15 cartridges per 28 days	Ovulation induction: first cycle starting dose of 75 IU daily for 14 days. If indicated by the ovarian response after the initial 14 days, adjust dose by up to 37.5 IU every 7 days. Continue until adequate ovarian response (up to 35 days). The initial dose in subsequent cycles is based on prior response. Maximum daily dose of 300 IU. ART: starting dose of 150 IU per day until adequate follicular development (up to 10 days of therapy in most cases). In patients whose endogenous gonadotropin levels are suppressed, starting dose of 150 IU per day (if under 35 years old) or 225 IU per day (if 35 years old or older). Adjust dose after 5 days based on response (adjust no more frequently than every 3-5 days by no more than 75-150 IU). Maximum daily dose of 450 IU. Source: Merck & Co., 2020
Follitropin alfa (Gonal-f RFF) 450/0.75ml pen injector	10 cartridges per 28 days	
Follitropin alfa (Gonal-f RFF) 900/1.5ml pen injector	7 cartridges per 28 days	
Follitropin alfa (Gonal-f RFF) 75 unit vial	60 vials per 28 days	
Follitropin beta (Follistim AQ) 150 unit cartridge	30 cartridges per 28 days	Ovulation induction: starting dose of 50 IU daily for at least 7 days, increased by 25 to 50 IU at weekly intervals until adequate ovarian response. Maximum daily dose of 250 IU. Controlled ovarian stimulation as part of an in vitro fertilization or intracytoplasmic sperm injection cycle: starting dose of 200 IU daily for at least 7 days with subsequent doses adjusted up or down based upon ovarian response. Maximum daily dose of 500 IU.
Follitropin beta (Follistim AQ) 300 unit cartridge	15 cartridges per 28 days	
Follitropin beta (Follistim AQ) 600 unit cartridge	10 cartridges per 28 days	

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

Follitropin beta (Follistim AQ) 900 unit cartridge	7 cartridges per 28 days	Source: Merck & Co., 2020
Menotropins for injection (Menopur) 75 IU vials	For ART: Do not administer daily doses of Menopur or Menopur in combination with Bravelle that exceed 450 IU. Therapy should not exceed 20 days.	Development of Multiple Follicles and Pregnancy in Ovulatory Women as Part of an ART Cycle: The recommended dosing scheme for persons undergoing IVF follows a stepwise approach and is individualized for each woman. The recommended initial dose of Menopur for women who have received a GnRH agonist for pituitary suppression is 225 IU. Menopur may be co-administered with Bravelle and the total initial dose when the products are combined should not exceed 225 IU (150 IU of Menopur and 75 IU of Bravelle or 75 IU of Menopur and 150 IU of Bravelle). Source: Ferring Pharmaceuticals, 2018
hCG subcutaneous injections (Initial CycleFootnotes for Initial Cycle*)		
Choriogonadotropin alfa (Ovidrel) prefilled syringe 250 mcg	1 prefilled syringe	Undergoing Ovulation Induction (OI): 250 µg should be administered one day following the last dose of the follicle stimulating agent. Undergoing Assisted Reproductive Technologies (ART): 250 µg should be administered one day following the last dose of the follicle stimulating agent.
hCG intramuscular injections (Initial CycleFootnotes for Initial Cycle*)		
Chorionic gonadotropin (Pregnyl) 5,000 U, 10,000 U	1 vial	Induction of ovulation and pregnancy in the anovulatory, infertile woman in whom the cause of anovulation is secondary and not due to primary ovarian failure and who has been appropriately pretreated with human menotropins. (See prescribing information for menotropins for dosage and administration for that drug product.): 5,000 to 10,000 USP units 1 day following the last dose of menotropins. (A dosage of 10,000 USP units is recommended in the labeling for menotropins.) Source: Merck & Co., 2015
Chorianic gonadotropin (Novarel) 5,000 U, 10,000 U	1 vial	Induction of ovulation and pregnancy in the anovulatory, infertile woman in whom the cause of anovulation is secondary and not due to primary ovarian failure and who has been appropriately pretreated with human menotropins (See prescribing information

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

		for menotropins for dosage and administration for that drug product): 5,000 to 10,000 USP Units one day following the last dose of menotropins. (A dosage of 10,000 USP Units is recommended in the labeling for menotropins). Source: Ferring Pharmaceuticals, 2018
GnRH Antagonists (Initial CycleFootnotes for Initial Cycle*)		
Ganirelix acetate, 250 ug/0.5 ml (prefilled syringe)		Inhibition of premature LH surges in women undergoing controlled ovarian hyperstimulation: After initiating FSH therapy on Day 2 or 3 of the cycle, Ganirelix Acetate Injection 250 mcg may be administered subcutaneously once daily during the mid to late portion of the follicular phase. By taking advantage of endogenous pituitary FSH secretion, the requirement for exogenously administered FSH may be reduced. Treatment with Ganirelix Acetate should be continued daily until the day of hCG administration. Source: Merck & Co., 2020
Cetrorelix acetate (Cetrotide), 0.25 mg		Inhibition of premature LH surges in women undergoing controlled ovarian stimulation: Cetrotide 0.25 mg may be administered subcutaneously once daily during the early- to mid-follicular phase. Cetrotide 0.25 mg is administered on either stimulation day 5 (morning or evening) or day 6 (morning) and continued daily until the day of hCG administration. Source: EMD Serono, 2018

Key: ART: advanced reproductive technology; BMI: body mass index; FSH: follicle stimulating hormone; HCG: human chorionic gonadotropin; IU: international units; MDV: multiple dose vial; PCOS: polycystic ovarian syndrome; PFS: prefilled syringe; U: units.

Notes:

Footnotes for Initial Cycle* Refills based upon documentation in cycle sheets.

Some plans exclude infertility services for ovarian failure; please check benefit plan descriptions.

Table: Male Gonadotropin Injectable Vial Management:

Medication	Strength	FDA-recommended Dosing	Length of approval
FollitropinsFootnotes for Follitropins*			

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

Follitropin alfa (Gonal-f)	450 unit or 1050 unit vial	Male hypogonadotropic hypogonadism: after normal serum testosterone levels are reached, the recommended dose of Gonal-f is 150 IU administered subcutaneously three times per week. and the recommended dose of hCG is 1,000 USP units (or the dose required to maintain serum testosterone levels within the normal range) three times a week. The lowest dose of Gonal-f which induces spermatogenesis should be utilized. Maximum dose of Gonal-f is 300 IU three times per week. Gonal-f may need to be administered for up to 18 months to achieve adequate spermatogenesis. Source: EMD Serono, 2018	12 months for hypogonadotropic hypogonadism
Follitropin beta (Follistim AQ)	150, 300, 600, 900 unit multi-dose cartridges	Induction of spermatogenesis in men with primary and secondary hypogonadotropic hypogonadism in whom the cause of infertility is not due to primary testicular failure: after normalization of serum testosterone levels, administer 450 IU per week (225 IU twice weekly or 150 IU three times weekly) of Follistim AQ Cartridge subcutaneously with the same pre-treatment hCG dose used to normalize testosterone levels. Source: Merck & Co., 2020	12 months for hypogonadotropic hypogonadism
hCG intramuscular injections			
Examples: Pregnyl, Novarel, hCG	10,000 unit 5,000 unit	Varies: Prepubertal cryptorchidism not due to anatomical obstruction (Therapy is usually instituted in children between the ages of 4 and 9): ▪ 4,000 USP Units three times weekly for three weeks. ▪ 5,000 USP Units every second day for four injections. ▪ 15 injections of 500 to 1,000 USP Units over a period of six weeks. ▪ 500 USP Units three times weekly for four to six weeks. If this course of treatment is not successful, another is begun one month later, giving 1,000 USP Units per injection. Selected cases of hypogonadotropic hypogonadism in males: ▪ 500 to 1,000 USP Units three times a week for three weeks, followed by the	6 months for prepubertal cryptorchidism; 12 months for hypogonadotropic hypogonadism

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

		same dose twice a week for three weeks. ▪ 4,000 USP Units three times weekly for six to nine months, following which the dosage may be reduced to 2,000 USP Units three times weekly for an additional three months. Source: Ferring Pharmaceuticals, 2018; Merck & Co., 2015	

Footnotes for follitropin* Concomitant recombinant follitropin and human chorionic gonadotropin therapy should be continued for at least 3 to 4 months before improvement in spermatogenesis can be expected.

Definitions

For purposes of this policy, the following definitions will be used:

Classification of ovulatory disorders:

Anovulation and oligo-ovulation are ovulatory disorders that are estimated to cause 21 % of female fertility problems. The World Health Organization classifies ovulation disorders into 3 groups.

- Group I: hypothalamic pituitary failure (hypothalamic amenorrhea or hypogonadotrophic hypogonadism).
- Group II: hypothalamic pituitary dysfunction (predominately polycystic ovary syndrome).
- Group III: ovarian failure.

Embryo Quality in ART Cycles:

An embryo is considered to be of reasonable quality (grade B or its equivalent) if it has less than 50 % fragmentation (see, e.g., Ebner, et al., 2001; Rhenman, et al., 2015; Shaw-Jackson, et al., 2013)..

Fertilization Rates in IVF Cycles:

Fertilization rates are considered poor if IVF cycles result in less than 50 % fertilization.

Ovarian Reserve in Response to Gonadotropin Stimulation:

Ovarian reserve is considered normal if 3 or more follicles develop and estrogen levels are greater than 500 mIU/ml following ovarian hyperstimulation with gonadotropins. Diminished ovarian reserve is indicated by peak estrogen levels less than 500 mIU/ml or fewer than 3 mature follicles are available at the time of stimulation and retrieval.

Semen Quality and Quantity:

Deficits in semen quantity are considered severe if there are less than 10 million total motile sperm (TMS) per ejaculate (unwashed specimen) or less than 3 million total motile sperm (washed specimen) on 2 separate occasions at least 2 weeks apart. TMS is calculated by multiplying the volume of ejaculate by the concentration (million sperm/ml) by the motility (% moving) (Hajder et al, 2016). Deficits in semen quality are considered severe if there are less than 4 % normal forms using Kruger strict morphology. In men who have met the definition of severe male factor infertility with abnormal sperm quality or quantity more than 2

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

weeks apart in the past; and then had a successful varicocelectomy resulting in normal sperm quality or quantity, ICSI is considered not medically necessary.

Semen Analysis: World Health Organization Reference Values

- pH: 7.2 to 8.0
- Sperm concentration: 16 million spermatozoa per ml or more
- Sperm morphology (percentage of normal forms): 4 % or more
- Semen volume: 1.4 ml or more
- Total motility (percentage of progressive motility and non-progressive motility): 42 % or more motile or 30 % or more with progressive motility
- Total sperm number: 39 million spermatozoa per ejaculate or more
- Immotile sperm percentage: 20 %
- Vitality: 54 % or more live spermatozoa

Source: Björndahl et al, 2022; Chung et al, 2023; Dhumal et al, 2021

Stages of Endometriosis

Surgically, endometriosis can be staged I–IV (Revised Classification of the American Society of Reproductive Medicine). The various stages show these findings:

Stage I (Minimal) -

Findings restricted to only superficial lesions and possibly a few filmy adhesions. Minimal disease is characterized by isolated implants and no significant adhesions.

Stage II (Mild) -

In addition, some deep lesions are present in the cul-de-sac. Mild endometriosis consists of superficial implants that are less than 5 cm in aggregate and are scattered on the peritoneum and ovaries. No significant adhesions are present.

Stage III (Moderate) -

As above, plus presence of endometriomas on the ovary and more adhesions. Moderate disease exhibits multiple implants, both superficial and deeply invasive. Peritubal and periovarian adhesions may be evident.

Stage IV (Severe) -

As above, plus large endometriomas, extensive adhesions. Severe disease is characterized by multiple superficial and deep implants, including large ovarian endometriomas. Filmy and dense adhesions are usually present.

Source: Adapted from ASRM, 1997; Schenken, 2025

References

The above policy is based on the following references:

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

1. Aboulghar M, Evers JH, Al-Inany H. Intra-venous albumin for preventing severe ovarian hyperstimulation syndrome. Cochrane Database Syst Rev. 2002:(2):CD001302.
2. Achilli C, Duran-Retamal M, Saab W, et al. The role of immunotherapy in in vitro fertilization and recurrent pregnancy loss: A systematic review and meta-analysis. Fertil Steril. 2018;110(6):1089-1100.
3. ACOG Committee on Practice Bulletins-Gynecology. ACOG Practice Bulletin. Clinical management guidelines for obstetrician-gynecologists number 34, February 2002.
4. Agarwal A, Deepinder F, Cocuzza M, et al. Efficacy of varicocelectomy in improving semen parameters: New meta-analytical approach. Urology. 2007;70(3):532-538.
5. Ahmad G, Watson A, Vandekerckhove P, Lilford R. Techniques for pelvic surgery in subfertility. Cochrane Database Syst Rev. 2006;(2):CD000221.
6. Ahmadian S, Mahdipour M, Pazhang M, et al. Effectiveness of stem cell therapy in treatment of ovarian disorders and female infertility: A systematic review. Curr Stem Cell Res Ther. 2020;15(2):173-186.
7. Albuquerque LE, Saconato H, Maciel MC. Depot versus daily administration of gonadotrophin releasing hormone agonist protocols for pituitary desensitization in assisted reproduction cycles. Cochrane Database Syst Rev. 2005;(1):CD002808.
8. Al-Inany H. Female infertility. In: BMJ Clinical Evidence. London, UK: BMJ Publishing Group; April 2004.
9. Al-Inany HG, Youssef M, Aboulghar M. Gonadotrophin-releasing hormone antagonists for assisted conception. Cochrane Database Syst Rev. 2011;(5):CD001750.
10. Al-Zebeidi J, Agdi M, Lary S, et al. Effect of empiric intravenous intralipid therapy on pregnancy outcome in women with unexplained recurrent implantation failure undergoing intracytoplasmic sperm injection-embryo transfer cycle: A randomized controlled trial. Gynecol Endocrinol. 2020;36(2):131-134.
11. American Association of Clinical Endocrinologists (AACE) Hypogonadism Task Force. American Association of Clinical Endocrinologists medical guidelines for clinical practice for the evaluation and treatment of hypogonadism in adult male patients--2002 update. Endocr Pract. 2002;8(6):439-456.
12. American Association of Clinical Endocrinologists (AACE). AACE medical guidelines for clinical practice for management of menopause. Endocrine Pract. 1999;5(6):354-366.
13. American Association of Clinical Endocrinologists. Medical guidelines for clinical practice for the evaluation and treatment of hypogonadism in adult male patients – 2002 Update. Endocr Pract. 2002:8:439-456.
14. American College of Obstetricians and Gynecologists (ACOG), Committee on Obstetric Practice. Ovarian reserve testing. ACOG Committee Opinion No. 618. Obstet Gynecol. 2015;125(1):268-273.
15. American College of Obstetricians and Gynecologists (ACOG). Infertility. ACOG Technical Bulletin No. 125. Washington, DC: ACOG; 1989.
16. American College of Obstetricians and Gynecologists (ACOG). Male infertility. ACOG Technical Bulletin No. 142. Washington, DC: ACOG; 1990.
17. American College of Obstetricians and Gynecologists (ACOG). Polycystic ovarian syndrome. ACOG Practice Bulletin No. 41. Washington, DC: ACOG; December 2002.
18. American College of Obstetricians and Gynecologists. ACOG practice bulletin no. 105: Bariatric surgery and pregnancy. Obstet Gynecol. 2009;113(6):1405-1413.
19. American College of Obstetricians and Gynecologists. ACOG Technology Assessment in Obstetrics and Gynecology No. 5: Sonohysterography. Obstet Gynecol. 2008;112(6):1467-1469.
20. American Society for Reproductive Medicine (ASRM). Definition of infertility: A committee opinion. Washington, DC: ASRM; 2023.
21. American Society for Reproductive Medicine (ASRM). The international glossary on infertility and fertility care (2017). Washington, DC: ASRM; 2017. Available at: https://www.asrm.org. Accessed June 5, 2025.
22. American Society for Reproductive Medicine (ASRM) and Society for Assisted Reproductive Technology (SART). Blastocyst transfer and production in clinically assisted reproduction. A Practice Committee Report. Birmingham, AL: ASRM; January 2001 (reviewed May 2003).
23. American Society for Reproductive Medicine (ASRM) and the Society for Assisted Reproductive Technology (SART). Preimplantation genetic diagnosis. Committee Opinion. Birmingham, AL: ASRM; June 2001.

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

24. American Society for Reproductive Medicine (ASRM) Practice Committee. Fertil Steril. 2003;80(5):1302-1308.
25. American Society for Reproductive Medicine (ASRM) Practice Committee. Salpingectomy for hydrosalpinx prior to IVF. Committee Report. Birmingham, AL: ASRM; July 2001 (reviewed May 2003).
26. American Society for Reproductive Medicine (ASRM), Ovarian tissue and oocyte cryopreservation. Practice Committee of the American Society for Reproductive Medicine. Fertil Steril. 2004;82:993-998.
27. American Society for Reproductive Medicine (ASRM). 2002 guidelines for gamete and embryo donation. ASRM Guidelines. Birmingham, AL: ASRM; June 2002.
28. American Society for Reproductive Medicine (ASRM). Aging and infertility in women. Committee Opinion. Birmingham, AL: ASRM; January 2002.
29. American Society for Reproductive Medicine (ASRM). Antiphospholipid antibodies do not affect IVF success. Committee Opinion. Birmingham, AL: ASRM; October 1999.
30. American Society for Reproductive Medicine (ASRM). Effectiveness and treatment for unexplained infertility. Educational Bulletin. Birmingham, AL: ASRM; September 2000 (reviewed May 2003).
31. American Society for Reproductive Medicine (ASRM). Electroejaculation (EEJ). Committee Opinion. Birmingham, AL: ASRM; August 18, 1995.
32. American Society for Reproductive Medicine (ASRM). Endometriosis and infertility. Educational Bulletin. Fertil Steril. 2004;81:1441-1446.
33. American Society for Reproductive Medicine (ASRM). Evidence-based treatments for couples with unexplained infertility: A guideline. Fertil Steril. 2020;113(2):305-322.
34. American Society for Reproductive Medicine (ASRM). Fact Sheet: Intracytoplasmic Sperm Injection (ICSI). Birmingham, AL: ASRM, 1996-1997.
35. American Society for Reproductive Medicine (ASRM). Guidelines for the provision of infertility dervices. ASRM Guidelines. Birmingham, AL: ASRM; July 1996.
36. American Society for Reproductive Medicine (ASRM). Guidelines on number of embryos transferred. Fertil Steril. 2004;82:773-774.
37. American Society for Reproductive Medicine (ASRM). Intracytoplasmic sperm injection (ICSI). Practice Committee Statement. Birmingham, AL: ASRM; November 5, 1994.
38. American Society for Reproductive Medicine (ASRM). Intravenous immunoglobulin (IVIG) and recurrent spontaneous pregnancy loss. Committee Opinion. Birmingham, AL: ASRM; October 1998.
39. American Society for Reproductive Medicine (ASRM). Microsurgical epididymal sperm aspiration (MESA). Practice Committee Statement. Birmingham, AL: ASRM; October 30, 1992.
40. American Society for Reproductive Medicine (ASRM). Multiple pregnancy associated with infertility therapy. Educational Bulletin. Birmingham, AL: ASRM; November 2000 (reviewed May 2003).
41. American Society for Reproductive Medicine (ASRM). Myomas and reproductive function. Educational Bulletin. Birmingham, AL: ASRM; November 2001 (reviewed May 2003).
42. American Society for Reproductive Medicine (ASRM). New techniques for sperm acquisition in obstructive azoospermia. Technical Bulletin. Birmingham, AL: ASRM; August 1999.
43. American Society for Reproductive Medicine (ASRM). Optimal evaluation of the infertile female. Committee Opinion. Birmingham, AL: ASRM; June 2000 (reviewed May 2003).
44. American Society for Reproductive Medicine (ASRM). Ovarian hyperstimulation syndrome. Educational Bulletin. Fertil Steril. 2003;80:1309-1314.
45. American Society for Reproductive Medicine (ASRM). Repetitive oocyte donation. Committee Opinion. Birmingham, AL: ASRM; November 2000 (reviewed May 2003).
46. American Society for Reproductive Medicine (ASRM). Round spermatid nucleus injection (ROSNI). Fertil Steril. 2003;80(3):687-689.
47. American Society for Reproductive Medicine (ASRM). Revised American Society for Reproductive Medicine classification of endometriosis: 1996. Fertil Steril. 1997;67(5):817-821.
48. American Society for Reproductive Medicine (ASRM). Smoking and infertility. Educational Bulletin. Fertil Steril. 2004;81:1181-1186.
49. American Society for Reproductive Medicine (ASRM). The role of assisted hatching in IVF: A review of the literature. Committee Opinion. Birmingham, AL: ASRM; August 2000 (reviewed May 2003).
50. American Society for Reproductive Medicine (ASRM). Use of insulin sensitizing agents in the treatment of polycystic ovary syndrome. Committee Opinion Birmingham, AL: ASRM; April 2000

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

(reviewed May 2003).
51. American Society for Reproductive Medicine (ASRM). Vasectomy reversal. Technical Bulletin. Birmingham, AL: ASRM; August 1999.
52. American Society for Reproductive Medicine. Fertility preservation and reproduction in cancer patients. Fertil Steril. 2005;83(6):1622-1628.
53. American Society for Reproductive Medicine/American Urological Association, Inc. Report on evaluation of the azoospermic male. Baltimore, MD: American Urological Association; April 2001.
54. American Society for Reproductive Medicine/American Urological Association, Inc. Report on management of obstructive azoospermia. Baltimore, MD: American Urological Association; April 2001.
55. American Society for Reproductive Medicine/American Urological Association, Inc. Report on optimal evaluation of the infertile male. Baltimore, MD: American Urological Association; April 2001.
56. American Society for Reproductive Medicine/American Urological Association, Inc. Report on varicocele and infertility. Baltimore, MD: American Urological Association; April 2001.
57. American Society of Health-System Pharmacists, Inc. American Hospital Formulary Service Drug Information. Bethesda, MD: American Society of Health-System Pharmacists; updated periodically.
58. Asgari R, Mansouri K, Bakhtiari M, et al. Association between the FAS/FASL variants and risk of male infertility in Asian populations: A systematic review and meta-analysis. Medicina (Kaunas). 2019;55(6).
59. Athaullah N, Proctor M, Johnson NP. Oral versus injectable ovulation induction agents for unexplained subfertility. Cochrane Database Syst Rev. 2002;(3):CD003052.
60. Attia AM, Al-Inany HG, Proctor ML. Gonadotrophins for idiopathic male factor subfertility. Cochrane Database Syst Rev. 2007:(4):CD005071.
61. Ball CM. Laparoscopic ablation or resection of endometriosis deposits for moderate and severe endometriosis. STEER: Succint and Timely Evaluated Evidence Reviews. Bazian, Ltd. eds. London, UK: Wessex Institute for Health Research and Development, University of Southampton; 2001;11(1).
62. Bamigboye AA, Morris J. Oestrogen supplementation, mainly diethylstilbestrol, for preventing miscarriages and other adverse pregnancy outcomes. Cochrane Database Syst Rev. 2003; (3):CD004353.
63. Bayram N, van Wely M, van der Veen F. Pulsatile gonadotrophin releasing hormone for ovulation induction in subfertility associated with polycystic ovary syndrome. Cochrane Database Syst Rev. 2003;(3):CD000412.
64. Bayram N, van Wely M, van der Veen F. Recombinant FSH versus urinary gonadotrophins or recombinant FSH for ovulation induction in subfertility associated with polycystic ovary syndrome. Cochrane Database Syst Rev. 2001;(2):CD002121.
65. Ben-Meir A, Aboo-Dia M, Revel A, The benefit of human chorionic gonadotropin supplementation throughout the secretory phase of frozen-thawed embryo transfer cycles. Fertil Steril. 2010;93(2):351-354.
66. Bensdorp A, Cohlen BJ, Heineman MJ, Vanderkerchove P. Intra-uterine insemination for male subfertility. Cochrane Database Syst Rev. 2007;(4):CD000360.
67. Bergin K, Eliner Y, Duvall DW, Jr. et al. The use of propensity score matching to assess the benefit of the endometrial receptivity analysis in frozen embryo transfers. Fertil Steril. 2021;116(2):396-403.
68. Besselink DE, Farquhar C, Kremer JAM, et al. Cervical insemination versus intra-uterine insemination of donor sperm for subfertility. Cochrane Database Syst Rev. 2008;(2):CD000317.
69. Björndahl L, Kirkman Brown J; other Editorial Board Members of the WHO Laboratory Manual for the Examination and Processing of Human Semen. The sixth edition of the WHO Laboratory Manual for the Examination and Processing of Human Semen: Ensuring quality and standardization in basic examination of human ejaculates. Fertil Steril. 2022;117(2):246-251.
70. Blake D, Proctor M, Johnson N, Olive D. Cleavage stage versus blastocyst stage embryo transfer in assisted conception. Cochrane Database Syst Rev. 2007;(4):CD002118.
71. Boomsma CM, Heineman MJ, Cohlen BJ, Farquhar C. Semen preparation techniques for intrauterine insemination. Cochrane Database Syst Rev. 2007;(4):CD004507.
72. Boomsma CM, Keay SD, Macklon NS. Peri-implantation glucocorticoid administration for assisted reproductive technology cycles. Cochrane Database Syst Rev. 2007;(1):CD005996.
73. Bordewijk EM, Weiss NS, Nahuis MJ, et al; M-ovin study group. Gonadotrophins versus clomiphene citrate with or without IUI in women with normogonadotropic anovulation and clomiphene failure: A

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

cost-effectiveness analysis. Hum Reprod. 2019;34(2):276-284.
74. Brannstrom M, Belfort MA, Ayoubi JM. Uterus transplantation worldwide: Clinical activities and outcomes. Curr Opin Organ Transplant. 2021;26(6):616-626.
75. Brannstrom M, Diaz-Garcia C. Uterus transplantation for absolute uterine factor infertility: Ethics, patient selection, and consent. UpToDate Inc., Waltham, MA. Last reviewed June 2021a.
76. Brannstrom M, Diaz-Garcia C. Uterus transplantation for absolute uterine factor infertility: Surgery, immunosuppression, and obstetric management. UpToDate Inc., Waltham, MA. Last reviewed June 2021b.
77. Brigham and Women's Hospital. Infertility. A guide to evaluation, treatment, and counseling. Boston, MA: Brigham and Women's Hospital; 2003.
78. Brown J, Farquhar C, Beck J, et al. Clomiphene and anti-oestrogens for ovulation induction in PCOS. Cochrane Database Syst Rev. 2009;(4):CD002249.
79. Brown JA, Buckingham K, Abou-Setta A, Buckett W. Ultrasound versus 'clinical touch' for catheter guidance during embryo transfer in women. Cochrane Database Syst Rev. 2007;(1):CD006107.
80. Burrows E. Is minimal or mild endometriosis associated with infertility? Evidence Centre Evidence Report. Clayton, VIC: Centre for Clinical Effectiveness (CCE); 2000.
81. Cahill DJ, Wardle PG. Management of infertility. BMJ. 2002;325(7354):28-32.
82. Callejo J, Salvador C, Miralles A, et al. Long-term ovarian function evaluation after autografting by implantation with fresh and frozen-thawed human ovarian tissue. J Clin Endocrinol Metab. 2001;86(9):4489-4494.
83. Cantineau AE, Janssen MJ, Cohlen BJ, Allersma T. Synchronised approach for intrauterine insemination in subfertile couples. Cochrane Database Syst Rev. 2014;12:CD006942.
84. Cantineau AEP, Cohlen BJ, Heineman MJ. Ovarian stimulation protocols (anti-oestrogens, gonadotrophins with and without GnRH agonists/antagonists) for intrauterine insemination (IUI) in women with subfertility. Cochrane Database Syst Rev. 2007;(2):CD005356.
85. Cantineau AEP, Cohlen BJ, Heineman MJ. Intrauterine insemination versus fallopian tube sperm perfusion for non tubal infertility. Cochrane Database Syst Rev. 2009;(2):CD001502.
86. Cantineau AEP, Heineman MJ, Cohlen BJ. Single versus double intrauterine insemination (IUI) in stimulated cycles for subfertile couples. Cochrane Database Syst Rev. 2003:(1):CD003854.
87. Cao D, Ren Z, Lu D, et al. Association between CYP1A1 rs4646903 T > C genetic variations and male infertility risk: A meta-analysis. Medicine (Baltimore). 2019;98(31):e16543.
88. Carpinello OJ, Jahandideh S, Yamasaki M, et al. Does ovarian stimulation benefit ovulatory women undergoing therapeutic donor insemination? Fertil Steril. 2021;115(3):638-645.
89. Carson DS, Bucci KK. Infertility in women: An update. J Am Pharm Assoc. 1998;38(4):480-486.
90. Cayli S, Jakab A, Ovari L, et al. Biochemical markers of sperm function: male fertility and sperm selection for ICSI. Reprod Biomed Online. 2003;7(4):462-468.
91. Check JH, Graziano V, Lee G, et al. Neither sildenafil nor vaginal estradiol improves endometrial thickness in women with thin endometria after taking oral estradiol in graduating dosages. Clin Exp Obstet Gynecol. 2004;31(2):99-102.
92. Check JH, Katsoff B. Successful pregnancy with spontaneous ovulation in a woman with apparent premature ovarian failure who failed to conceive despite four transfers of embryos derived from donated oocytes. Clin Exp Obstet Gynecol. 2006;33(1):13-15.
93. Check ML, Check JH, Kaplan H. Pregnancy despite imminent ovarian failure and extremely high endogenous gonadotropins and therapeutic strategies: Case report and review. Clin Exp Obstet Gynecol. 2004;31(4):299-301.
94. Chen SJ, Allam JP, Duan YG, Haidl G. Influence of reactive oxygen species on human sperm functions and fertilizing capacity including therapeutical approaches. Arch Gynecol Obstet. 2013;288(1):191-199.
95. Chighizola CB, de Jesus GR. Antiphospholipid antibodies and infertility. Lupus. 2014;23(12):1232-1238.
96. Chuang AT, Howards SS. Male infertility: Evaluation and nonsurgical therapy. Urologic Clin N Am. 1998;24(4):703-713.
97. Chung E, Atmoko W, Saleh R, et al. Sixth edition of the World Health Organization laboratory manual of semen analysis: Updates and essential take away for busy clinicians. Arab J Urol. 2023;22(2):71-74.
98. Cimadomo D, Vaiarelli A, Colamaria S, et al. Luteal phase anovulatory follicles result in the production of competent oocytes: Intra-patient paired case-control study comparing follicular versus

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

luteal phase stimulations in the same ovarian cycle. Hum Reprod. 2018;33(8):1442-1448.
99. Conaghan J, Chen AA, Willman SP, et al. Improving embryo selection using a computer-automated time-lapse image analysis test plus day 3 morphology: Results from a prospective multicenter trial. Fertil Steril. 2013;100(2):412-419.
100. Corabian P, Hailey D. The efficacy and adverse effects of in vitro fertilization and embryo transfer. Int J Technol Assess Health Care. 1999;15(1):66-85.
101. Corabian P, Scott A. Ovulation induction drug therapy for anovulatory infertility associated with polycystic ovary syndrome. Health Technology Assessment 33. Edmonton, AB: Alberta Heritage Foundation for Medical Research (AHFMR); 2004.
102. Craciunas L, Tsampras N, Raine-Fenning N, Coomarasamy A. Intrauterine administration of human chorionic gonadotropin (hCG) for subfertile women undergoing assisted reproduction. Cochrane Database Syst Rev. 2018;10:CD011537.
103. Dakhly DMR, Bayoumi YA, Sharkawy M, et al. Intralipid supplementation in women with recurrent spontaneous abortion and elevated levels of natural killer cells. Int J Gynaecol Obstet. 2016135(3):324-327.
104. Damario MA, Hammitt DG, Galanits TM, et al. Pronuclear stage cryopreservation after intracytoplasmic sperm injection and conventional IVF: implications for timing of the freeze. Fertil Steril. 1999;72(6):1049-1054.
105. D'Angelo A, Amso N. Coasting (withholding gonadotrophins) for preventing ovarian hyperstimulation syndrome. Cochrane Database Syst Rev. 2002;(3):CD002811.
106. D'Angelo A, Amso N. Embryo freezing for preventing ovarian hyperstimulation syndrome. Cochrane Database Syst Rev. 2007;(3):CD002806.
107. Daolio J, Palomba S, Paganelli S, et al. Uterine transplantation and IVF for congenital or acquired uterine factor infertility: A systematic review of safety and efficacy outcomes in the first 52 recipients. PLoS One. 2020;15(4):e0232323.
108. Das S, Blake D, Farquhar C, Seif MMW. Assisted hatching on assisted conception (IVF & ICSI). Cochrane Database Syst Rev. 2009;(2):CD001894.
109. Daya S, Gunby JL, Ogah J. Luteal phase support in assisted reproduction cycles. Cochrane Database Syst Rev. 2008;(3):CD004830.
110. Daya S, Maheshwari A, Siristatidis CS, Bhattacharya S, Gibreel AF. Gonadotrophin-releasing hormone agonist protocols for pituitary desensitization in in vitro fertilization and gamete intrafallopian transfer cycles. Cochrane Database of Syst Rev. 2000;(1):CD001299.
111. Daya S. Follicle-stimulating hormone and human menopausal gonadotropin for ovarian stimulation in assisted reproduction cycles. Cochrane Database Syst Rev. 1996;(1):CD000061.
112. Derks RS, Farquhar C, Mol BWJ, et al. Techniques for preparation prior to embryo transfer. Cochrane Database Syst Rev. 2009;(4):CD007682.
113. Dhumal SS, Naik P, Dakshinamurthy S, Sullia K. Semen pH and its correlation with motility and count - A study in subfertile men. JBRA Assist Reprod. 2021;25(2):172-175.
114. Diamond MP, Legro RS, Coutifaris C, et al; NICHD Reproductive Medicine Network. Letrozole, gonadotropin, or clomiphene for unexplained infertility. N Engl J Med. 2015;373(13):1230-1240.
115. Dohle GR, Arver S, Bettocchi C, et al. Guidelines on male hypogonadism. Arnhem, The Netherlands: European Association of Urology; February 2012.
116. Dreyer K, Out R, Hompes PG, Mijatovic V. Hysterosalpingo-foam sonography, a less painful procedure for tubal patency testing during fertility workup compared with (serial) hysterosalpingography: A randomized controlled trial. Fertil Steril. 2014;102(3):821-825.
117. Duffy JM, Ahmad G, Mohiyiddeen L, et al. Growth hormone for in vitro fertilization. Cochrane Database Syst Rev. 2010;(1):CD000099.
118. Duffy JMN, Johnson N, Ahmad G, Watson A. Postoperative procedures for improving fertility following pelvic reproductive surgery. Cochrane Database Syst Rev. 2009;(2):CD001897.
119. Ebner T, Yaman C, Moser M, et al. Embryo fragmentation in vitro and its impact on treatment and pregnancy outcome. Fertil Steril. 2001;76(2):P281-P285.
120. Eisman LE, Pisarska MD, Wertheimer S, et al. Clinical utility of the endometrial receptivity analysis in women with prior failed transfers. J Assist Reprod Genet. 2021;38(3):645-650.
121. Evers JLH, Collins J, Clarke J. Surgery or embolisation for varicoceles in subfertile men. Cochrane Database Syst Rev. 2009;(1):CD000479.

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

122. ExEm Foam Inc. ExEm Foam, the only FDA-approved contrast agent for ultrasound tubal patency testing, now available in the U.S. Press Release. Nashville, TN; ExEm Foam; February 17, 2021.
123. Expert Panel on Women's Imaging, Wall DJ, Reinhold C, et al. ACR appropriateness criteria female infertility. J Am Coll Radiol. 2020;17(5S):S113-S124.
124. Fabbri R, Porcu E, Marsella T, et al. Human oocyte cryopreservation: new perspectives regarding oocyte survival. Hum Reprod. 2001;16(3):411-416.
125. Farquhar C, Brown J. Laparoscopic 'drilling' by diathermy or laser for ovulation induction in anovulatory polycystic ovary syndrome. Cochrane Database Syst Rev. 2012;(6):CD001122.
126. Forti G, Krausz C. Clinical review 100: Evaluation and treatment of the infertile couple. J Clin Endocrinol Metab. 1998;83(12):4177-4188.
127. Fronek J, Kristek J, Chlupac J, et al. Human uterus transplantation from living and deceased donors: The interim results of the first 10 cases of the Czech trial. J Clin Med. 2021;10(4):586.
128. George K, Nair R, Tharyan P. Ovulation triggers in anovulatory women undergoing ovulation induction. Cochrane Database Syst Rev. 2008;(3):CD006900.
129. Gerig NE, Meacham RB, Ohl DA. Use of electroejaculation in the treatment of ejaculatory failure secondary to diabetes mellitus. Urology. 1997;49(2):239-242.
130. Ghosh I, Bharadwaj A, Datta K. Reduction in the level of hyaluronan binding protein 1 (HABP1) is associated with loss of sperm motility. J Reprod Immunol. 2002;53(1-2):45-54.
131. Gleicher N, Kushnir VA, Barad DH. Preimplantation genetic screening (PGS) still in search of a clinical application: A systematic review. Reprod Biol Endocrinol. 2014;12:22.
132. Gleicher N. Autoantibodies in infertility: Current opinion. Hum Reprod Update. 1998;4(2):169-176.
133. Glujovsky D, Pesce R, Fiszbajn G, et al. Endometrial preparation for women undergoing embryo transfer with frozen embryos or embryos derived from donor oocytes. Cochrane Database Syst Rev. 2010;(1):CD006359.
134. Goldman MB, Thornton KL, Ryley D, et al. A randomized clinical trial to determine optimal infertility treatment in older couples: The Forty and Over Treatment Trial (FORT-T). Fertil Steril. 2014;101(6):1574-1581.e1-e2.
135. Goldstein M. Surgical management of male infertility and other scrotal disorders. In: Campbell's Urology. 7th ed. PC Walsh, AB Retnik, ED Vaughan, AJ Wein, eds. Philadelphia, PA: W.B. Saunders Co.; 1998; Ch. 44:1331-1377.
136. Griffiths A, D'Angelo A, Amso N. Surgical treatment of fibroids for subfertility. Cochrane Database Syst Rev. 2006;(3):CD003857.
137. Guercini F, Pajoncini C, Bard R, et al. Echoguided drug infiltration in chronic prostatitis: Results of a multi-centre study. Arch Ital Urol Androl. 2005;77(2):87-92.
138. Gutarra-Vilchez RB, Bonfill Cosp X, Glujovsky D, et al. Vasodilators for women undergoing fertility treatment. Cochrane Database Syst Rev. 2018;10:CD010001.
139. Gutarra-Vilchez RB, Urrútia G, Glujovsky D, et al. Vasodilators for women undergoing fertility treatment. Cochrane Database Syst Rev. 2014;10:CD010001.
140. Hajder M, Hajder E, Husic A. The effects of total motile sperm count on spontaneous pregnancy rate and pregnancy after IUI treatment in couples with male factor and unexplained infertility. Med Arch. 2016;70(1):39-43.
141. Hammond CB, Stillman RJ. Infertility and assisted reproduction. In: Danforth's Obstetrics and Gynecology. 8th ed. Ch. 41. JR Scott, PJ DiSaia, CB Hammond, WN Spellacy, eds. Philadelphia, PA: Lippincott Williams & Wilkins; 1999:649-667.
142. Hannoun A, Abu-Musa A. Gamete intrafallopian transfer (GIFT) in the treatment of severe male factor infertility. Int J Gynaecol Obstet. 1998;61(3):293-295.
143. Hanson MA, Dumesic DA. Initial evaluation and treatment of infertility in a primary-care setting. Mayo Clin Proc. 1998;73(7):681-685.
144. Hassan SN, Alfozan H, Hart R, Qadri H. Ovarian cyst aspiration prior to in vitro fertilization treatment for subfertility [Protocol for Cochrane Review]. Cochrane Database Syst Rev. 2006;(2):CD005999.
145. Heineman MJ, Beerendonk CC, Kaandorp CJ. National guideline 'Cryopreservation of ovarian tissue'. Ned Tijdschr Geneeskd. 2008;152(45):2452-2455.
146. Helmerhorst FM, Van Vliet HAAM, Gornas T, et al. Intra-uterine insemination versus timed intercourse for cervical hostility in subfertile couples. Cochrane Database Syst Rev. 2005;(4):CD002809.

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

147. Hong X, Ma J, Yin J, et al. The association between vaginal microbiota and female infertility: A systematic review and meta-analysis. Arch Gynecol Obstet. 2020;302(3):569-578.
148. Huang EY, Shandley LM, Mehta A, et al. Fertility preservation for iatrogenic infertility: Patient barriers and opportunities for the reproductive medicine workforce. Urol Pract. 2025;12(2):194-201.
149. Hughes E, Brown J, Collins J, Vandekerckhove P. Clomiphene citrate for unexplained subfertility in women. Cochrane Database Syst Rev. 2010;(1):CD000057.
150. Hughes E, Brown J, Collins JJ, et al. Ovulation suppression for endometriosis for women with subfertility. Cochrane Database Syst Rev. 2007;(3):CD000155.
151. Hughes E, Brown J, Tiffin G, Vandekerckhove P. Danazol for unexplained subfertility. Cochrane Database Syst Rev. 2007;(1):CD000069.
152. Hughes E, Collins J, Vandekerckhove P. Bromocriptine for unexplained subfertility in women. Cochrane Database Syst Rev. 1996;(4):CD000044.
153. Hughes E, Collins J, Vandekerckhove P. Clomiphene citrate for ovulation induction in women with oligo-amenorrhoea. Cochrane Database Syst Rev. 1996;(1):CD000056.
154. Hughes E, Collins J, Vandekerckhove P. Gonadotrophin-releasing hormone analogue as an adjunct to gonadotropin therapy for clomiphene-resistant polycystic ovarian syndrome. Cochrane Database Syst Rev. 1996;(1):CD000097.
155. Hughes E, Collins J, Vandekerckhove P. Ovulation induction with urinary follicle stimulating hormone versus human menopausal gonadotropin for clomiphene-resistant polycystic ovary syndrome. Cochrane Database Syst Rev. 1996;(1):CD000087.
156. Hull MGR, Cahill DJ. Gonadal disorders. Female infertility. Endocrinol Metab Clin N Am. 1998;27(4):851-876.
157. Huszar G, Ozenci CC, Cayli S, et al. Hyaluronic acid binding by human sperm indicates cellular maturity, viability, and unreacted acrosomal status. Fertil Steril. 2003;79 Suppl 3:1616-1624.
158. Illions EH, Valley MT, Kaunitz AM. Infertility: A clinical guide for the internist. Women's health issues, part II. Med Clin N Am. 1998;82(2):271-295.
159. Institute for Clinical Systems Improvement (ICSI). Diagnosis and management of infertility. Bloomington, MN: ICSI; revised July 2004.
160. Iwami N, Komiya S, Asada Y, et al. Shortening time to pregnancy in infertile women by personalizing treatment of microbial imbalance through Emma & Alice: A multicenter prospective study. Reprod Med Biol. 2025;24:e12634.
161. Jacobson TZ, Duffy J, Barlow DH, et al. Laparoscopic surgery for subfertility associated with endometriosis. Cochrane Database Syst Rev. 2010;(1):CD001398.
162. Jarow JP, Wright WW, Brown TR, Bioactivity of androgens within the testes and serum of normal men. J Androl. 2005;26(3):343-348.
163. Johal JK, Gardner RM, Vaughn SJ, et al. Pregnancy success rates for lesbian women undergoing intrauterine insemination. F S Rep. 2021;2(3):275-281.
164. Johnson N, Vandekerckhove P, Watson A, et al. Tubal flushing for subfertility. Cochrane Database Syst Rev. 2007;(3):CD003718.
165. Johnson NP, van Voorst S, Sowter MC, et al. Surgical treatment for tubal disease in women due to undergo in vitro fertilisation. Cochrane Database Syst Rev. 2010;(1):CD002125.
166. Jones BP, Ranaei-Zamani N, Vali S, et al. Options for acquiring motherhood in absolute uterine factor infertility; adoption, surrogacy and uterine transplantation. Obstet Gynaecol. 2021;23(2):138-147.
167. Jungwirth A, Diemer T, Dohle GR, et al. Guidelines on male infertility. Arnhem, The Netherlands: European Association of Urology (EAU); February 2012.
168. Kamischke A, Nieschlag E. Conventional treatments of male infertility in the age of evidence-based andrology. Hum Reprod. 1998;13 Suppl 1:62-75.
169. Kaser DJ, Goldman MB, Fung JL, et al. When is clomiphene or gonadotropin intrauterine insemination futile? Results of the Fast Track and Standard Treatment Trial and the Forty and Over Treatment Trial, two prospective randomized controlled trials. Fertil Steril. 2014;102(5):1331-1337.
170. Ko EY, Sabanegh ES Jr, Agarwal A. Male infertility testing: Reactive oxygen species and antioxidant capacity. Fertil Steril. 2014;102(6):1518-1527.
171. Kuohung W, Hornstein MD. Evaluation of female infertility. UpToDate [online serial], Waltham, MA: UpToDate; reviewed January 2015b; Janaury 2021.
172. Kuohung W, Hornstein MD. Female infertility: Evaluation. UpToDate [online serial]. Waltham, MA: UpToDate; reviewed May 2024.

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

173. Kuohung W, Hornstein MD. Overview of infertility. UpToDate [online serial]. Waltham, MA: UpToDate; reviewed January 2015a.
174. Kuohung W, Hornstein MD. Overview of treatment of female infertility. UpToDate [online serial]. Waltham, MA: UpToDate; reviewed January 2014; January 2017.
175. Kuohung W, Hornstein MD. Treatments for female infertility. UpToDate [online serial]. Waltham, MA: UpToDate; reviewed January 2020; January 2022
176. Kwan I, Bhattacharya S, McNeil A, van Rumste MME. Monitoring of stimulated cycles in assisted reproduction (IVF and ICSI). Cochrane Database Syst Rev. 2008;(2):CD005289.
177. La Sala GB, Montanari R, Dessanti L, et al. The role of diagnostic hysteroscopy and endometrial biopsy in assisted reproductive technologies. Fertil Steril. 1998;70(2):378-380.
178. Lai R. Understanding infertility in patients with sickle cell disease [website]. Rare Disease Advisor. Vancouver, British Columbia, Canada; August 18, 2022.
179. Lee E, Illingworth P, Wilton L, Chambers GM. The clinical effectiveness of preimplantation genetic diagnosis for aneuploidy in all 24 chromosomes (PGD-A): Systematic review. Hum Reprod. 2015;30(2):473-483.
180. Legro RS, Arslanian SA, Ehrmann DA, et al. Diagnosis and Treatment of Polycystic Ovarian Syndrome (PCOS): A Endocrine Society Clinical Practice Guideline. Chevy Chase, MD: Endocrine Society; December 2013.
181. Lepine S, McDowell S, Searle LM, et al. Advanced sperm selection techniques for assisted reproduction. Cochrane Database Syst Rev. 2019;7(7):CD010461.
182. Lin Y, Qi J, Sun Y. Platelet-rich plasma as a potential new strategy in the endometrium treatment in assisted reproductive technology. Front Endocrinol (Lausanne). 2021;12:707584.
183. Linara-Demakakou E, Bodri D, Wang J, et al. Cumulative live birth rates following insemination with donor spermatozoa in single women, same-sex couples and heterosexual patients. Reprod Biomed Online.2020;41(6):1007-1014.
184. Ludwin I., Ludwin, A. et al. Accuracy of hysterosalpingo-foam sonography in comparison to hysterosalpingo-contrast sonography with air/saline and to laparoscopy with dye. Human Reproduction 2017, 32(4):758-769.
185. Ly KD, Aziz N, Safi J, Agarwal A. Evidence-based management of infertile couples with repeated implantation failure following IVF. Current Women's Health Reviews. 2010;6(3):200-218.
186. Maheshwari A, Gibreel A, Siristatidis CS, Bhattacharya S. Gonadotrophin-releasing hormone agonist protocols for pituitary suppression in assisted reproduction. Cochrane Database Syst Rev. 2011; (8):CD006919.
187. Malasevskaia I, Al-Awadhi AA. A new approach for treatment of woman with absolute uterine factor infertility: A traditional review of safety and efficacy outcomes in the first 65 recipients of uterus transplantation. Cureus. 2021;13(1):e12772.
188. Maleki-Hajiagha A, Razavi M, Rouholamin S, et al. Intrauterine infusion of autologous platelet-rich plasma in women undergoing assisted reproduction: A systematic review and meta-analysis. J Reprod Immunol. 2020;137:103078.
189. Malinova M, Abouyta T, Krasteva M. The effect of vaginal sildenafil citrate on uterine blood flow and endometrium in the infertile women. Akush Ginekol (Sofiia). 2013;52 Suppl 1:26-30.
190. Management of infertility caused by ovulatory dysfunction. American College of Obstetricians and Gynecologists. Obstet Gynecol. 2002;99(2):347-358.
191. Martin C, Woodland E. Sperm selection technology in ART. Semin Reprod Med. 2021;39(5-06):200-206.
192. Martin MC. Infertility. Section VI. Reproductive endocrinology and infertility. In: Current Obstetric & Gynecologic Diagnosis & Treatment. AH DeCherney, ML Pernoll, eds. New York, NY: McGraw-Hill; 2000.
193. Massin N. New stimulation regimens: Endogenous and exogenous progesterone use to block the LH surge during ovarian stimulation for IVF. Hum Reprod Update. 2017;23(2):211-220.
194. McDonnell R, Marjoribanks J, Hart RJ. Ovarian cyst aspiration prior to in vitro fertilization treatment for subfertility. Cochrane Database Syst Rev. 2014;12:CD005999.
195. McGrath E, Ryan EJ, Lynch L, et al. Changes in endometrial natural killer cell expression of CD94, CD158a and CD158b are associated with infertility. Am J Reprod Immunol. 2009;61(4):265-276.
196. Merative L.P. Follitropin. Merative Micromedex. Ann Arbor, MI: Merative; 2025. Available at: https://www.micromedexsolutions.com. Accessed May 8, 2025.

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

197. Metelev AY, Bogdanov AB, Ivkinl EV, et al. Hyperbaric oxygen therapy in the treatment of male infertility associated with increased sperm DNA fragmentation and reactive oxygen species in semen. Urologiia. 2015;(5):74-76.
198. Miller D, Pavitt S, Sharma V, et al. Physiological, hyaluronan-selected intracytoplasmic sperm injection for infertility treatment (HABSelect): A parallel, two-group, randomised trial. Lancet. 2019;393(10170):416-422.
199. Momeni M, Rahbar MH, Kovanci E. A meta-analysis of the relationship between endometrial thickness and outcome of in vitro fertilization cycles. J Hum Reprod Sci. 2011;4(3):130-137.
200. Morell V. Women's health. Basic infertility assessment. Primary Care Clin N Am. 1997;24(1):195-204.
201. Morley LC, Simpson N, Tang T. Human chorionic gonadotrophin (hCG) for preventing miscarriage. Cochrane Database Syst Rev. 2013;1:CD008611.
202. Morshedi M, Duran HE, Taylor S, Oehninger S. Efficacy and pregnancy outcome of two methods of semen preparation for intrauterine insemination: a prospective randomized study. Fertil Steril. 2003;79 Suppl 3:1625-1632.
203. Mouanness M, Ali-Bynom S, Jackman J, et al. Use of intra-uterine injection of platelet-rich plasma (PRP) for endometrial receptivity and thickness: A literature review of the mechanisms of action. Reprod Sci. 2021;28(6):1659-1670.
204. Myers ER, McCrory DC, Mills AA, et al. Effectiveness of assisted reproductive technology (ART). Evidence Report/Technology Assessment No.167. Rockville, MD: Agency for Healthcare Research and Quality (AHRQ); 2008.
205. Nakajima ST, Molloy MH, Oi RH, et al. Clinical evaluation of luteal function. Obstet Gynecol. 1994;84:219-221.
206. Nardo LG, Gelbaya TA, Wilkinson H, et al. Circulating basal anti-Müllerian hormone levels as predictor of ovarian response in women undergoing ovarian stimulation for in vitro fertilization. Fertil Steril. 2009;92(5):1586-1593.
207. National Collaborating Centre for Women's and Children's Health. Fertility. Assessment and treatment for people with fertility problems. Clinical Guideline. London, UK: RCOG Press; February 2004.
208. National Institute for Clinical Excellence (NICE). Fallopian tube recanalisation by guidewire. Interventional Procedure Guidance 71. London, UK: NICE; 2004.
209. National Institute for Clinical Excellence (NICE). Falloposcopy with coaxial catheter. Interventional Procedure Guidance 62. London, UK: NICE; 2004.
210. National Institute for Clinical Excellence (NICE). Fertility: Assessment and treatment for people with fertility problems. Clinical Guideline No. 11. London, UK: NICE; February 2004.
211. National Institute of Health and Care Excellence (NICE). Fertility problems: Assessment and treatment. NICE Clinical Guideline 156. London, UK: NICE; February 2013.
212. Nelson SM, Yates RW, Lyall H, et al. Anti-Müllerian hormone-based approach to controlled ovarian stimulation for assisted conception. Hum Reprod. 2009;24(4):867-875.
213. No authors listed. ACOG Committee Opinion No. 405: Ovarian tissue and oocyte cryopreservation. Obstet Gynecol. 2008;111(5):1255-1256.
214. No authors listed. DuoStim in cases of PGT: Comparison of embryo quantity and embryonic quality using MitoScore. ClinicalTrials.gov. Bethesda, MD: National Library of Medicine; updated September 3, 2018.
215. North American Menopause Society (NAMS). Clinical challenges of perimenopause: Consensus opinion of The North American Menopause Society. Menopause. 2000;7(1):5-13.
216. Nugent D, Vandekerckhove P, Hughes E, et al. Gonadotrophin therapy for ovulation induction in subfertility associated with polycystic ovary syndrome. Cochrane Database Syst Rev. 2000; (3):CD000410.
217. Oatway C, Gunby J, Daya S. Day three versus day two embryo transfer following in vitro fertilization or intracytoplasmic sperm injection. Cochrane Database Syst Rev. 2004;(2):CD004378.
218. Ohl DA. Electroejaculation. Urol Clin N Amer. 1993;20(1):181-188.
219. Ontario Ministry of Health and Long-Term Care, Medical Advisory Secretariat (MAS). In vitro fertilization and multiple pregnancies. Health Technology Literature Review. Toronto, ON: MAS; 2006.
220. Ozkan ZS, Deveci D, Kumbak B, et al. What is the impact of Th1/Th2 ratio, SOCS3, IL17, and IL35 levels in unexplained infertility? J Reprod Immunol. 2014;103:53-58.

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

221. Pandian Z, Akande VA, Harrild K, Bhattacharya S. Surgery for tubal infertility. Cochrane Database Syst Rev. 2008;(3):CD006415.
222. Pandian Z, Bhattacharya S, Ozturk O, et al. Number of embryos for transfer following in-vitro fertilisation or intra-cytoplasmic sperm injection. Cochrane Database Syst Rev. 2009;(2):CD003416.
223. Pandian Z, Gibreel A, Bhattacharya S. In vitro fertilisation for unexplained subfertility. Cochrane Database Syst Rev. 2012;(4):CD003357.
224. Pandian Z, McTavish AR, Aucott L, S, et al. Interventions for 'poor responders' to controlled ovarian hyperstimulation (COH) in in-vitro fertilisation (IVF). Cochrane Database Syst Rev. 2010; (1):CD004379.
225. Paulson R. In vitro fertilization. UpToDate [online serial]. Waltham, MA: UpToDate; reviewed January 2019.
226. Pecker LH, Cameron K. Sickle cell disease and infertility risks: Implications for counseling and care of affected girls and women. Expert Rev Hematol. 2024;17(8):493-504.
227. Pfeifer S, Goldberg J, McClure D, et al.; Practice Committees of the American Society for Reproductive Medicine and the Society for Assisted Reproductive Technology. Mature oocyte cryopreservation: A guideline. Fertil Steril. 2013;99(1):37-43.
228. Physicians' Desk Reference (PDR). Montvale, NJ: Thomson PDR; updated periodically.
229. Piccioni MG, Riganelli L, Filippi V, et al. Sonohysterosalpingography: Comparison of foam and saline solution. J Clin Ultrasound. 2017;45(2):67-71.
230. Pichon-Riviere A, Augustovski F, Garcia Marti S, et al. Serum anti-mullerian hormone (AMH) as ovarian reserve marker. Summary. IRR No. 164. Buenos Aires, Argentina: Institute for Clinical Effectiveness and Health Policy (IECS); 2009.
231. Polyzos NP, Tzioras S, Mauri D, Tatsioni A. Double versus single intrauterine insemination for unexplained infertility: A meta-analysis of randomized trials. Fertil Steril. 2010;94(4):1261-1266.
232. Poustie VJ, Dodd S, Drakeley AJ. Low-dose aspirin for in vitro fertilisation. Cochrane Database Syst Rev. 2007;(4):CD004832.
233. Pouwer AW, Farquhar C, Kremer JAM. Long-acting FSH versus daily FSH for women undergoing assisted reproduction. Cochrane Database Syst Rev. 2012; (6):CD009577.
234. Practice Committee of the American Society for Reproductive Medicine and the Practice Committee for the Society for Assisted Reproductive Technologies. Electronic address: ASRM@asrm.org. Guidance on the limits to the number of embryos to transfer: A committee opinion. Fertil Steril. 2021;116(3):651-654.
235. Practice Committee of American Society for Reproductive Medicine (ASRM). Definitions of infertility and recurrent pregnancy loss: A committee opinion. Fertil Steril. 2013;99(1):63.
236. Practice Committee of the American Society for Reproductive Medicine. Evidence-based treatments for couples with unexplained infertility: A guideline. Fertil Steril. 2020;113(2):305-322.
237. Practice Committee of American Society for Reproductive Medicine; Practice Committee of Society for Assisted Reproductive Technology. Ovarian tissue and oocyte cryopreservation. Fertil Steril. 2008;90(5 Suppl):S241-S246.
238. Practice Committee of American Society for Reproductive Medicine; Practice Committee of Society for Assisted Reproductive Technology. Criteria for number of embryos to transfer: A committee opinion. Fertil Steril. 2013;99(1):44-46.
239. Practice Committee of the American Society for Reproductive Medicine; Practice Committee of the Society for Assisted Reproductive Technology. Guidelines on number of embryos transferred. Fertil Steril. 2009;92(5):1518-1519.
240. Practice Committee of the American Society for Reproductive Medicine. Testing and interpreting measures of ovarian reserve: A committee opinion. Fertil Steril. 2015;103(3):e9-e17.
241. Practice Committee of the American Society for Reproductive Medicine (ASRM). Guidance on the limits to the number of embryos to transfer: A committee opinion. Fertil Steril. 2017;107(4):901-903.
242. Practice Committee of the American Society of Reproductive Medicine. Use of clomiphene citrate in infertile women: A committee opinion. Fertil & Steril. 2013;100:341-348.
243. Practice Committees of the American Society for Reproductive Medicine and the Society for Assisted Reproductive Technology. In vitro maturation: A committee opinion. Fertil Steril. 2013;99(3):663-666.
244. Quetrell H, Vaughan DA, Dodge LE, et al. Intrauterine insemination (IUI): More is not always better. Abstract O-126. Fertil Steril. 2023:120(4 Suppl):E53.

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

245. Rafael ZB. Endometrial receptivity analysis (ERA) test: An unproven technology. Hum Reprod Open. 2021;2021(2):hoab010.
246. Ranganathan S, Ganguly AK, Datta K. Evidence for presence of hyaluronan binding protein on spermatozoa and its possible involvement in sperm function. Mol Reprod Dev. 1994;38(1):69-76.
247. Rein MS, Barbieri RL. The infertile couple -- part I. In: Kistner's Gynecology & Women's Health. 7th ed. KJ Ryan, RS Berkowitz, RL Barbieri, A Dunaif, eds. St. Louis, MO: Mosby, Inc.; 1999; Ch. 13: 325-354.
248. Rein MS, Barbieri RL. The infertile couple -- part II. In: Kristner's Gynecology & Women's Health. 7th ed. KJ Ryan, RS Berkowitz, RL Barbieri, A Dunaif, eds. St. Louis, MO: Mosby, Inc.; 1999; Ch. 14: 355-365.
249. Reindollar RH, Regan MM, Neumann PJ, et al. A randomized clinical trial to evaluate optimal treatment for unexplained infertility: The fast track and standard treatment (FASTT) trial . Fertil Steril. 2010;94(3):888-899.
250. Rhenman A, Berglund L, Brodin T, et al. Which set of embryo variables is most predictive for live birth? A prospective study in 6252 single embryo transfers to construct an embryo score for the ranking and selection of embryos. Hum Reprod. 2015;30(1):28-36.
251. Ricci S, Bennett C, Falcone T. Uterine transplantation: Evolving data, success, and clinical importance. J Minim Invasive Gynecol. 2021;28(3):502-512.
252. Riestenberg C, Kroener L, Quinn M, et al. Routine endometrial receptivity array in first embryo transfer cycles does not improve live birth rate. Fertil Steril. 2021;115(4):1001-1006.
253. Riganelli L, Casorelli A, Caccetta J, et al. Ultrasonography reappraisal of tubal patency in assisted reproduction technology patients: comparison between 2D and 3D-sonohysterosalpingography. A pilot study. Minerva Ginecol. 2018;70(2):123-128.
254. Rosen M. Intracytoplasmic sperm injection. UpToDate [online serial]. Waltham, MA: UpToDate; reviewed August 2024.
255. Royal College of Obstetricians and Gynaecologists. The initial investigation and management of the infertile couple. Evidence-Based Clinical Guidelines No. 2. London, UK: RCOG Press; February 1998.
256. Royal College of Obstetricians and Gynaecologists. The management of infertility in secondary care. Evidence-Based Clinical Guidelines No. 3. London, UK: RCOG Press; February 1998.
257. Royal College of Obstetricians and Gynaecologists. The management of infertility in tertiary care. Evidence-Based Clinical Guidelines No. 6. London, UK: RCOG Press; January 2000.
258. Sabeur K, Cherr GN, Yudin AI, Overstreet JW. Hyaluronic acid enhances induction of the acrosome reaction of human sperm through interaction with the PH-20 protein. Zygote. 1998;6(2):103-111.
259. Sallam HN, Garcia-Velasco JA, Dias S, Arici A. Long-term pituitary down-regulation before in vitro fertilization (IVF) for women with endometriosis. Cochrane Database Syst Rev. 2006;(1):CD004635.
260. Schenken R. Endometriosis in adults: Clinical features, evaluation, and diagnosis. UpToDate [online serial]. Waltham, MA: UpToDate; reviewed February 2025.
261. Schlegel PN, Sigman M, Collura B, et al. Diagnosis and treatment of infertility in men: AUA/ASRM guideline part I. J Urol. 2021;205(1):36-43.
262. Schlegel PN, Sigman M, Collura B, et al. Diagnosis and treatment of infertility in men: AUA/ASRM guideline part II.J Urol. 2021;205(1):44-51.
263. Schriock ED. Practical aspects of pulsatile gonadotropin-releasing hormone administration. Am J Obstet Gynecol. 1990;163(5):1765-1770.
264. Shaker AG, Zosmer A, Dean N, et al. Comparison of intravenous albumin and transfer of fresh embryos with cryopreservation of all embryos for subsequent transfer in prevention of ovarian hyperstimulation syndrome. Fertil Steril. 1996;65(5):992-996.
265. Sharara FI, Scott RT Jr, Seifer DB. The detection of diminished ovarian reserve in infertile women. Am J Obstet Gynecol. 1998;179(3 Pt 1):804-812.
266. Sharma H, Mavuduru RS, Singh SK, Prasad R. Heterogeneous spectrum of mutations in CFTR gene from Indian patients with congenital absence of the vas deferens and their association with cystic fibrosis genetic modifiers. Mol Hum Reprod. 2014;20(9):827-835.
267. Shaw-Jackson C, Bertrand E, Becker B, et al. Vitrification of blastocysts derived from fair to poor quality cleavage stage embryos can produce high pregnancy rates after warming. J Assist Reprod Genet. 2013;30(8):1035-1042.

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

268. Showell MG, Mackenzie-Proctor R, Brown J, et al. Antioxidants for male subfertility. Cochrane Database Syst Rev. 2014;12:CD007411.
269. Sigman M, Howards SS. Male infertility. In: Campbell's Urology. 7th ed. PC Walsh, AB Retnik, ED Vaughan, AJ Wein, eds. Philadelphia, PA: WB Saunders Co.; 1998; Ch. 43:1287-1330.
270. Simon C, Gomez C, Cabanillas S, et al; ERA-RCT Study Consortium Group. A 5-year multicentre randomized controlled trial comparing personalized, frozen and fresh blastocyst transfer in IVF. Reprod Biomed Online. 2020;41(3):402-415.
271. Sinawat S, Buppasiri P, Lumbiganon P, Pattanittum P. Long versus short course treatment with metformin and clomiphene citrate for ovulation induction in women with PCOS. Cochrane Database Syst Rev. 2008;(1):CD006226.
272. Siristatidis CS, Maheshwari A, Bhattacharya S. In vitro maturation in sub fertile women with polycystic ovarian syndrome undergoing assisted reproduction. Cochrane Database Syst Rev. 2009; (1):CD006606.
273. Siristatidis CS, Vrachnis N, Creatsa M, et al. In vitro maturation in subfertile women with polycystic ovarian syndrome undergoing assisted reproduction. Cochrane Database Syst Rev. 2013;10:CD006606.
274. Smulders B, van Oirschot SM, Farquhar C, et al. Oral contraceptive pill, progestogen or estrogen pre-treatment for ovarian stimulation protocols for women undergoing assisted reproductive techniques. Cochrane Database Syst Rev. 2010;(1):CD006109.
275. Society of Obstetricians and Gynaecologists of Canada (SOGC). The Canadian Consensus Conference on Menopause and Osteoporosis. 2002 Update. Ottawa, ON: SOGC; October 2002.
276. Soliman GM, Fetih G, Abbas AM. Thermosensitive bioadhesive gels for the vaginal delivery of sildenafil citrate: I vitro characterization and clinical evaluation in women using clomiphene citrate for induction of ovulation. Drug Dev Ind Pharm. 2017;43(3):399-408.
277. Son WY, Park SE, Lee KA, et al. Effects of 1,2-propanediol and freezing-thawing on the in vitro developmental capacity of human immature oocytes. Fertil Steril. 1996;66(6):995-999.
278. Sonksen J, Sommer P, Biering-Sorensen F, et al. Pregnancy after assisted ejaculation procedures in men with spinal cord injury. Arch Phys Med Rehabil. 1997;78(10):1059-1061.
279. Sonmezer M, Oktay K. Fertility preservation in patients undergoing gonadotoxic treatment or gonadal resection. UpToDate [online serial]. Waltham, MA: UpToDate; reviewed February 2014.
280. Soria M, Pradillo G, García J, et al. Pregnancy predictors after intrauterine insemination: Analysis of 3012 cycles in 1201 couples. J Reprod Infertil. 2012;13(3):158-166.
281. Steiner AZ, Herring AH, Kesner JS, et al. Antimüllerian hormone as a predictor of natural fecundability in women aged 30-42 years. Obstet Gynecol. 2011;117(4):798-804.
282. Steiner AZ. Clinical implications of ovarian reserve testing. Obstet Gynecol Surv. 2009;64(2):120-128.
283. Su HI, Sammel MD, Green J, et al. Antimullerian hormone and inhibin B are hormone measures of ovarian function in late reproductive-aged breast cancer survivors. Cancer. 2010;116(3):592-599.
284. Sunkara SK, Tuthill J, Khairy M, et al. Pituitary suppression regimens in poor responders undergoing IVF treatment: A systematic review and meta-analysis. Reprod Biomed Online. 2007;15(5):539-546.
285. Swedish Council on Technology Assessment in Health Care (SBU). Children born from in vitro fertilization (IVF). Stockholm, Sweden: SBU; 2000.
286. Swerdloff RS, Wang C. Evaluation of male infertility. UpToDate [online serial]. Waltham, MA: UpToDate; reviewed February 2014.
287. Swerdloff RS, Wang C. Evaluation of male infertility. UpToDate [online serial]. Waltham, MA: UpToDate; reviewed January 2017. .
288. Tang T, Lord JM, Norman RJ, et al. Insulin-sensitising drugs (metformin, rosiglitazone, pioglitazone, D-chiro-inositol) for polycystic ovary syndrome. Cochrane Database Syst Rev. 2010;(1):CD003053.
289. The Practice Committee of the American Society for Reproductive Medicine. Testing and interpreting measures of ovarian reserve: A committee opinion. Fertil Steril. 2012;98(6):1407-1415.
290. Thessaloniki ESHRE/ASRM-Sponsored PCOS Consensus Workshop Group. Consensus on infertility treatment related to polycystic ovary syndrome. Fertil Steril. 2008;89(3):505-522.
291. Thomas K, Thomson A, Wood S, Endometrial integrin expression in women undergoing in vitro fertilization and the association with subsequent treatment outcome. Fertil Steril. 2003;80(3):502-507.
292. Toledo AA, Tucker MJ, Bennett JK, et al. Electroejaculation in combination with in vitro fertilization and gamete micromanipulation for treatment of anejaculatory male infertility. Am J Obstet Gynecol. 1992;167:322-325, discussion 325-326.

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

293. Toth TL, Baka SG, Veeck LL, et al. Fertilization and in vitro development of cryopreserved human prophase I oocytes. Fertil Steril. 1994;61(5):891-894.
294. Toulis KA, Iliadou PK, Venetis CA, et al. Inhibin B and anti-Mullerian hormone as markers of persistent spermatogenesis in men with non-obstructive azoospermia: A meta-analysis of diagnostic accuracy studies. Hum Reprod Update. 2010;16(6):713-724.
295. Tso LO, Costello MF, Albuquerque LE, et al. Metformin treatment before and during IVF or ICSI in women with polycystic ovary syndrome. Cochrane Database Syst Rev. 2009;(2):CD006105.
296. Turek PJ. Male infertility. In: Smith's General Urology. EA Tanagho, JW McAninch, eds. Stamford, CT: Appleton & Lange; 2000.
297. Twisk M, Mastenbroek S, van Wely M, et al. Preimplantation genetic screening for abnormal number of chromosomes (aneuploidies) in in vitro fertilisation or intracytoplasmic sperm injection. Cochrane Database Syst Rev. 2006;(1):CD005291.
298. U.S. Food and Drug Administration (FDA). FDA announces voluntary withdrawal of pergolide products. Agency working with product manufacturers. FDA News. P07-54. Rockville, MD: FDA; March 29, 2007.
299. Vaiarelli A, Cimadomo D, Trabucco E, et al. Double stimulation in the same ovarian cycle (DuoStim) to maximize the number of oocytes retrieved from poor prognosis patients: A multicenter experience and SWOT analysis. Front Endocrinol (Lausanne). 2018;9:317.
300. Vaiarelli A, Cimadomo D, Ubaldi N, et al. What is new in the management of poor ovarian response in IVF? Curr Opin Obstet Gynecol. 2018;30(3):155-162.
301. van der Linden M, Buckingham K, Farquhar C, et al. Luteal phase support for assisted reproduction cycles. Cochrane Database Syst Rev. 2011;(10):CD009154.
302. Van Peperstraten A, Proctor ML, Johnson NP, Philipson G. Techniques for surgical retrieval of sperm prior to ICSI for azoospermia. Cochrane Database Syst Rev. 2008;(2):CD002807.
303. van Rumste MME, Evers JLH, Farquhar CM. Intra-cytoplasmic sperm injection versus conventional techniques for oocyte insemination during in vitro fertilisation in patients with non-male subfertility. Cochrane Database Syst Rev. 2003;(2):CD001301.
304. van Welie N, van Rijswijk J, Dreyer K, et al. Can hysterosalpingo-foam sonography replace hysterosalpingography as first-choice tubal patency test? A randomized non-inferiority trial. Hum Reprod. 2022;37(5):969-979.
305. van Wely M, Kwan I, Burt AL, et al. Recombinant versus urinary gonadotrophin for ovarian stimulation in assisted reproductive technology cycles. Cochrane Database Syst Rev. 2011; (2):CD005354.
306. Van Wely M, Westergaard LG, Bossuyt PMM, Van der Veen F. Human menopausal gonadotropin versus recombinant follicle stimulation hormone for ovarian stimulation in assisted reproductive cycles. Cochrane Database Syst Rev. 2003;(1):CD003973.
307. Vandekerckhove P, Lilford R, Vail A, et al. Clomiphene or tamoxifen for idiopathic oligo/asthenospermia. Cochrane Database Syst Rev. 1996;(4):CD000151.
308. Vandekerckhove P, Lilford R, Vail A, Hughes E. Androgens versus placebo or no treatment for idiopathic oligo/asthenospermia. Cochrane Database Syst Rev. 1996;(4):CD000150.
309. Vandekerckhove P, Lilford R, Vail A, Hughes E. Bromocriptine for idiopathic oligo/asthenospermia. Cochrane Database Syst Rev. 1996;(4):CD000152.
310. Vandekerckhove P, Lilford R, Vail A, Hughes E. Kinin-enhancing drugs for unexplained subfertility in men. Cochrane Database Syst Rev. 1996;(4):CD000153.
311. Vasilopoulos E, Fragkiadaki P, Kalliora C, et al. The association of female and male infertility with telomere length (Review). Int J Mol Med. 2019;44(2):375-389.
312. Verajankorva E, Laato M, Pollanen P. Analysis of 508 infertile male patients in south-western Finland in 1980-2000: Hormonal status and factors predisposing to immunological infertility. Eur J Obstet Gynecol Reprod Biol. 2003;111(2):173-178.
313. Verhulst SM, Cohlen BJ, Hughes E, et al. Intra-uterine insemination for unexplained subfertility. Cochrane Database Syst Rev. 2006;(4):CD001838.
314. VerMilyea MD, Tan L, Anthony JT, et al. Computer-automated time-lapse analysis results correlate with embryo implantation and clinical pregnancy: A blinded, multi-centre study. Reprod Biomed Online. 2014;29(6):729-736.
315. Wang C, Swerdloff RS. Treatment of male infertility. UpToDate [online serial]. Waltham, MA: UpToDate; reviewed January 2017.

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

316. WebMD. Glossary of fertility terms [website]. 2025. Available at: https://www.webmd.com/baby/glossary-of-fertility-terms. Accessed June 5, 2025.
317. Winger EE, Reed JL. Low circulating CD4(+) CD25(+) Foxp3(+) T regulatory cell levels predict miscarriage risk in newly pregnant women with a history of failure. Am J Reprod Immunol. 2011;66(4):320-328.
318. Wongtra-ngan S, Vutyavanich T, Brown J. Follicular flushing during oocyte retrieval in assisted reproductive techniques. Cochrane Database Syst Rev. 2010;(9):CD004634.
319. World Health Organization (WHO). Current Practices and Controversies in Assisted Reproduction. Report of a Meeting on 'Medical, Ethical and Social Aspects of Assisted Reproduction', WHO Headquarters, Geneva, Switzerland, 17–21 September 2001. E Vayena, PJ Rowe, PD Griffin, eds. Geneva, Switzerland: WHO; 2002.
320. World Health Organization (WHO). WHO Laboratory Manual for the Examination of Human Semen and Sperm-Cervical Mucus Interaction. 4th ed. Cambridge, UK: Cambridge University Press; May 1999.
321. World Health Organization (WHO). WHO laboratory manual for the examination and processing of human semen. 5th ed. Geneva, Switzerland; World Health Organization; 2010.
322. World Health Organization (WHO). WHO Manual for the Standardized Investigation, Diagnosis and Management of the Infertile Male. Cambridge, UK: Cambridge University Press; 2000.
323. Yang XJ, Yuan P, Wu X, et al. Detection of CFTR gene mutations in azoospermia patients with congenital unilateral absence of the vas deferens. Zhonghua Nan Ke Xue. 2015;21(3):229-233.
324. Yossry M, Aboulghar M, D'Angelo A, Gillett W. In vitro fertilisation versus tubal reanastmosis (sterilisation reversal) for subfertility after tubal sterilisation. Cochrane Database Syst Rev. 2006; (3):CD004144.
325. Youssef M, Al-Inany HG, Aboulghar M, et al. Recombinant versus urinary human chorionic gonadotrophin for ovulation induction in assisted conception. Cochrane Database Syst Rev. 2011; (4):CD003719.
326. Yu J, Chen Z, Ni Y, Li Z. CFTR mutations in men with congenital bilateral absence of the vas deferens (CBAVD): A systemic review and meta-analysis. Hum Reprod. 2012;27(1):25-35.
327. Zhang J, Jiang W, Zhou Q, et al. CAG-repeat polymorphisms in the polymerase γ gene and male infertility: A meta-analysis. Andrologia. 2016;48(9):882-889.
328. Zinger M, Liu JH, Thomas MA. Successful use of vaginal sildenafil citrate in two infertility patients with Asherman's syndrome. J Womens Health (Larchmt). 2006;15(4):442-444.

Policy History

- Last Review opens in a new browser pop-up window 12/18/2025

  Effective: 05/20/1999

  Next Review: 07/23/2026

- Review History opens in a new browser pop-up window

- Definitions opens in a new browser pop-up window

Additional Information

close popup

- Clinical Policy Bulletin Notes opens in a new browser pop-up window

You are now leaving the Aetna website.

Links to various non-Aetna sites are provided for your convenience only. Aetna Inc. and its subsidiary companies are not responsible or liable for the content, accuracy, or privacy practices of linked sites, or for products or services described on these sites.

Continue opens in new window

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF